

TIG Stack

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TIG Stack





influxdb

InfluxDB

- **InfluxDB** is a high-performance, open-source **Time Series Database (TSDB)** developed by InfluxData.
- It is specifically designed and optimized for handling, storing, and retrieving **time series data**—data points that are indexed in time order.
- Time series data is characterized by its temporal nature, often having a high volume of entries recorded at regular or irregular intervals.

Architecture and Data Model

- **Optimized for Time Series:** Unlike relational databases, InfluxDB organizes data as time series, where each entry consists of a timestamp, measurement name, one or more fields (values), and tags (key-value metadata for indexing).
- **Line Protocol:** Its ingestion protocol records data as measurement-name tag-set field-set timestamp, supporting efficient write operations at high volumes and fast, indexed lookups using tags.
- **Precision and Storage:** Timestamps can be stored at nanosecond resolution, making it suitable for high-frequency applications (e.g., finance, scientific experiments).
- **Columnar Storage with Compression:** Data blocks are stored sequentially per field and tagset, enabling fast aggregate queries and efficient compression, which slash storage costs by up to 90% on the latest versions.

Features and Capabilities

- **High Throughput:** Handles millions of data points per second with minimal latency, making it ideal for automation, sensor monitoring, and large-scale analytics.
- **Continuous Queries and Downsampling:** Supports automated aggregation and summarization over time (e.g., hourly/daily averages) to maintain long-term trends while optimizing storage.
- **Retention Policies:** Automatically control data lifecycle, cleaning up or downsampling data that is no longer needed at its original resolution.
- **Horizontal Scalability:** Can be deployed as single-node or clustered across several servers for high availability and distributed workloads.
- **Integration:** Connects easily with visualization/monitoring tools like Grafana, integrates with ETL pipelines, and has SDKs for many languages.

Industrial and IoT Relevance

- InfluxDB is central in environments where sensor or event data is generated continuously, supporting both real-time operational dashboards and historical analysis for predictive maintenance and anomaly detection.
- Its open-source core ensures flexibility, while enterprise and cloud offerings provide managed scalability and advanced security for mission-critical deployments.
- InfluxDB enables efficient processing and analysis of time series data, making it a backbone for modern telemetry, analytics, and intelligent automation workflows.

Common Use Cases

InfluxDB is an excellent choice for any scenario where data changes over time and needs to be monitored, analyzed, or visualized in real-time or near real-time.

- **Infrastructure Monitoring:** Collecting and analyzing server metrics (CPU, RAM, network I/O), application performance, and logging data (DevOps/Observability).
- **Internet of Things (IoT):** Storing and analyzing massive streams of sensor data from industrial equipment, smart devices, and edge deployments.
- **Real-time Analytics:** Gaining quick insights for decision-making in areas like financial market analysis (stock tickers) or e-commerce tracking.
- **Energy Management:** Monitoring smart grid data and utility usage.

Token and Bucket

- The concepts of **Token** and **Bucket** are fundamental to InfluxDB, particularly in version 2.x and InfluxDB Cloud.
- They represent the two core functions of **security/authentication** and **data organization**, respectively.

Token (API Token)

The **API Token** is the primary method of **authentication and authorization** in modern InfluxDB. It is a secure, encrypted string that replaces the older username/password method for API interactions.

Key Roles of a Token:

- **Authentication:** The token proves the identity of the client (a user, a script, or an application like Telegraf) making the request. It is included in the Authorization header of every API call.
- **Authorization/Permissions:** A token is not just a key; it defines what actions the holder is permitted to perform and on which resources.

A single token can be granted:

- **Read access** to specific buckets.
- **Write access** to specific buckets.
- Access to other InfluxDB resources (e.g., tasks, dashboards, or users).

- **Types:**
 - **Read/Write Token:** Scoped to grant specific read and/or write permissions on one or more buckets.
 - **All Access Token:** Grants full read and write access to all resources within a single **Organization**.
 - **Operator Token:** Grants full access to all resources across *all* organizations (used for administrative tasks).

Bucket

A **Bucket** is the named location where time-series data is permanently stored in InfluxDB 2.x. It serves as the primary organizational unit for data and is a critical container that controls data life-cycle policies.

Key Roles of a Bucket:

- **Data Storage:** A bucket is the repository for all your time-series data points (which are composed of a measurement, tags, fields, and a timestamp).
- **Data Organization:** It is the highest-level grouping of data within an **Organization**. Conceptually, a bucket is the combination of a **Database** and a **Retention Policy** from older InfluxDB 1.x versions.

Key Roles of a Bucket Cond..

- **Retention Policy:** Every bucket has an associated **retention period** (e.g., 30 days, 1 year, or infinite). This policy automatically deletes data that is older than the specified duration, which is crucial for managing the size and cost of the time-series database.
- **Access Control Target:** Buckets are the resources to which **Tokens** grant access. A token is useless unless it has been explicitly granted read or write permissions on the target bucket.

Feature	Token	Bucket
Purpose	Authentication & Authorization	Data Storage & Organization
Definition	A secure, encrypted string used as an API Key.	A named location that holds time-series data.
Controls	Access permissions (Read/Write) to data and resources.	Data retention policy (how long data is kept).
Relationship	A Token must be created to grant access to a Bucket.	Data must be written to a specific Bucket.



Grafana

- Grafana is an open-source **analytics and interactive visualization web application**.
- Its primary purpose is to allow users to **query, visualize, alert on, and understand metrics, logs, and traces** (collectively known as *observability data*) regardless of where that data is stored.
- In simple terms, it's a powerful tool for creating beautiful, customizable, and interactive **dashboards** that help you monitor the performance and health of your applications, infrastructure, and business.

Key Features and Concepts

Visualization

Grafana provides a wide array of visualization options, known as **Panels**, which can be arranged in a **Dashboard**.

These include:

- **Time-series Graphs:** For showing data changes over time (the most common use).
- **Heatmaps and Histograms:** For analyzing data distribution.
- **Gauge and Stat Panels:** For displaying single, key performance indicators (KPIs).
- **Tables and Text Panels:** For displaying raw data or descriptive information.

Data Sources (The Power of Connectivity)

Grafana does **not** collect or store the data itself; it acts as a universal frontend. It connects to various **Data Sources** to retrieve and display information. This flexibility is its greatest strength.

- **Time Series Databases (TSDBs): Prometheus, InfluxDB** (which you asked about previously), Graphite, etc.
- **Logging Systems:** Elasticsearch, Loki (Grafana Labs' own log aggregation system).
- **Relational/SQL Databases:** PostgreSQL, MySQL, Microsoft SQL Server.
- **Cloud Monitoring:** AWS CloudWatch, Google Cloud Monitoring, Azure Monitor.

Alerting

- Grafana allows you to define rules based on the data displayed in your dashboards.
- When a condition is met (e.g., CPU usage exceeds 90% for five minutes), Grafana can trigger an **alert** and send notifications via various channels like email, Slack, PagerDuty, etc.

Plugins and Extensibility

- Grafana is highly extensible through a robust plugin system.
- Plugins can add support for new data sources, new types of visualization panels, and integrate with third-party tools.

Grafana in the Monitoring Stack

- Grafana is commonly used as the "V" (Visualization) component in modern monitoring stacks, complementing data collection tools:

Component	Role	Example Tool
Collector	Gathers metrics, logs, or traces.	Telegraf, Prometheus Scraper
Storage	Persists the time-series data or logs.	InfluxDB, Prometheus (TSDB), Loki (Log Storage)
Visualization	Connects to storage, queries data, and displays dashboards.	Grafana

Integration with InfluxDB

- Grafana connects directly to InfluxDB as a data source, supporting both InfluxQL and Flux query languages depending on the database version.
- With Grafana 12.2+, users get a redesigned InfluxDB source plugin that includes support for SQL, InfluxQL, and improved query building.
- Dashboards can be built to visualize time series stored in InfluxDB as graphs, tables, gauges, or more complex composites, with full configurability of color, axis, thresholds, and data grouping.
- Alerts, notifications, and pre-built templates for InfluxDB monitoring and system diagnostics are available, including visualizations of logs and performance metrics.

Typical Workflow

- Add InfluxDB as a data source in the Grafana UI, providing the database URL, token, organization name, and selecting the desired query language.
- Use the query editor to select measurements (tables), fields, time ranges, and apply aggregation/grouping for dynamic, real-time visual displays.
- Grafana dashboards integrate seamlessly into industrial IoT and monitoring setups, enabling powerful insight into real-time and historic trends from diverse data streams.

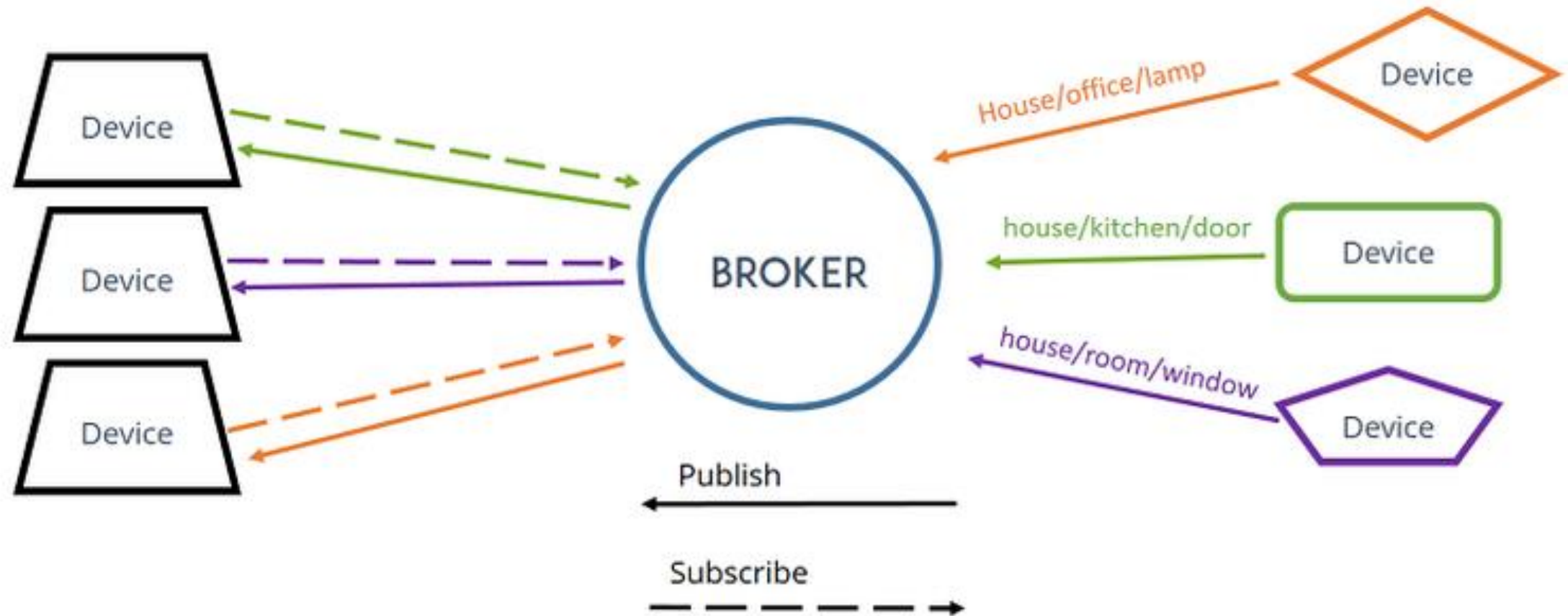
Pairing InfluxDB with Grafana is standard practice in IIoT, industrial automation, and IT monitoring due to their combined efficiency, extensibility, and flexibility for operational analytics and alerting.



MQTT Broker

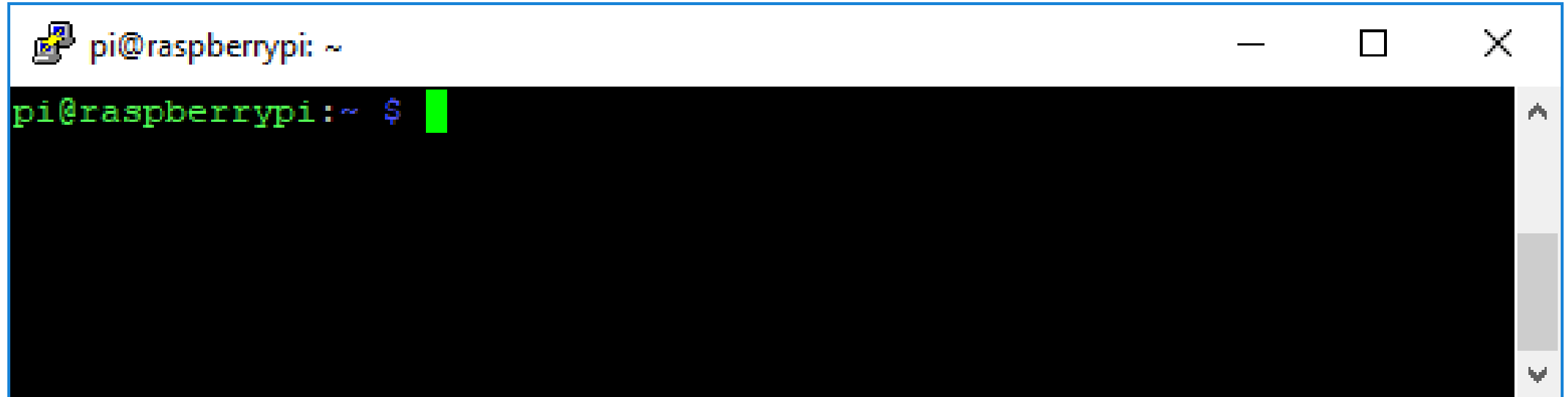
- MQTT stands for **M**essage **Q**ueuing **T**elemetry **T**ransport. MQTT is a simple messaging protocol, designed for constrained devices with low bandwidth. So, it's the perfect solution to exchange data between multiple IoT devices.
- MQTT communication works as a *publish* and *subscribe* system. Devices *publish* messages on a specific topic.
- All devices that are *subscribed* to that topic receive the message.
- The **MQTT broker** is responsible for **receiving** all messages, **filtering** the messages, **deciding** who is interested in them, and then **publishing** the message to all subscribed clients.

MQTT Broker



Installing Mosquitto Broker on Raspberry Pi OS

- Open a new Raspberry Pi terminal window



- Run the following command to upgrade and update your system:

```
sudo apt update && sudo apt upgrade
```

- Press **Y** and **Enter**. It will take some time to update and upgrade (in my case, it took approximately 10 minutes).

- To install the Mosquitto Broker enter these next commands:

```
sudo apt install -y mosquitto mosquitto-clients
```

- To make Mosquitto auto start when the Raspberry Pi boots, you need to run the following command (this means that the Mosquitto broker will automatically start when the Raspberry Pi starts):

```
sudo systemctl enable mosquitto.service
```

- Now, test the installation by running the following command:

```
mosquitto -v
```

- This returns the Mosquitto version that is currently running in your Raspberry Pi. It will be 2.0.11 or above.



pi@raspberrypi: ~



```
Setting up libdlt2:armhf (2.18.6-1) ...
Setting up libwebsockets16:armhf (4.0.20-2) ...
Setting up mosquitto (2.0.11-1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/mosquitto.service →
/lib/systemd/system/mosquitto.service.
Processing triggers for man-db (2.9.4-2) ...
Processing triggers for libc-bin (2.31-13+rpt2+rpil+deb11u2) ...
pi@raspberrypi:~ $ sudo systemctl enable mosquitto.service
Synchronizing state of mosquitto.service with SysV service script with /lib/syst
emd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mosquitto
pi@raspberrypi:~ $ mosquitto -v
1639226740: mosquitto version 2.0.11 starting
1639226740: Using default config.
1639226740: Starting in local only mode. Connections will only be possible from
clients running on this machine.
1639226740: Create a configuration file which defines a listener to allow remote
access.
1639226740: For more details see https://mosquitto.org/documentation/authenticat
ion-methods/
1639226740: Opening ipv4 listen socket on port 1883.
1639226740: Error: Address already in use
1639226740: Opening ipv6 listen socket on port 1883.
1639226740: Error: Address already in use
```

- When starting Mosquitto (version **2.x and later**), you may see this message:
 - “Starting in local only mode. Connections will only be possible from clients running on this machine. Create a configuration file which defines a listener to allow remote access.”
- This indicates that **remote connections are disabled by default**. In Mosquitto **2.0 and above**, you must explicitly configure the broker to allow external devices to connect by creating a configuration file and defining a listener.
- According to the official Mosquitto documentation:
 - “In Mosquitto 2.0 and newer, you must explicitly configure authentication and listener settings before clients can connect. In older versions, the default behavior allowed clients to connect without authentication.”

Enable Remote Access/ Authentication

Mosquitto Broker Enable Remote Access (No Authentication)

- Run the following command to open the mosquitto.conf file.
`sudo nano /etc/mosquitto/mosquitto.conf`
- Move to the end of the file using the arrow keys and paste the following two lines:

```
listener 1883  
allow_anonymous true
```



pi@raspberrypi: ~



GNU nano 5.4

/etc/mosquitto/mosquitto.conf *

Place your local configuration in /etc/mosquitto/conf.d/

#

A full description of the configuration file is at

/usr/share/doc/mosquitto/examples/mosquitto.conf.example

pid_file /run/mosquitto/mosquitto.pid

persistence true

persistence_location /var/lib/mosquitto/

log_dest file /var/log/mosquitto/mosquitto.log

include_dir /etc/mosquitto/conf.d

listener 1883

allow_anonymous true

^G Help

^O Write Out

^W Where Is

^K Cut

^T Execute

^C Location

^X Exit

^R Read File

^_ Replace

^U Paste

^J Justify

^_ Go To Line

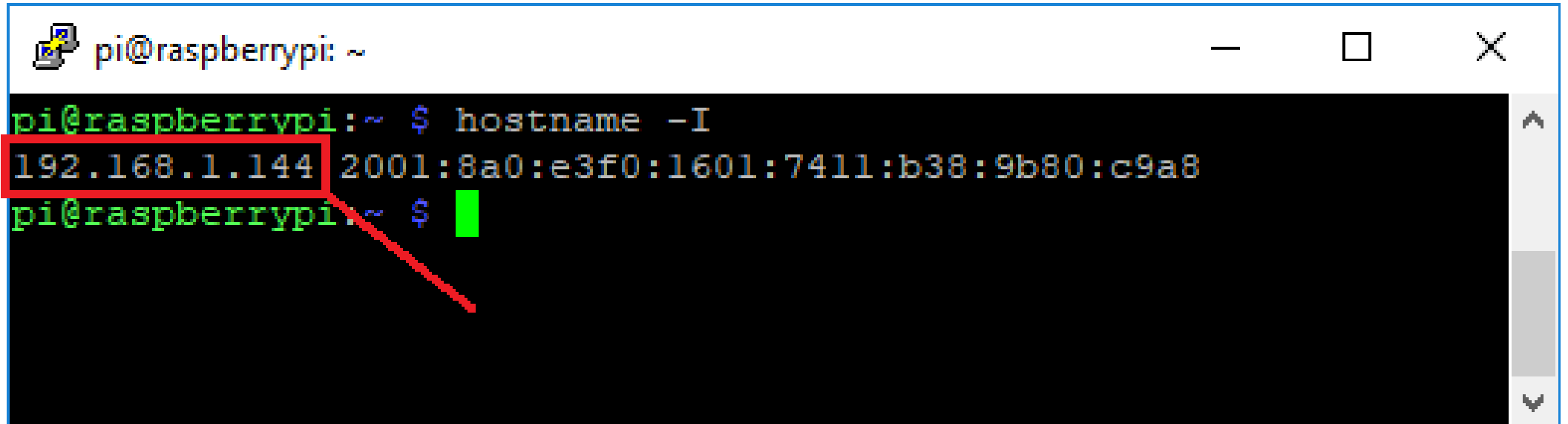
- Then, press **CTRL-X** to exit and save the file. Press **Y** and **Enter**.
- Restart Mosquitto for the changes to take effect.

```
sudo systemctl restart mosquitto
```

Raspberry Pi IP Address

- To use Mosquitto broker later in your projects, you'll need to know the Raspberry Pi IP address.
- To retrieve your Raspberry Pi IP address, type the next command in your Pi Terminal window:

hostname -I

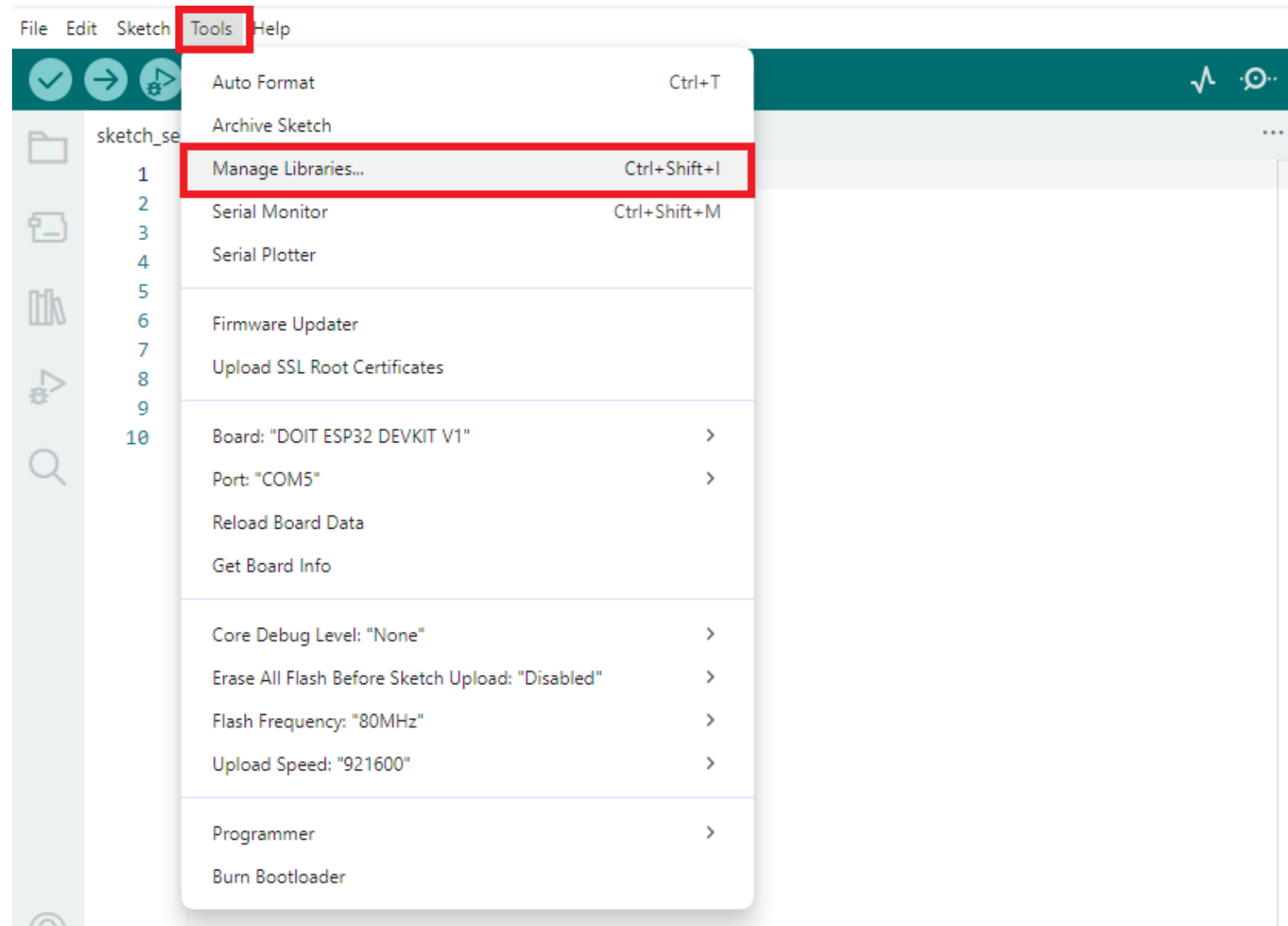


```
pi@raspberrypi: ~  
pi@raspberrypi:~ $ hostname -I  
192.168.1.144 2001:8a0:e3f0:1601:7411:b38:9b80:c9a8  
pi@raspberrypi:~ $
```

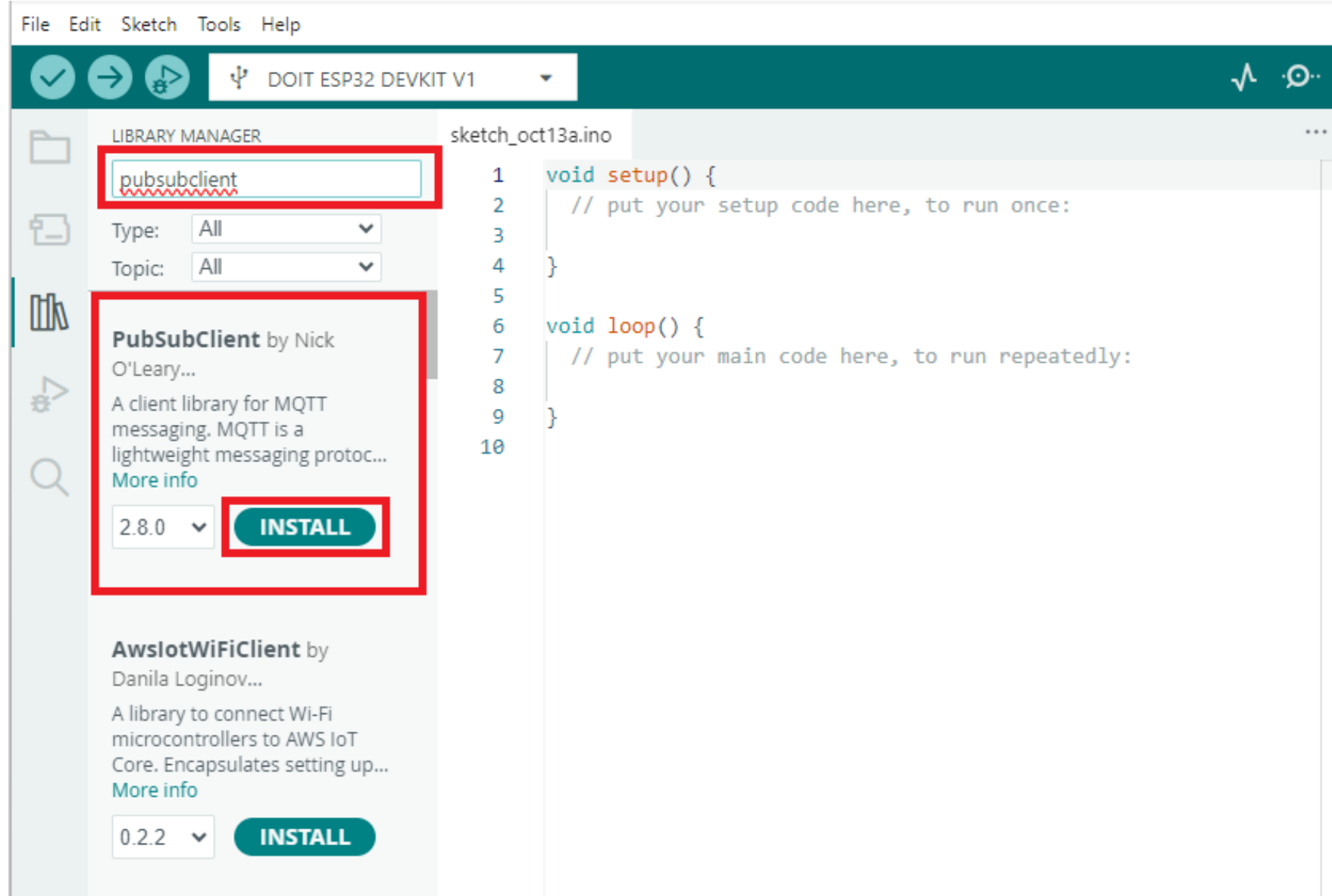


Library Installation - PubSubClient (If required, install the PubSubClient library)

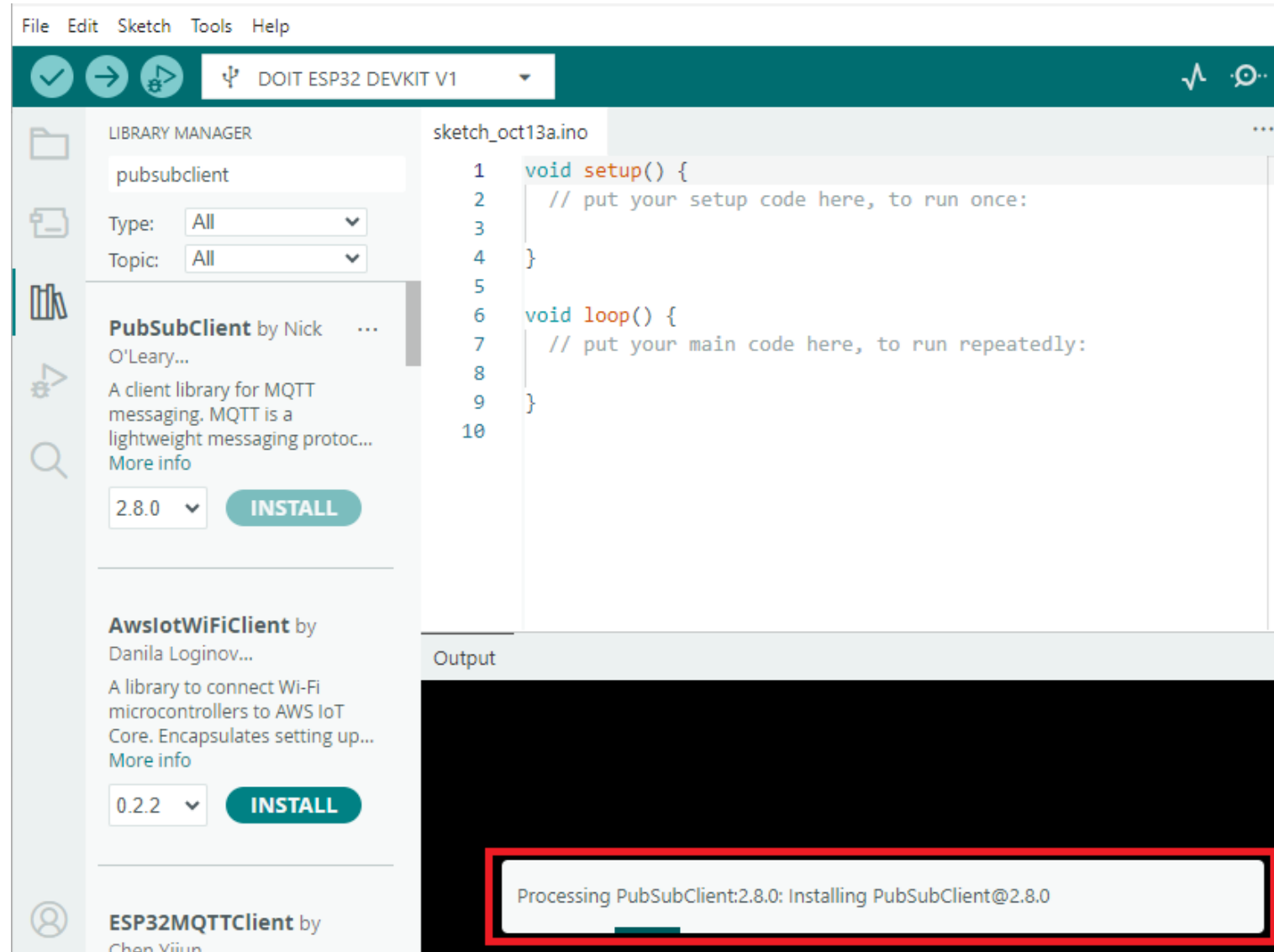
- In the Arduino IDE, go to the Tools menu, then select Manage Libraries... to open the Library Manager.



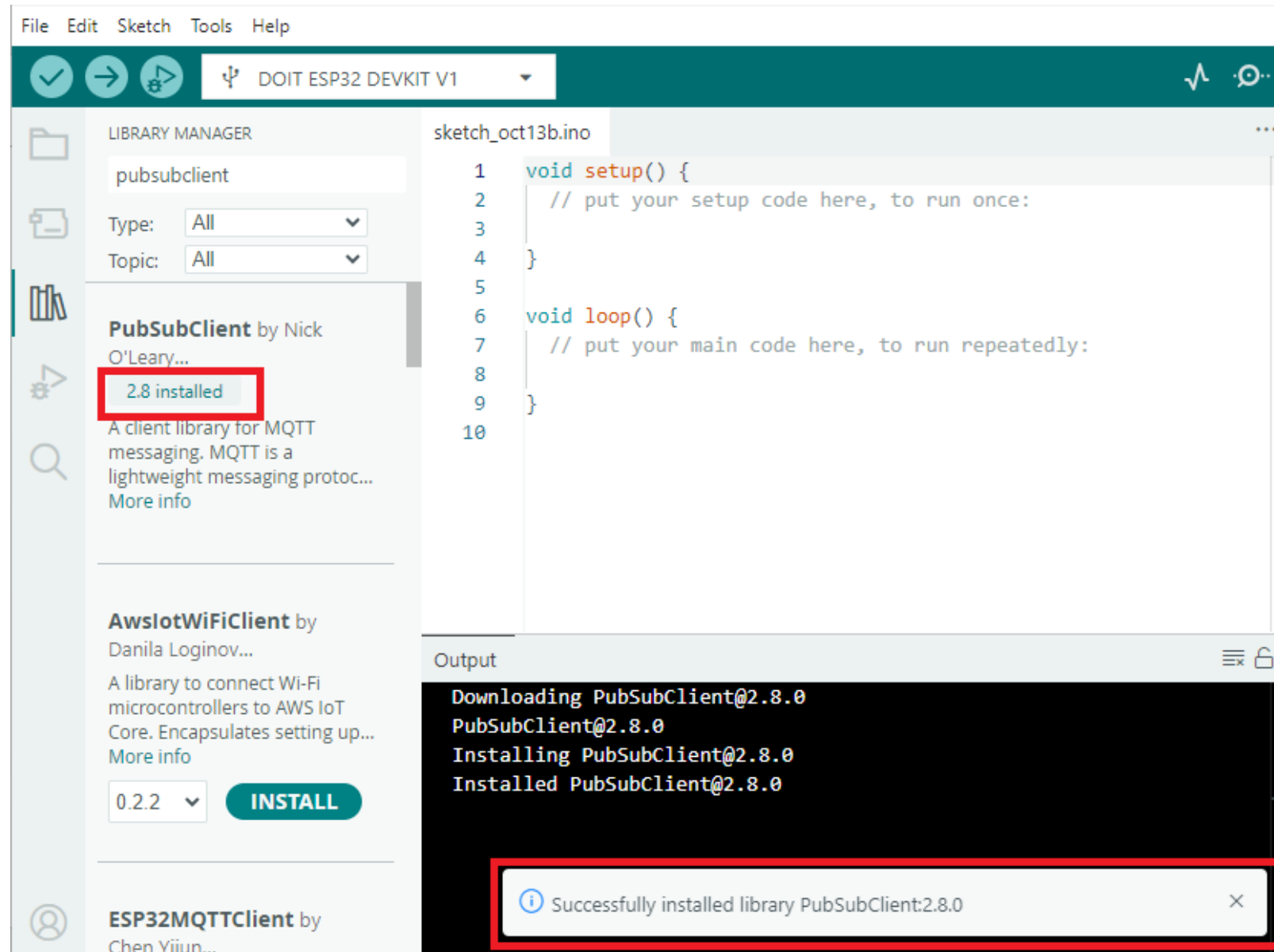
- In the Library Manager search box, type **PubSubClient (by Nick O'Leary)** , then select the **PubSubClientLibrary** and click Install



- The PubSubClient library installation process is in progress. Wait until the installation completes; once finished, you can include the library in your Arduino sketch and start using the MQTT protocol

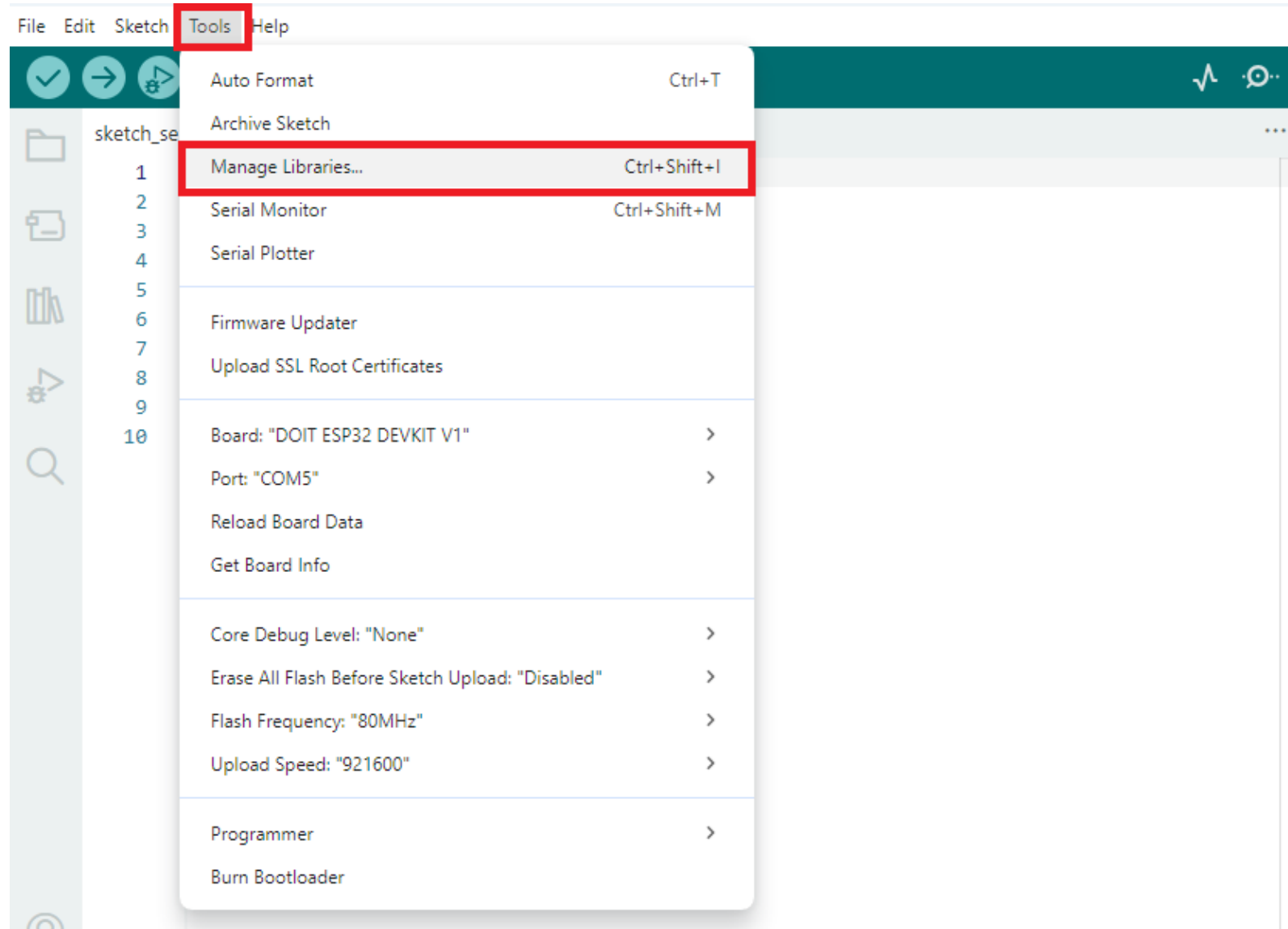


- After the installation is complete, It shows Installed.

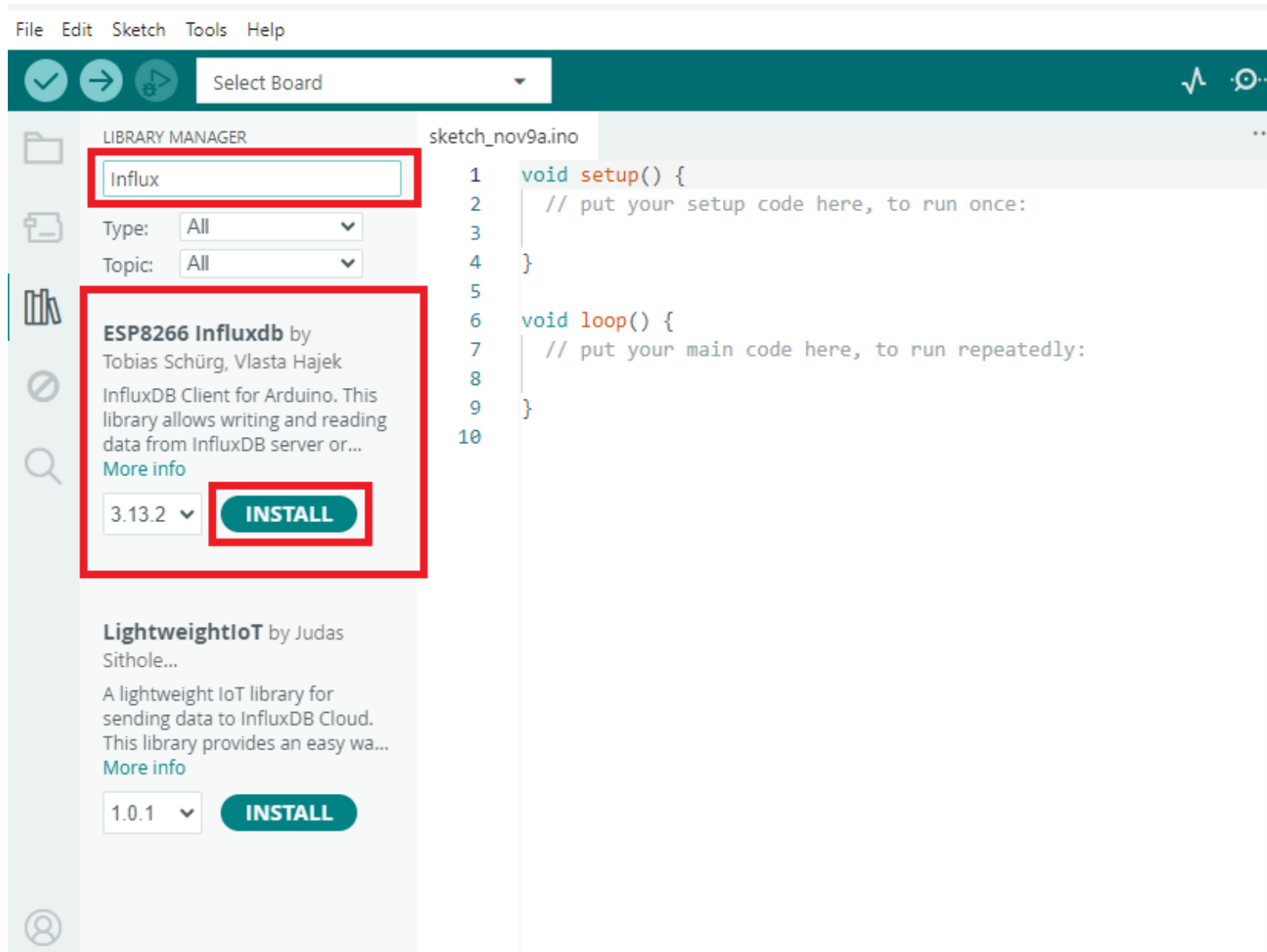


Library Installation InfluxDB

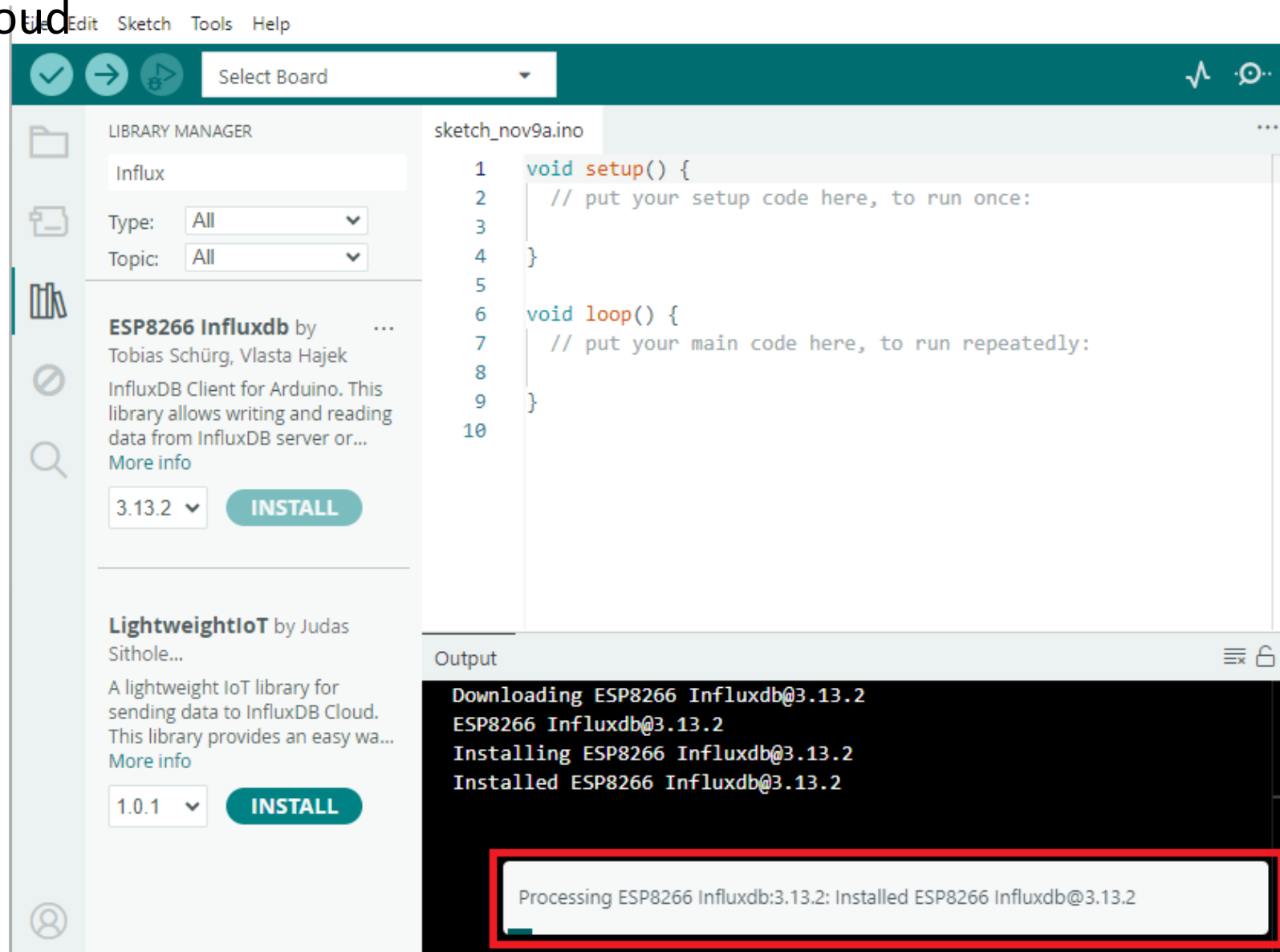
- In the Arduino IDE, go to the Tools menu, then select Manage Libraries... to open the Library Manager.



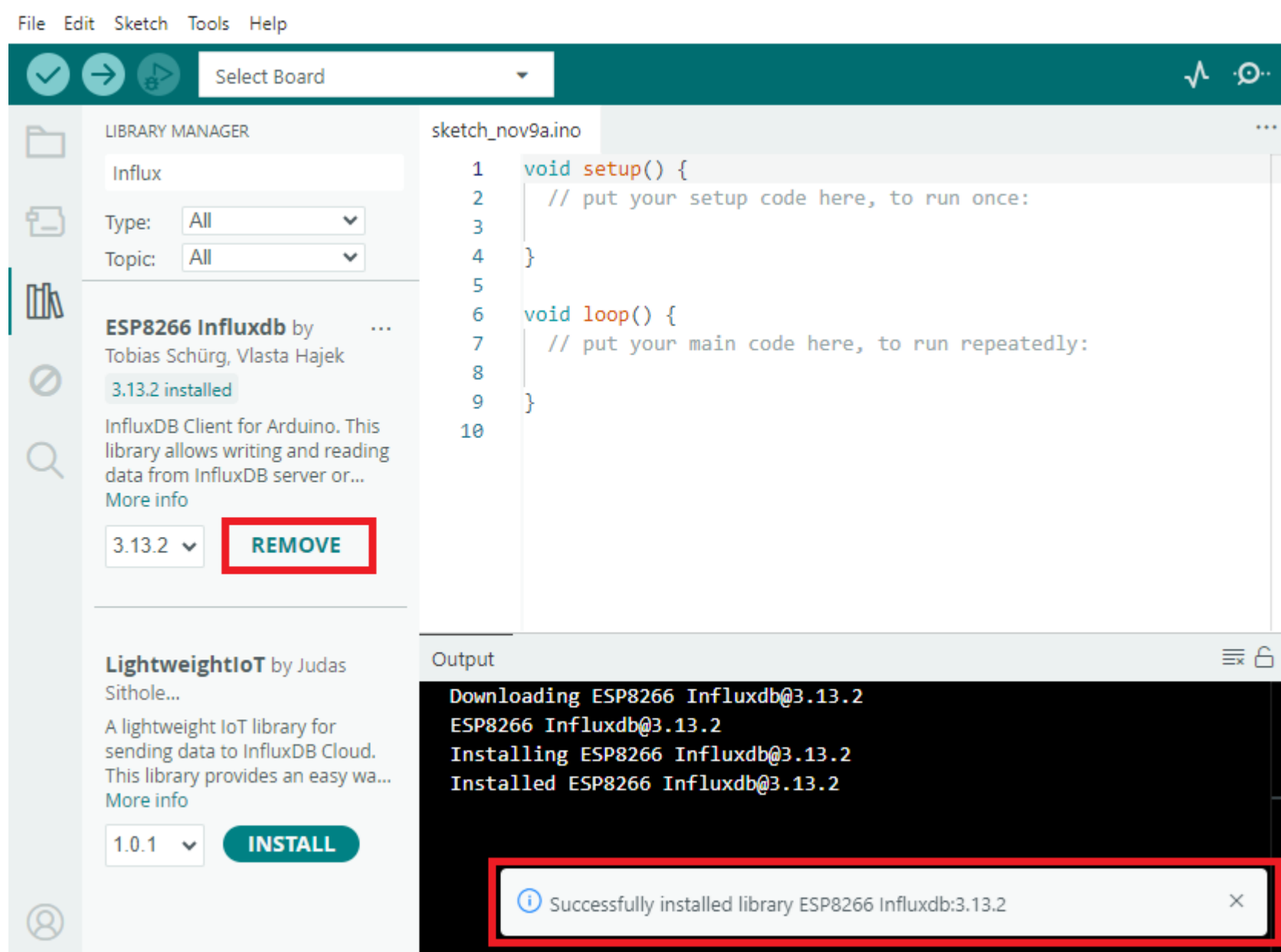
- In the Library Manager search box, type **Influx**, then select the ESP8266 influxdb and click Install



- The **ESP8266 influx** library installation process is in progress. Wait until the installation completes; once finished, you can include the library in your Arduino sketch and start using the Influxdb Cloud



- After the installation is complete, the 'Install' button changes to 'Remove'



Arduino Code

```
// ----- HEADER FILES -----  
#include <WiFi.h>  
#include <WiFiClientSecure.h>    // Required for HTTPS (InfluxDB Cloud)  
#include <InfluxDbClient.h>  
#include <InfluxDbCloud.h>  
#include <PubSubClient.h>  
  
// ----- WIFI -----  
#define WIFI_SSID      "Pradeep"  
#define WIFI_PASSWORD  "123456789"  
  
// ----- INFLUXDB CLOUD SETTINGS -----  
#define INFLUXDB_URL    "https://us-east-1-1.aws.cloud2.influxdata.com"  
#define  
INFLUXDB_TOKEN  "9htuKmvvtH5ynU9mznTsFc_nN5ctz0oJ47VUHNkhEkjvWUVxMjpe686moPWTn7W9GtDWNB6Z1zQF  
9oQ0aLIwFw=="  
#define INFLUXDB_ORG    "tektork"  
#define INFLUXDB_BUCKET "SensorData"  
#define INFLUXDB_MEASUREMENT "esp32_sensor_data"
```

```
// Time Zone (IST = UTC + 5:30)
#define TZ_INFO "IST-5:30"

// ----- MQTT (Raspberry Pi - Mosquitto) -----
#define MQTT_BROKER      "192.168.95.75"
#define MQTT_PORT        1883
#define MQTT_CLIENT_ID   "esp32_sensor_client"

// Topics
#define MQTT_TOPIC_LDR    "home/esp32/ldr"
#define MQTT_TOPIC_POT    "home/esp32/pot"
#define MQTT_TOPIC_JSON   "home/esp32/sensors"

// ----- SENSOR PINS -----
const int LDR_PIN = 39;    // GPIO36
const int POT_PIN = 36;    // GPIO34

// Reading interval (10 seconds)
const int READ_INTERVAL_MS = 10000;
```

```
// ----- OBJECTS -----
```

```
// Secure Client for HTTPS
```

```
WiFiClientSecure wifiClient;
```

```
// InfluxDB client without Secure client in constructor
```

```
InfluxDBClient client(INFLUXDB_URL, INFLUXDB_ORG, INFLUXDB_BUCKET, INFLUXDB_TOKEN);
```

```
// Data structure for point
```

```
Point sensorPoint(INFLUXDB_MEASUREMENT);
```

```
// MQTT
```

```
WiFiClient mqttWiFiClient;
```

```
PubSubClient mqttClient(mqttWiFiClient);
```

```
// ----- FUNCTIONS -----
```

```
void connectWiFi()  
{  
    Serial.print("Connecting to WiFi");  
    WiFi.begin(WIFI_SSID, WIFI_PASSWORD);  
  
    while (WiFi.status() != WL_CONNECTED)  
    {  
        Serial.print(".");  
        delay(400);  
    }  
    Serial.println("\n WiFi Connected");  
    Serial.print("IP Address: ");  
    Serial.println(WiFi.localIP());  
}
```



```
void connectMQTT()
{
    Serial.print("Connecting to MQTT...");
    while (!mqttClient.connected())
    {
        if (mqttClient.connect(MQTT_CLIENT_ID))
        {
            Serial.println("Connected.");
            mqttClient.publish("home/esp32/status", "ESP32 is online");
        }
    }
    else
    {
        Serial.print("Failed, state=");
        Serial.println(mqttClient.state());
        delay(2000);
    }
}
```

```
// ----- SETUP -----  
void setup() {  
    Serial.begin(115200);  
    pinMode(LDR_PIN, INPUT);  
    pinMode(POT_PIN, INPUT);  
    connectWiFi();  
    // Allow HTTPS communication without certificate  
    wifiClient.setInsecure();  
    client.setInsecure();  
  
    timeSync(TZ_INFO, "pool.ntp.org", "time.nis.gov");  
  
    sensorPoint.addTag("device", "esp32-01");  
    sensorPoint.addTag("location", "home_lab");  
  
    if (client.validateConnection()) {  
        Serial.println(" Connected to InfluxDB Cloud");  
    } else {  
        Serial.print(" InfluxDB connection failed: ");  
        Serial.println(client.getLastErrorMessage());  
    }  
    mqttClient.setServer(MQTT_BROKER, MQTT_PORT);  
    connectMQTT();  
}
```

```
// ----- LOOP -----
```

```
void loop() {  
  
    if (WiFi.status() != WL_CONNECTED) connectWiFi();  
    if (!mqttClient.connected()) connectMQTT();  
  
    sensorPoint.clearFields();  
  
    int ldrRaw = analogRead(LDR_PIN);  
    int potRaw = analogRead(POT_PIN);  
  
    float lightPercent = map(ldrRaw, 0, 4095, 100, 0);  
    float potPercent    = map(potRaw, 0, 4095, 0, 100);  
  
    sensorPoint.addField("ldr_raw", ldrRaw);  
    sensorPoint.addField("pot_raw", potRaw);  
    sensorPoint.addField("light_percent", lightPercent);  
    sensorPoint.addField("pot_percent", potPercent);  
}
```

```
// ----- SEND TO INFLUXDB CLOUD -----
```

```
if (!client.writePoint(sensorPoint)) {  
    Serial.print(" InfluxDB write failed: ");  
    Serial.println(client.getLastErrorMessage());  
} else {  
    Serial.printf(" InfluxDB write OK → LDR=%d  POT=%d\n", ldrRaw, potRaw);  
}
```

```
// ----- SEND TO MQTT (Raspberry Pi) -----
```

```
char jsonPayload[128];  
snprintf(jsonPayload, sizeof(jsonPayload),  
    "{\"ldr_raw\":%d,\"pot_raw\":%d,\"light_percent\":%.2f,\"pot_percent\":%.2f}\",  
    ldrRaw, potRaw, lightPercent, potPercent);
```

```
mqttClient.publish(MQTT_TOPIC_JSON, jsonPayload);  
mqttClient.publish(MQTT_TOPIC_LDR, String(ldrRaw).c_str());  
mqttClient.publish(MQTT_TOPIC_POT, String(potRaw).c_str());
```

```
Serial.println(" MQTT Published: ");  
Serial.println(jsonPayload);  
mqttClient.loop();  
delay(READ_INTERVAL_MS);
```

```
}
```

Serial monitor Output



The screenshot shows a serial monitor window titled "Output Serial Monitor X". It features a text input field at the top with the placeholder "Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM5')". To the right of the input field are two dropdown menus: "No Line Ending" and "115200 baud". The main area of the window displays a series of log messages from a device. A red rectangular box highlights the text from "Connecting to WiFi..." down to the JSON object. The status bar at the bottom is dark teal and displays "Ln 91, Col 24" followed by "DOIT ESP32 DEVKIT V1 on COM5", a bell icon, and a window icon.

```
Connecting to WiFi...
WiFi Connected
IP Address: 192.168.95.77
Syncing time
Synchronized time: Sat Nov  8 16:17:51 2025

Connected to InfluxDB Cloud
Connecting to MQTT...Connected.
InfluxDB write OK → LDR=1300  POT=1791
MQTT Published:
{"ldr_raw":1300,"pot_raw":1791,"light_percent":69.00,"pot_percent":43.00}
```

Ln 91, Col 24 DOIT ESP32 DEVKIT V1 on COM5

Influxdb Configuration

- Open your web browser and go to- <https://www.influxdata.com/index/>

The screenshot shows the InfluxData website homepage. At the top, a green banner reads "Meet us at AWS re:Invent: Learn what is new with InfluxDB and get a demo." with a "Learn More" link. The navigation bar includes the InfluxData logo, links for Products, Use Cases, Developers, and Pricing, and buttons for Contact Us, Sign In, and Start Now. The main content area features the headline "Real-world AI starts with time series." followed by "Time series starts with InfluxDB." and "Join the millions of developers building real-time systems with InfluxDB, the leading time series database." Below this are "Get InfluxDB" and "Contact Sales" buttons. A large image on the right shows a "Configuration" sidebar and a "Asset Monitoring" dashboard with various charts like "Humidity & Temperature", "Co Levels by Sensor", "Sensor Connection", and "Vibration mm/s". A "Run Query" dialog is also visible. The footer displays logos for Honeywell, Rocket Lab, Intuit, NVIDIA, Capital One, Leolabs, Unity, and Siemens Energy, with the text "Developers choose InfluxDB" at the bottom.

Meet us at AWS re:Invent: Learn what is new with InfluxDB and get a demo. [Learn More](#)

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Real-world AI starts with time series.

Time series starts with InfluxDB.

Join the millions of developers building real-time systems with InfluxDB, the leading time series database.

[Get InfluxDB](#) [Contact Sales](#)

Configuration

- System Overview
- Configure
- Manage Databases
- Manage Tokens
- Manage Plugins
- Write Data
- Query Data
- Dashboards
- Get Help

Asset Monitoring

Plant 10.26 - DB95

Humidity & Temperature

Co Levels by Sensor

Sensor Connection

Vibration mm/s

Run Query

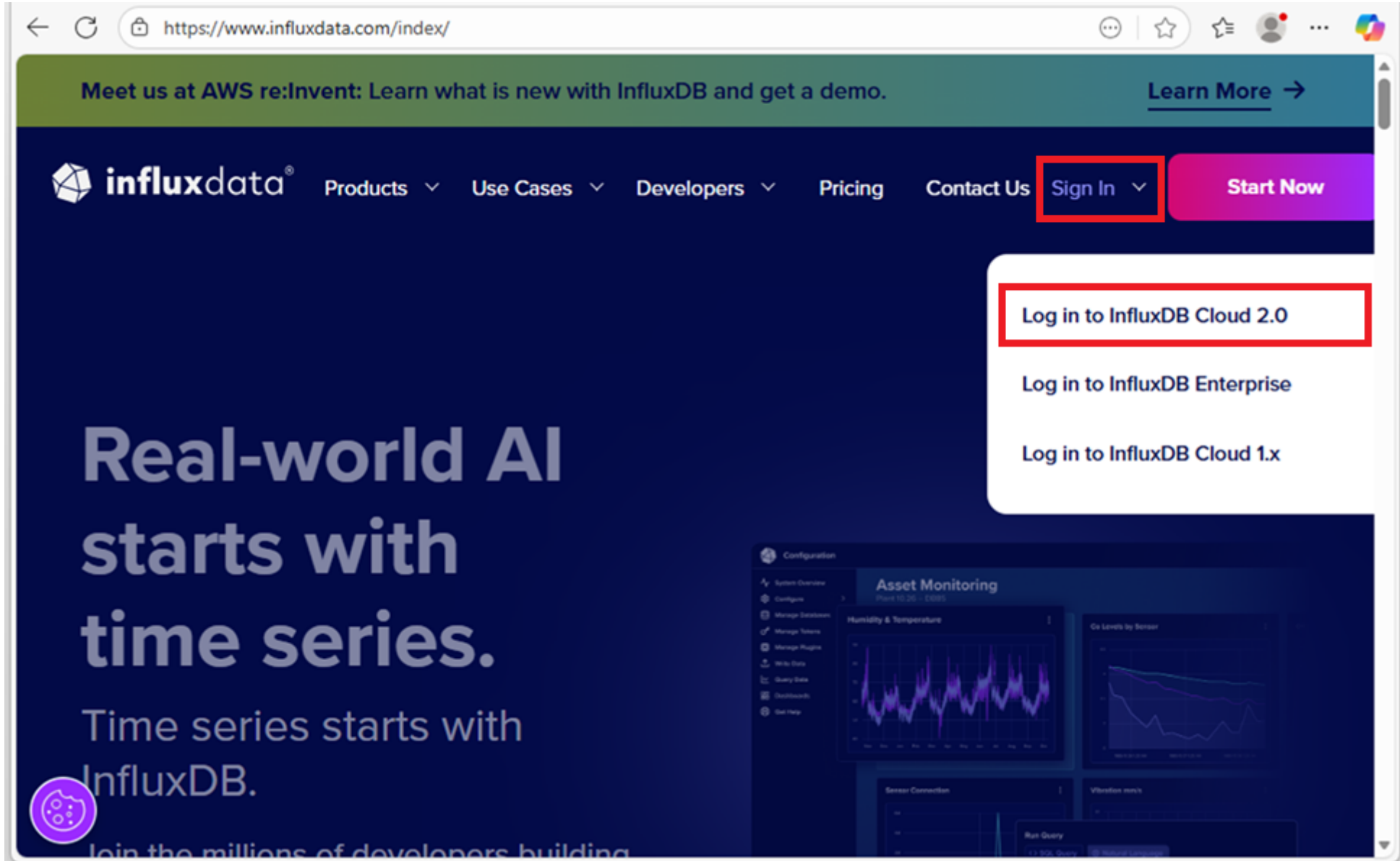
[SQL Query](#) [Natural Language](#)

How has compressor vibration trended at the Dayton plant over the last 7 hours?

Honeywell ROCKETLAB intuit NVIDIA Capital One LEOLABS Unity SIEMENS energy

Developers choose InfluxDB

- First, click on the 'Sign In' option, then choose 'Log in to InfluxDB Cloud 2.0'



The screenshot shows the InfluxData website homepage. The navigation bar includes links for Products, Use Cases, Developers, Pricing, Contact Us, and a Sign In dropdown menu. The Sign In dropdown menu is open, showing three options: Log in to InfluxDB Cloud 2.0, Log in to InfluxDB Enterprise, and Log in to InfluxDB Cloud 1.x. The main content area features a large heading 'Real-world AI starts with time series.' and a subheading 'Time series starts with InfluxDB.' Below the heading is a small circular icon with a brain-like pattern. The bottom of the page shows a preview of the InfluxDB dashboard with various charts and graphs.

Meet us at AWS re:Invent: Learn what is new with InfluxDB and get a demo. [Learn More →](#)

influxdata Products ▾ Use Cases ▾ Developers ▾ Pricing Contact Us **Sign In ▾** [Start Now](#)

Real-world AI starts with time series.

Time series starts with InfluxDB.

Log in to InfluxDB Cloud 2.0

Log in to InfluxDB Enterprise

Log in to InfluxDB Cloud 1.x

Join the millions of developers building

- Here In the Sign In window, select the *Google* option to continue with your Google account.

← ↻ <https://cloud2.influxdata.com/signup> ☆ ☆ 👤 ...

influxdb cloud

Create your Free InfluxDB Cloud Serverless Account

No credit card required

Continue with

 **GOOGLE**

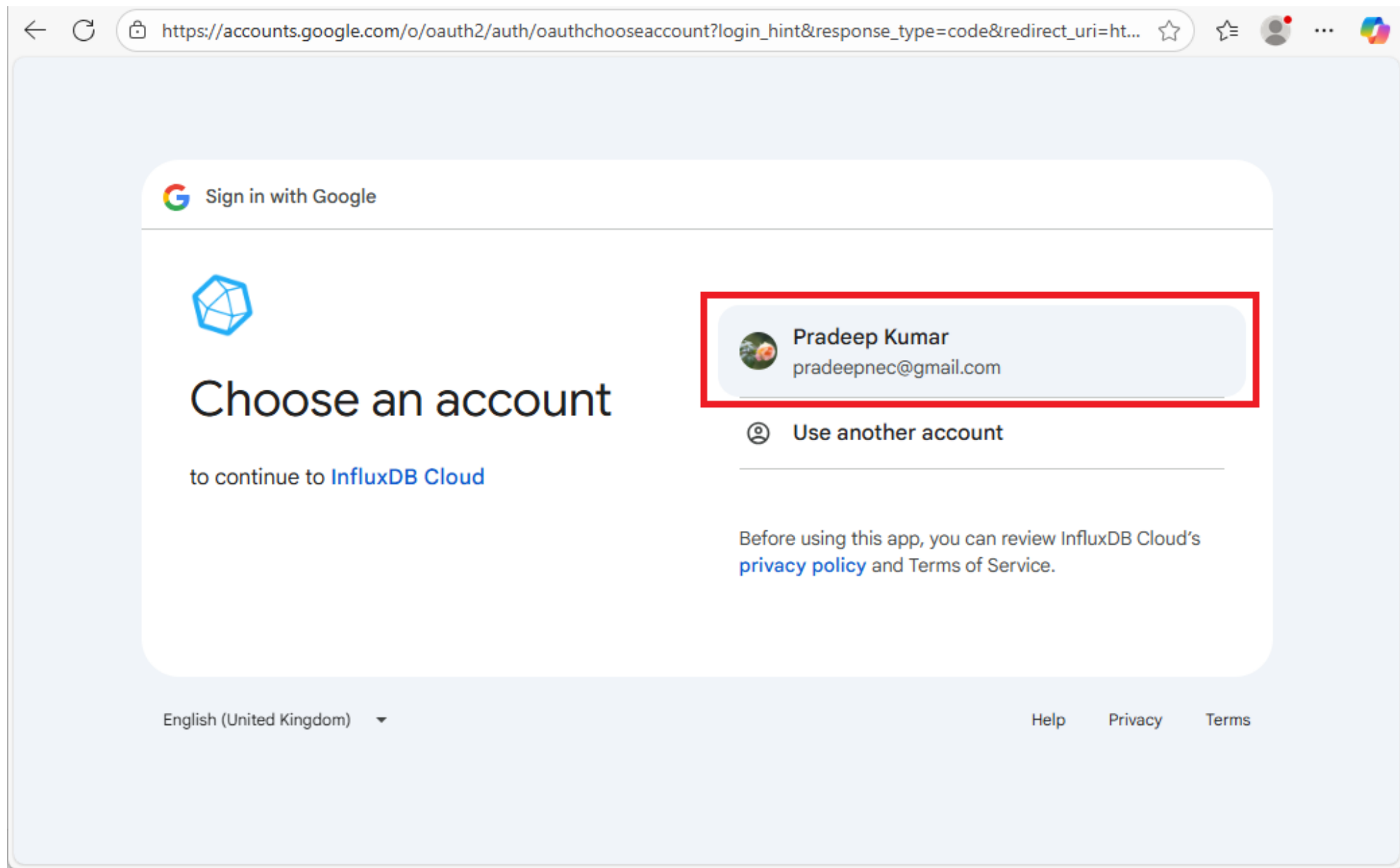
 **MICROSOFT**

OR

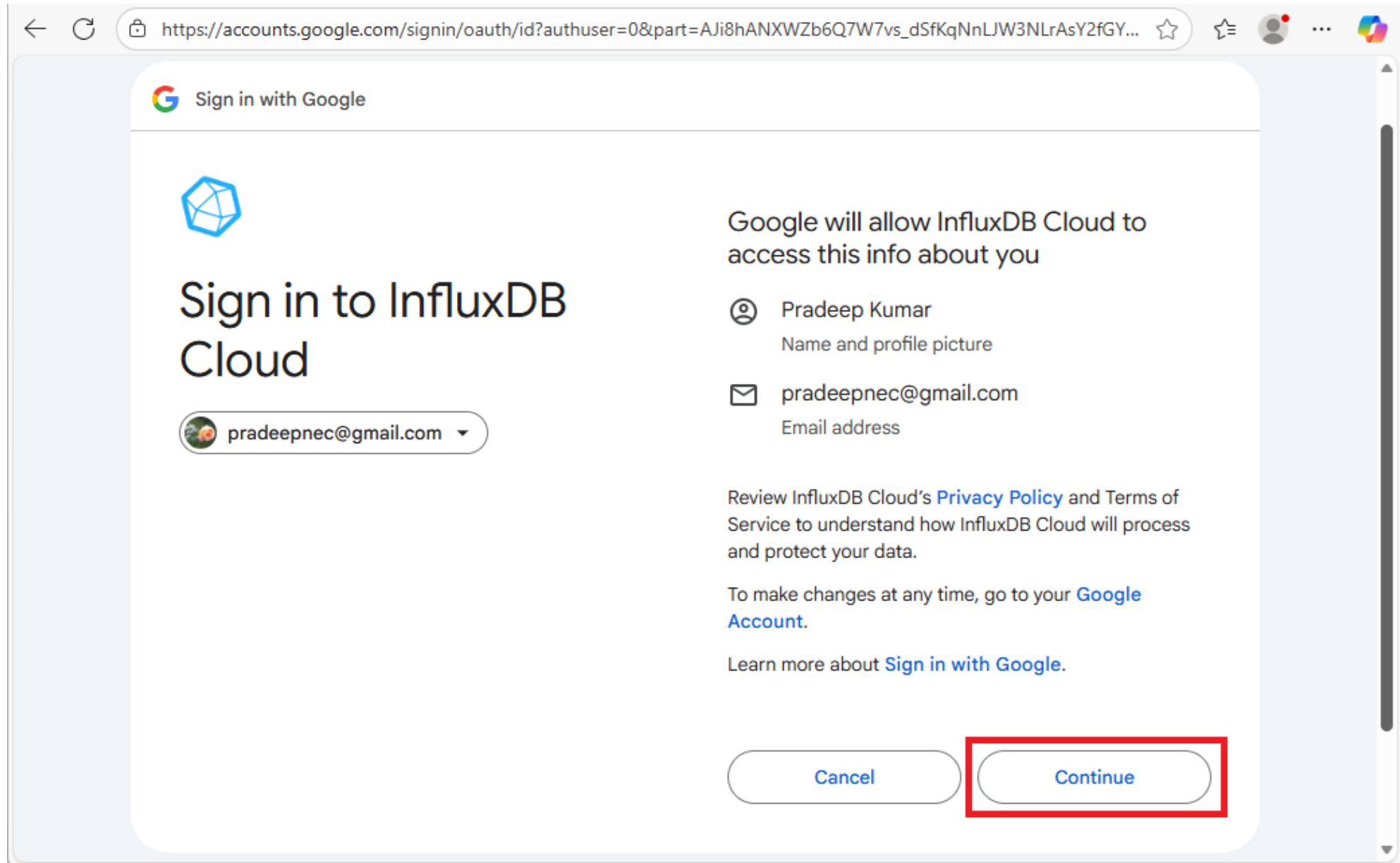
LOG IN **SIGN UP**

First Name* Last Name*

- Then choose your Google account if you are already logged in to Google.



- After selecting your Google account, click *Continue* to proceed.



- Enter the workspace details, accept the license agreement, and finally click the 'Continue' button.

← ↻ https://cloud2.influxdata.com/splash

Welcome to InfluxData

Just a few things before we get started. No credit card required.

Workspace Details

Account

Your company name

Tektork

Organization

Your first organization or project in your account

Tektork

Storage Provider

This is where we will store your time series data.

Amazon Web Services

aws

US East (N. Virginia)

Don't see the region you need? [Let us know.](#)

☒ I have viewed and agree to the [InfluxDB Cloud 2.0 Services Subscription Agreement](#) and [InfluxData Global Data Processing Agreement](#).

CONTINUE

- Select the desired account type. Click the '**Keep**' button to confirm your selection.

← ↻ <https://cloud2.influxdata.com/plans> 🔍 ☆ ⌵ 👤 ⋮

influxdb cloud™

Want a plan with more?
You can upgrade later at any time.

Usage-Based	Free	Annual
I know my workload	I'm just trying this out	Enterprise scale
		
Unlimited Storage Retention	30 days Storage	Unlimited Storage Retention
All Alert Handlers	Slack Only	All Alert Handlers
10,000 kb/s Reads 1,000 kb/s Writes Up to 1,000,000 Series	1,000 kb/s Reads 17 kb/s Writes Up to 10,000 Series	Scale Reads As Needed Scale Writes As Needed Scale Series As Needed
Billed monthly	No credit card required	Yearly Contract
UPGRADE NOW	KEEP	CONTACT SALES

- Once your account is created, The **Resource Center** window will automatically open.

The screenshot shows the InfluxDB Resource Center dashboard in a web browser. The browser's address bar displays the URL: `https://us-east-1-1.aws.cloud2.influxdata.com/orgs/e058e35b63bb4cfa`. The dashboard has a dark theme with a sidebar on the left containing navigation icons. The main content area is titled "Resource Center" and features a section "What would you like to do?" with four interactive cards: "Manage Databases & Security" (with a database icon), "Add Data" (with an upload icon), "Query Data" (with a code icon), and "Visualize & Alert" (with a chart icon). To the right, there is a "What's New" section featuring a banner for "INFLUXDB UNIVERSITY" and a new training article titled "New Training: Flux to InfluxQL, SQL, and Python". Below this, a "Latest Blogs" section displays a blog post from November 6th titled "How to Visualize Time Series Data with InfluxDB 3 & Apache Superset". The top right of the dashboard includes a user profile icon labeled "PK", a "Get \$250 free credit" message, and a green "UPGRADE NOW" button.

← ↻ `https://us-east-1-1.aws.cloud2.influxdata.com/orgs/e058e35b63bb4cfa` ☆ ☆ 👤 ... 🌐

Tektork > Tektork Get \$250 free credit **UPGRADE NOW** PK

Resource Center

What would you like to do?

-  **Manage Databases & Security**
Create and manage your buckets (databases) & access tokens.
-  **Add Data**
Write data into your database with our reporting agent, programmatically, API, CLI, or upload a CSV or Line Protocol File.
-  **Query Data**
Query your data with the UI, programmatically, or integrate with 3rd party tools.
-  **Visualize & Alert**
Integrate with 3rd party tools to visualize your data or set up alerts.

What's New

**INFLUXDB UNIVERSITY**

New Training: Flux to InfluxQL, SQL, and Python

Learn how to translate Flux queries to InfluxQL, SQL, and Python in this technical tutorial. We cover some basic Flux queries and their translations, discuss the limitations of SQL and InfluxQL, and make the case for using Pandas and Polars. Discover the advantages and limitations of each language and get examples of Flux to InfluxQL, SQL, and Python translations.

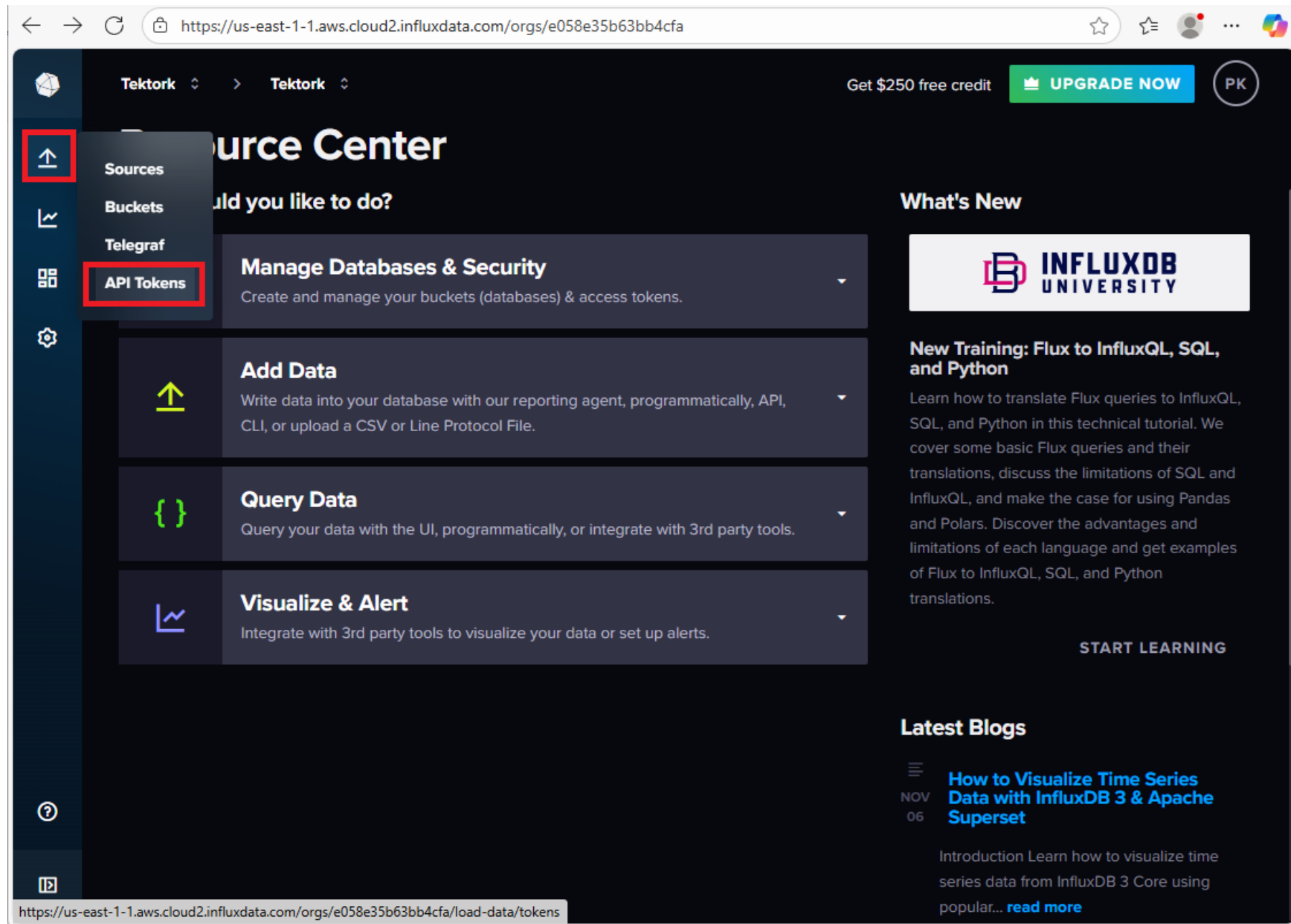
START LEARNING

Latest Blogs

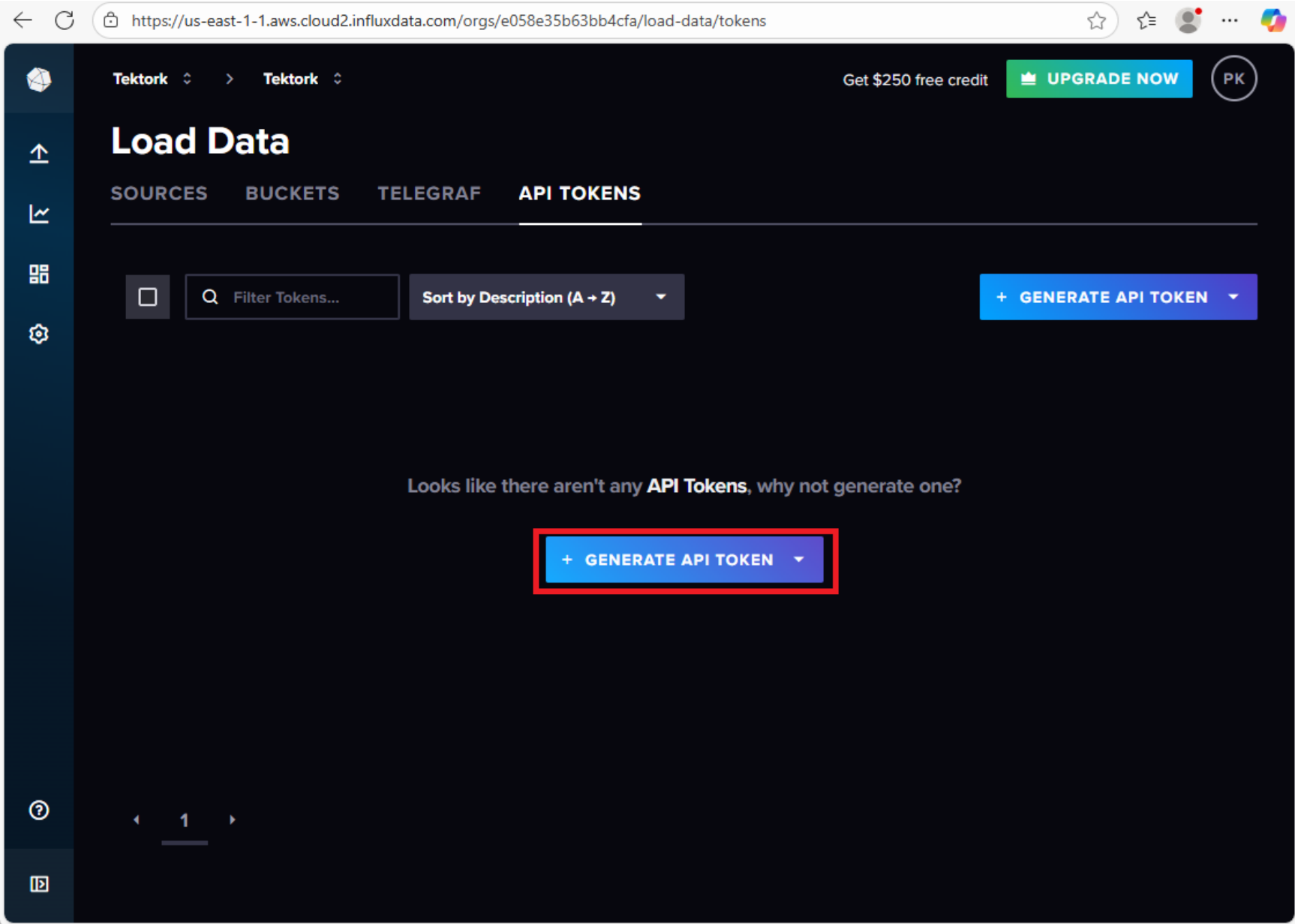
NOV 06 **How to Visualize Time Series Data with InfluxDB 3 & Apache Superset**

Introduction Learn how to visualize time series data from InfluxDB 3 Core using popular... [read more](#)

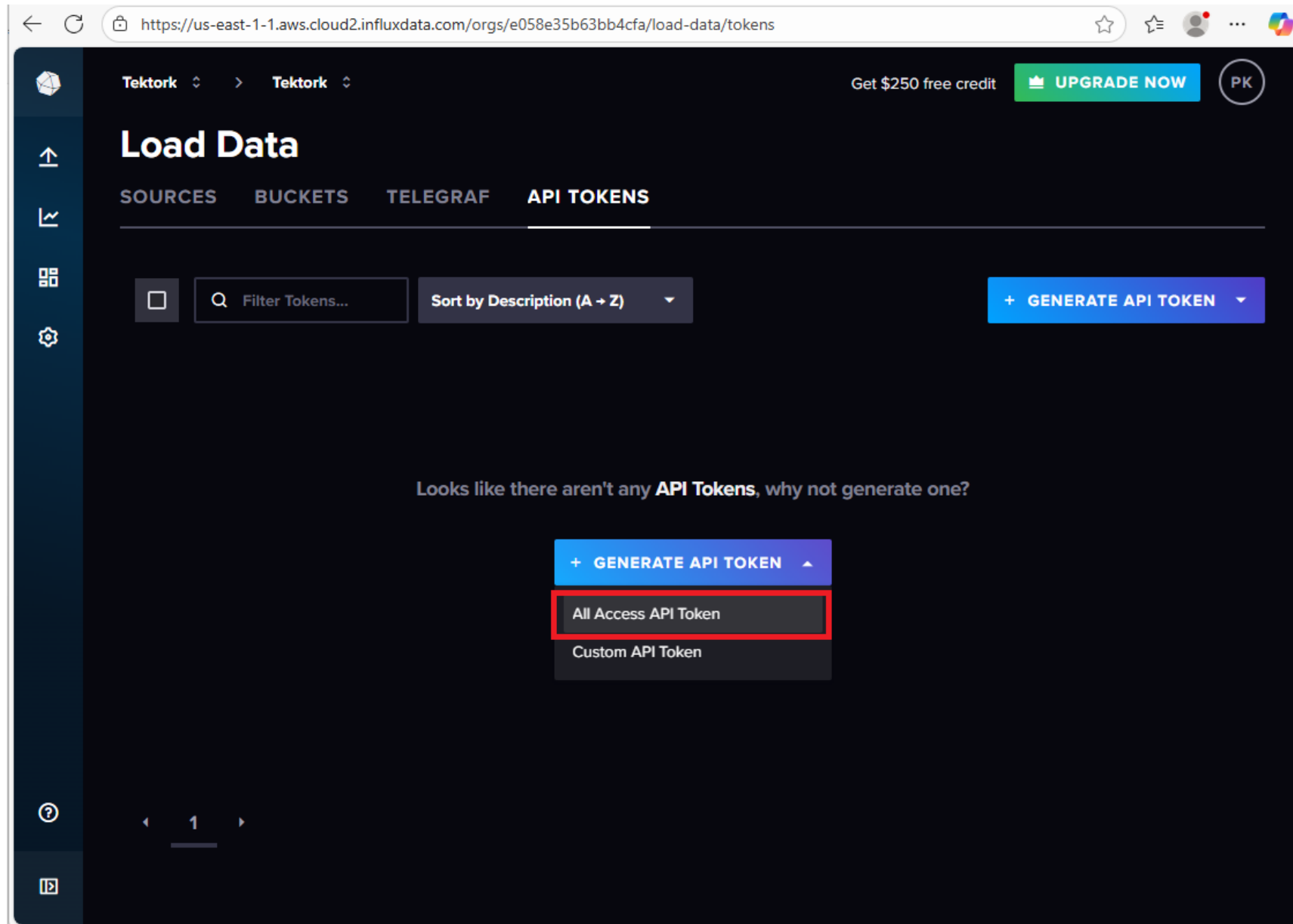
- In the left-side panel of the InfluxDB window, select the 'API Tokens' option under 'Load Data



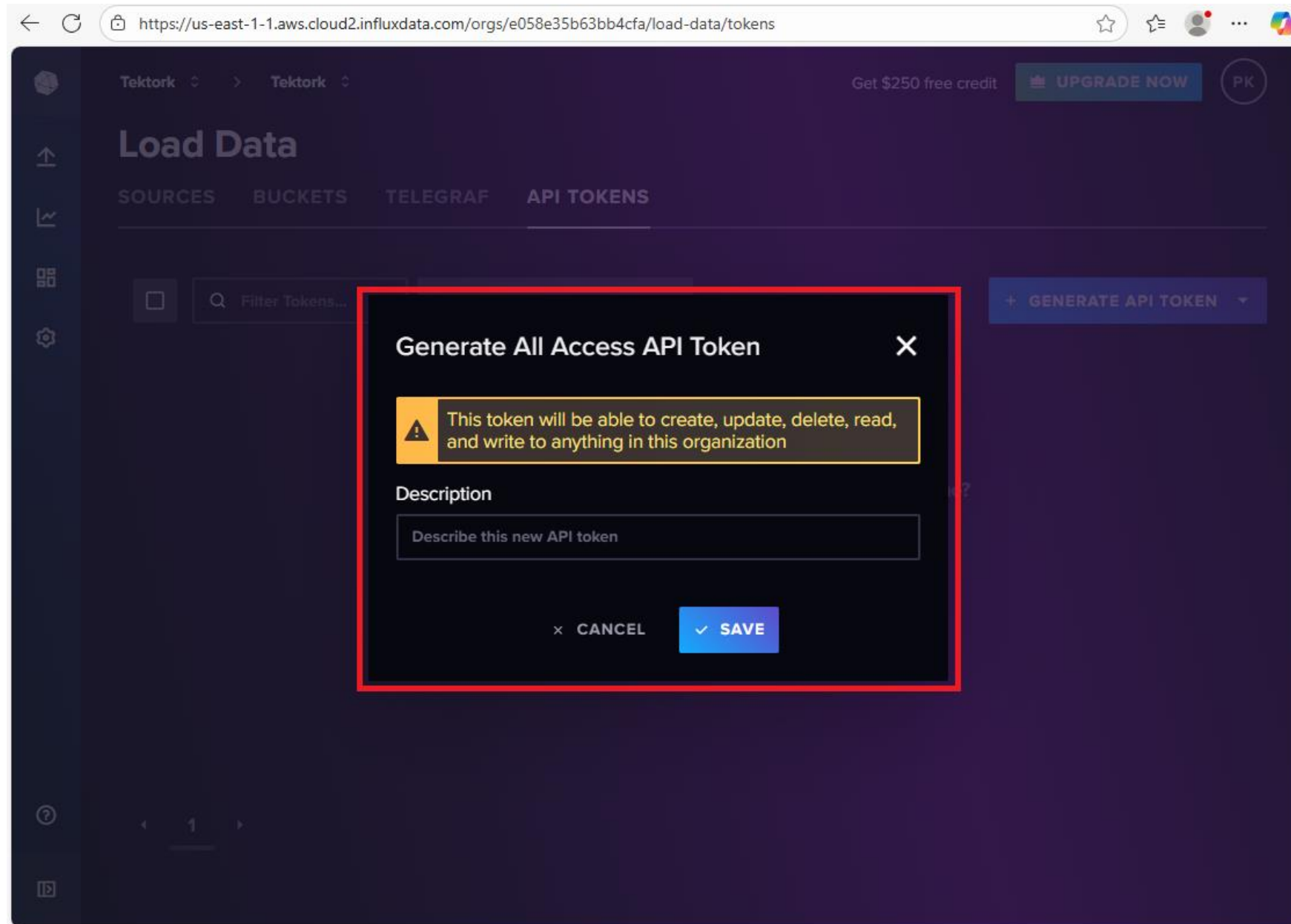
- Click the 'Generate API Token' button in the 'API Tokens' section.



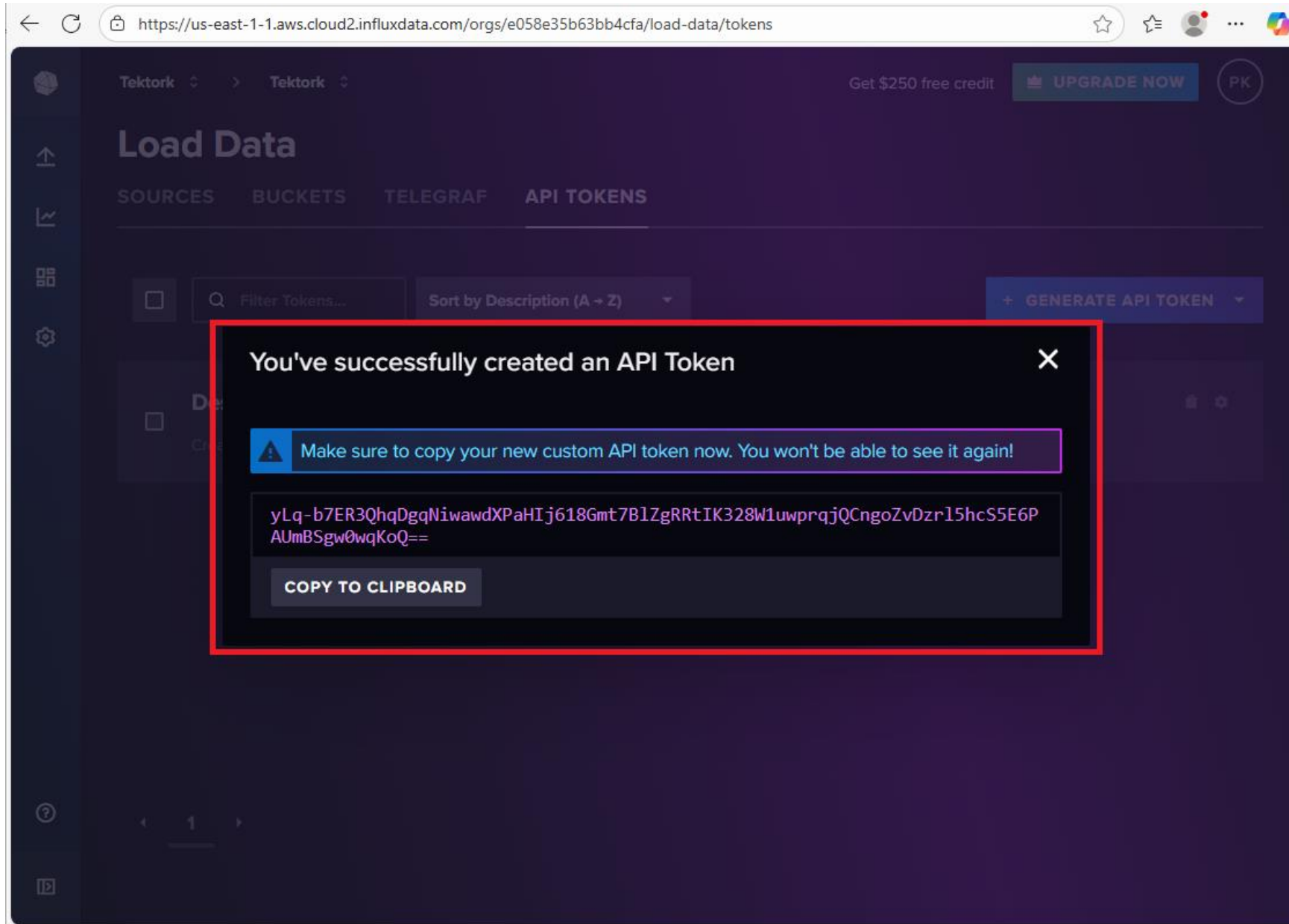
- Click on '**Generate API Token**' in the API Tokens section. Select '**All-Access API Token**' as the token type.



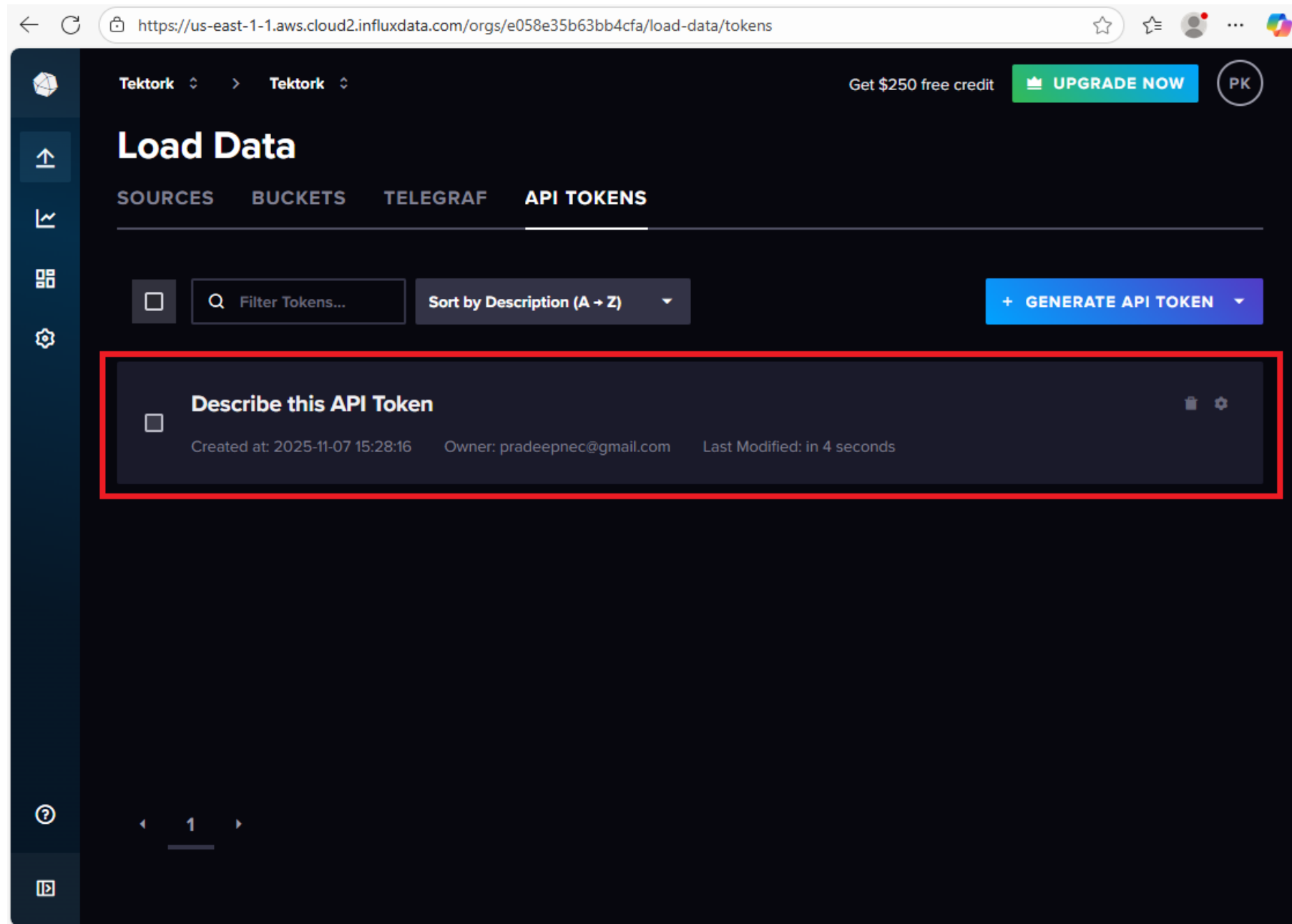
- Wait while the API token is being generated



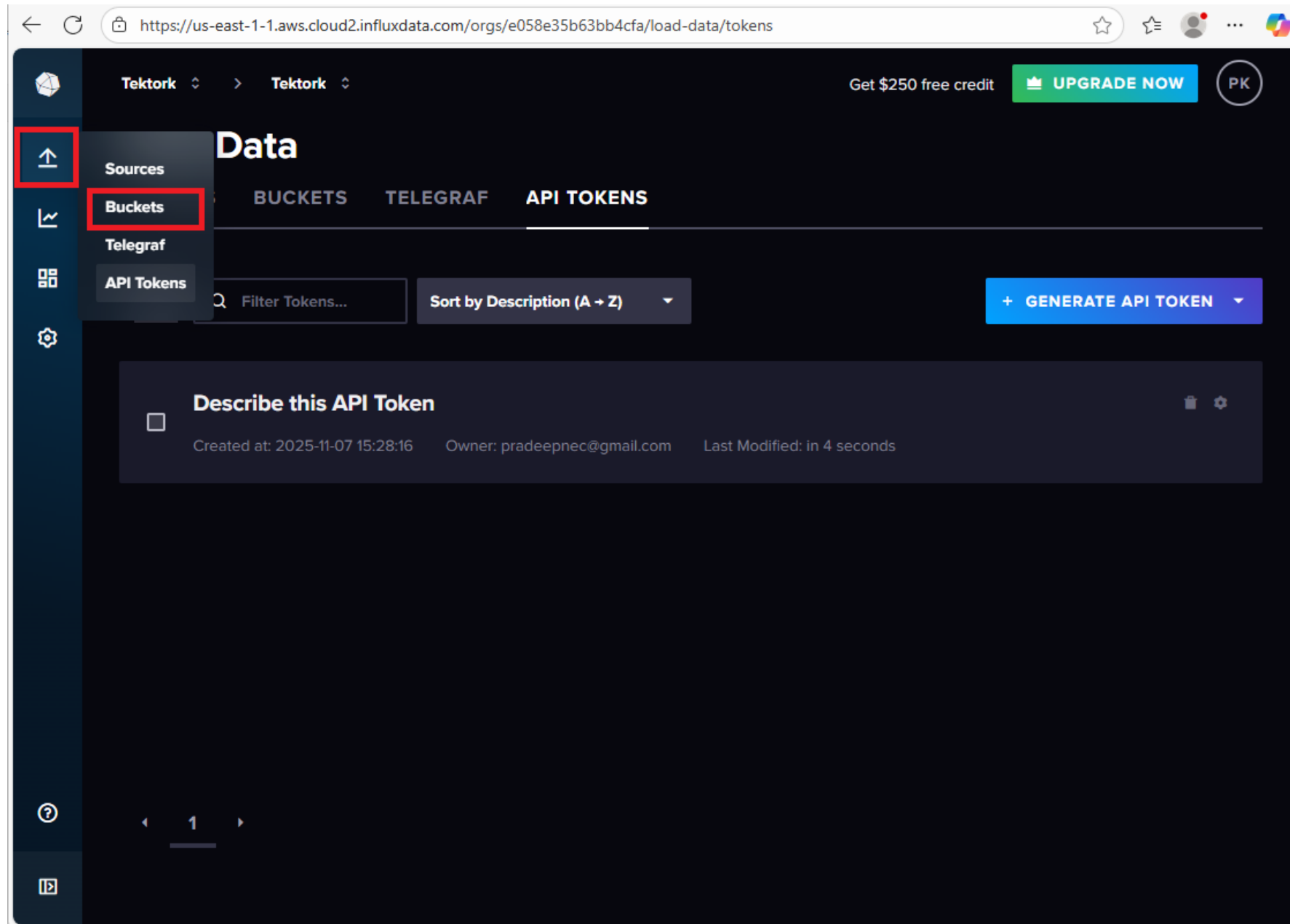
- Once the API token is generated, copy the token and save it in a Notepad or secure location.



- API token generation is complete.



- In the left-side panel of the InfluxDB window, select the 'Buckets' option under 'Load Data'



- Click the 'Create Bucket' button in the Buckets section

The screenshot shows the InfluxDB 'Load Data' interface. The 'BUCKETS' tab is selected, displaying a list of buckets. A red box highlights the '+ CREATE BUCKET' button in the top right corner. The interface includes a sidebar with navigation icons, a top navigation bar with the 'Teknork' logo and a 'Get \$250 free credit' button, and a main content area with a search bar and a sort dropdown. The bucket list shows two system buckets: '_monitoring' and '_tasks'. A sidebar on the right provides information about what a bucket is and how to write data into it.

URL: <https://us-east-1-1.aws.cloud2.influxdata.com/orgs/e058e35b63bb4cfa/load-data/buckets>

Get \$250 free credit **UPGRADE NOW** PK

Load Data

SOURCES **BUCKETS** TELEGRAF API TOKENS

Filter buckets... Sort by Name (A → Z)

+ CREATE BUCKET

Bucket Name	System Bucket	Retention	Schema Type	ID
_monitoring	System Bucket	Retention: 7 days	Schema Type: Implicit	ID: 81679dbf6ee9e131
_tasks	System Bucket	Retention: 3 days	Schema Type: Implicit	ID: 17a9280a857ccd29

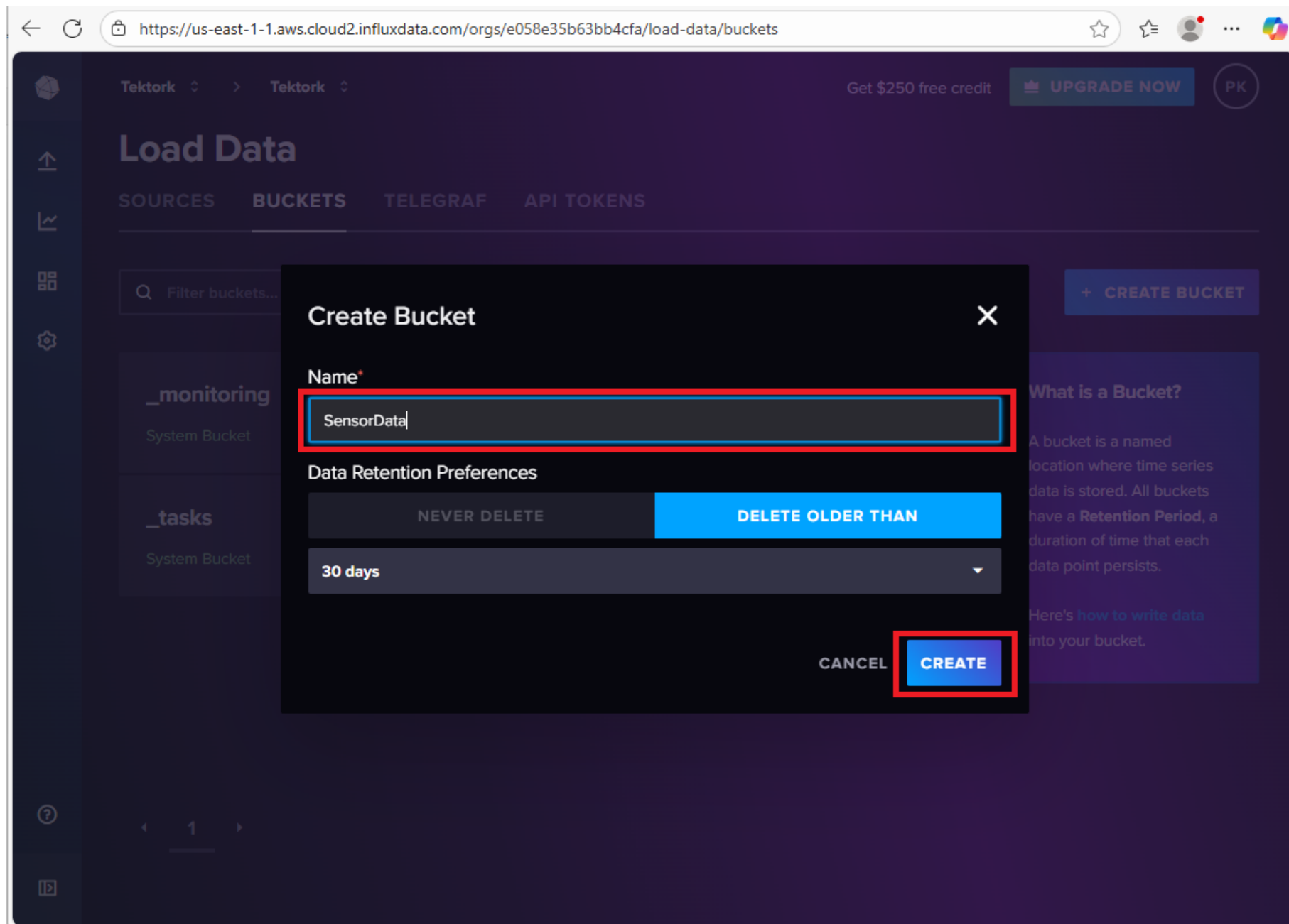
What is a Bucket?

A bucket is a named location where time series data is stored. All buckets have a **Retention Period**, a duration of time that each data point persists.

Here's [how to write data](#) into your bucket.

1

- Enter the bucket name, then click the 'Create' button



- Bucket has been successfully created.

The screenshot shows the InfluxDB 'Load Data' interface. The browser address bar displays the URL: `https://us-east-1-1.aws.cloud2.influxdata.com/orgs/e058e35b63bb4cfa/load-data/buckets`. The page header includes the 'Tektok' logo, a 'Get \$250 free credit' link, an 'UPGRADE NOW' button, and a user profile icon labeled 'PK'. The main navigation tabs are 'SOURCES', 'BUCKETS', 'TELEGRAF', and 'API TOKENS', with 'BUCKETS' currently selected. Below the tabs, there is a search bar labeled 'Filter buckets...', a sort dropdown set to 'Sort by Name (A → Z)', and a '+ CREATE BUCKET' button. The main content area lists three buckets: 'SensorData', '_monitoring', and '_tasks'. The 'SensorData' bucket is highlighted with a red border and contains the following details: Retention: 30 days, Schema Type: Implicit, ID: ef4550fa4b2a21ac, and buttons for '+ Add a label', '+ ADD DATA', and 'SETTINGS'. The '_monitoring' bucket is a 'System Bucket' with Retention: 7 days, Schema Type: Implicit, and ID: 81679dbf6ee9e131. The '_tasks' bucket is also a 'System Bucket' with Retention: 3 days, Schema Type: Implicit, and ID: 17a9280a857ccd29. On the right side, a blue sidebar titled 'What is a Bucket?' explains that a bucket is a named location for time series data, mentions the 'Retention Period', and provides a link to 'how to write data'.

Tektok > Tektok Get \$250 free credit **UPGRADE NOW** PK

Load Data

SOURCES **BUCKETS** TELEGRAF API TOKENS

Filter buckets... Sort by Name (A → Z) + CREATE BUCKET

SensorData

Retention: 30 days Schema Type: Implicit ID: ef4550fa4b2a21ac

+ Add a label + ADD DATA SETTINGS

_monitoring

System Bucket Retention: 7 days Schema Type: Implicit ID: 81679dbf6ee9e131

_tasks

System Bucket Retention: 3 days Schema Type: Implicit ID: 17a9280a857ccd29

What is a Bucket?

A bucket is a named location where time series data is stored. All buckets have a **Retention Period**, a duration of time that each data point persists.

Here's [how to write data](#) into your bucket.

Grafana Configuration

Open the Grafana Cloud - Go to: <https://grafana.com>

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EN ▾

Actually useful AI

No panel? No data? This is what it sounds like when devs cry. Good thing Grafana Assistant, an LLM agent built directly into Grafana Cloud, can fix it for you. (Or let's go crazy: make it purple.)

Learn more →



🤖 Fun

😐 Serious

<

||

>

The actually useful free plan

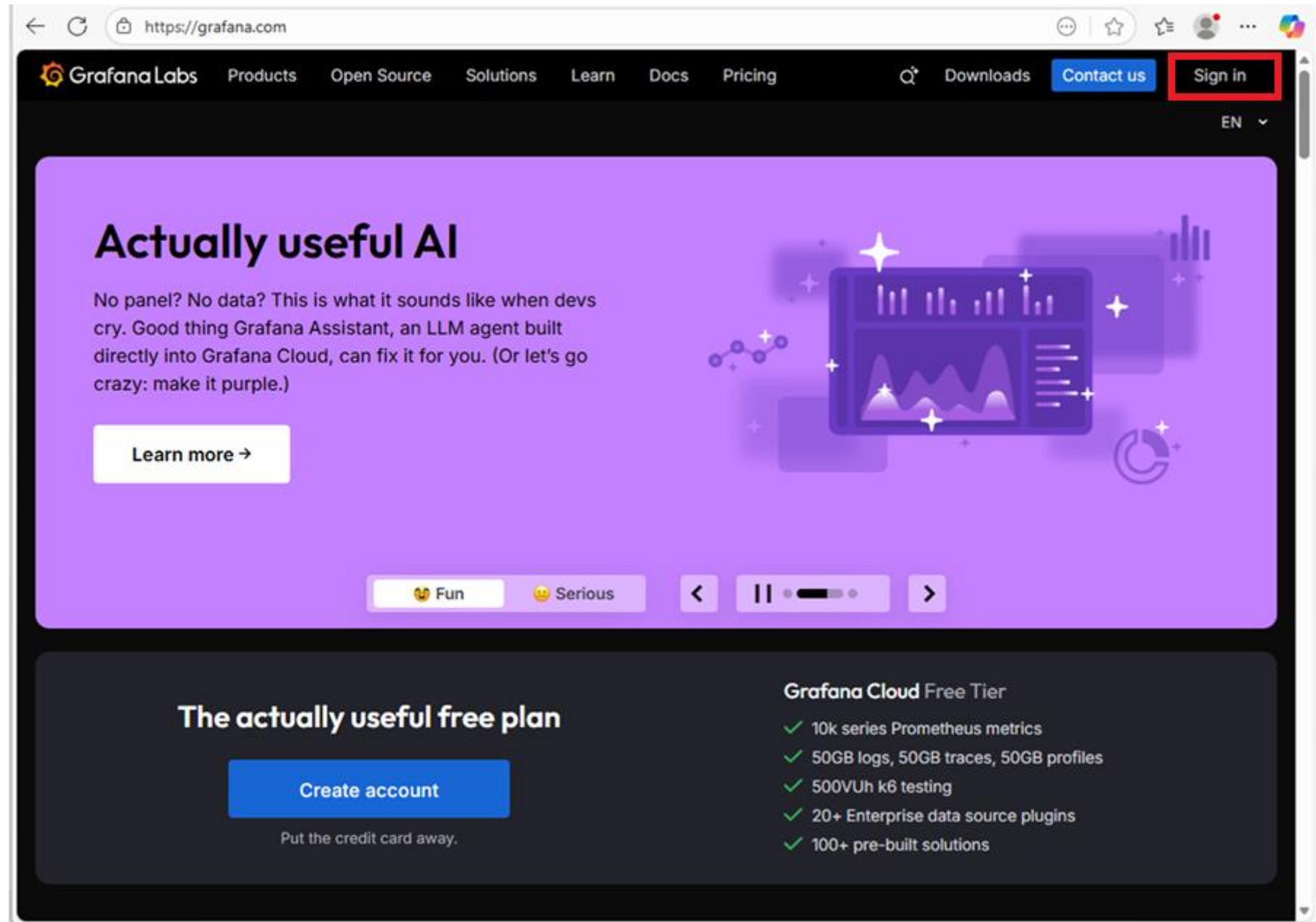
Create account

Put the credit card away.

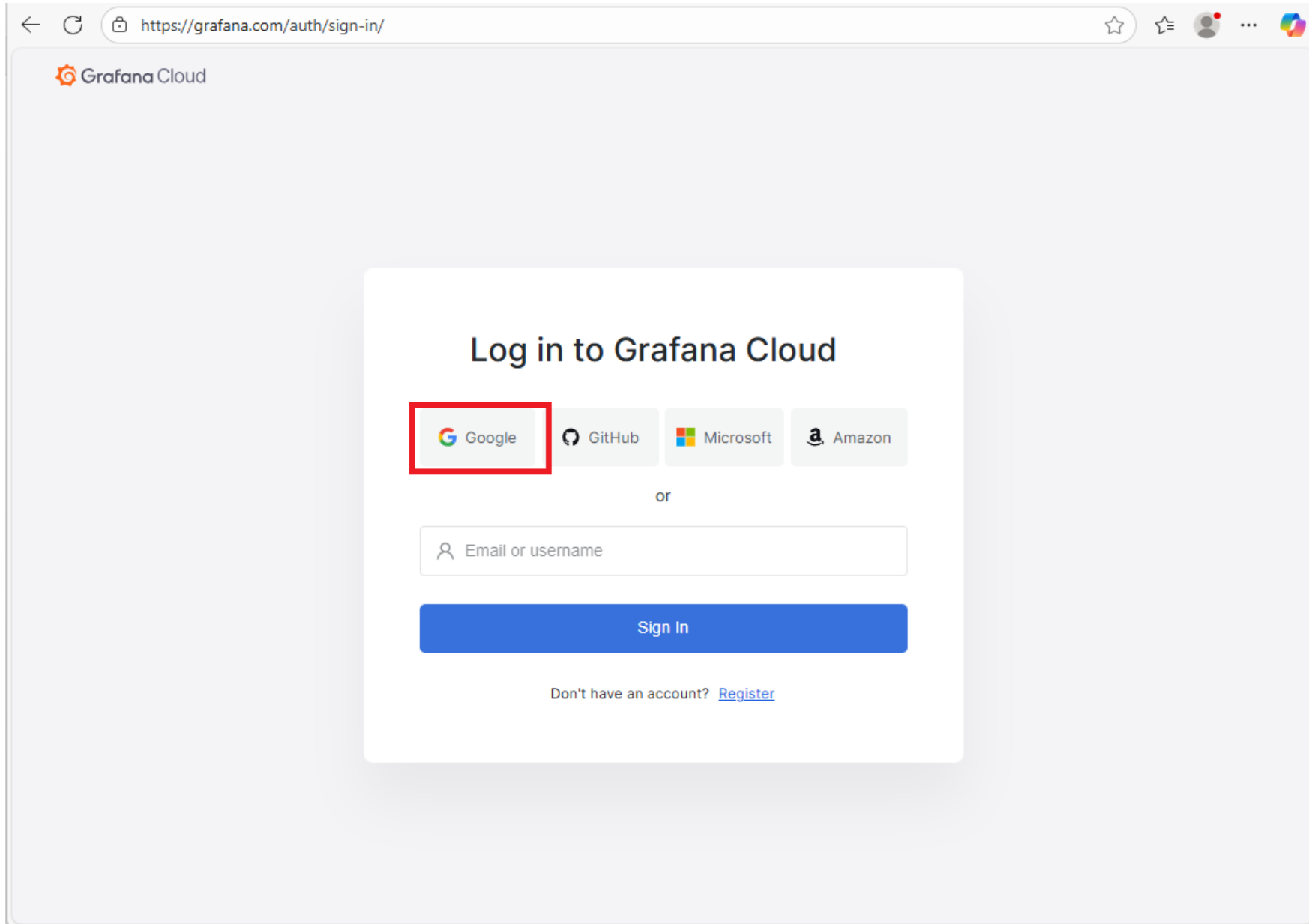
Grafana Cloud Free Tier

- ✓ 10k series Prometheus metrics
- ✓ 50GB logs, 50GB traces, 50GB profiles
- ✓ 500VUh k6 testing
- ✓ 20+ Enterprise data source plugins
- ✓ 100+ pre-built solutions

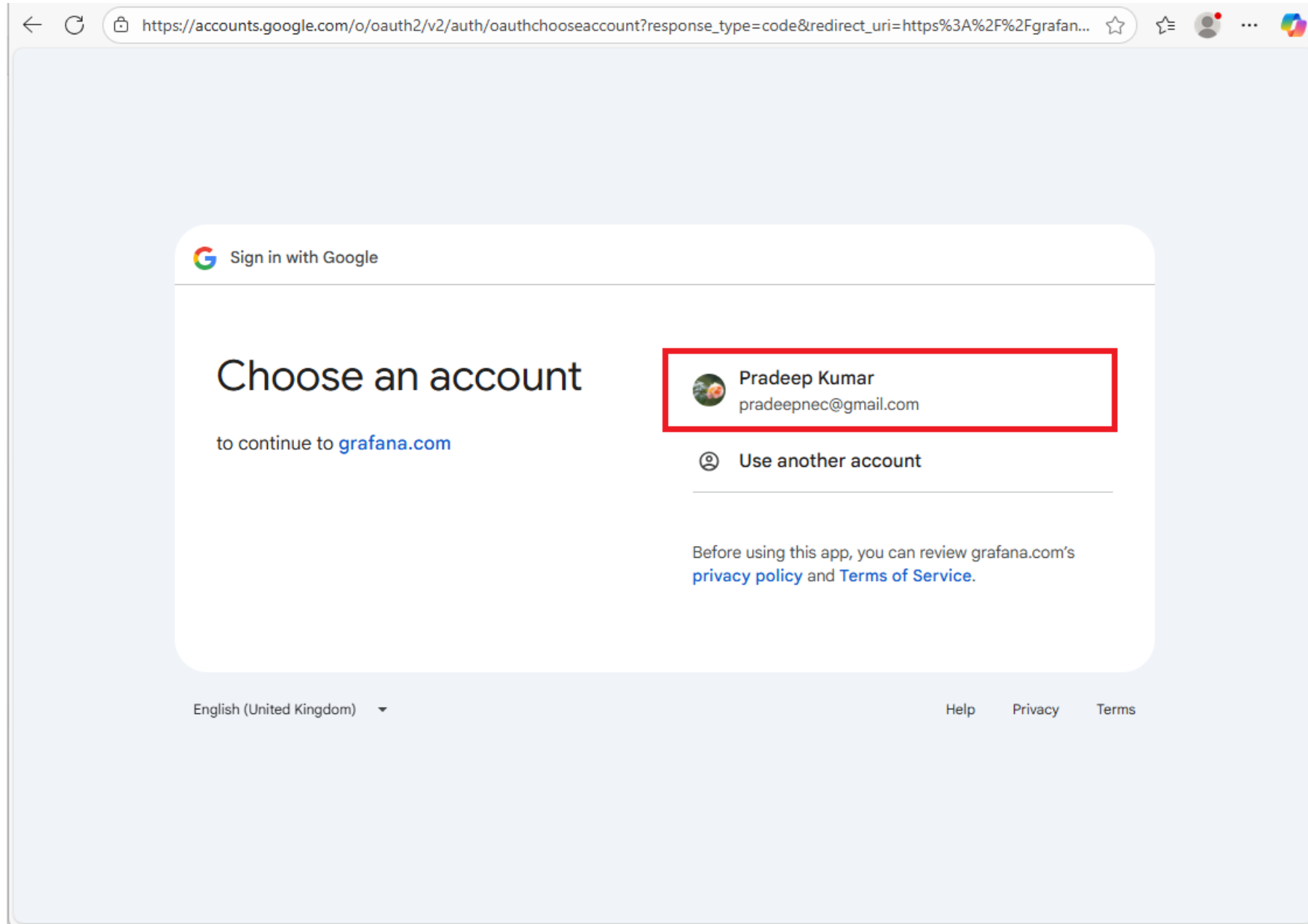
- Click on the *Sign In* button located at the top-right corner of the page.



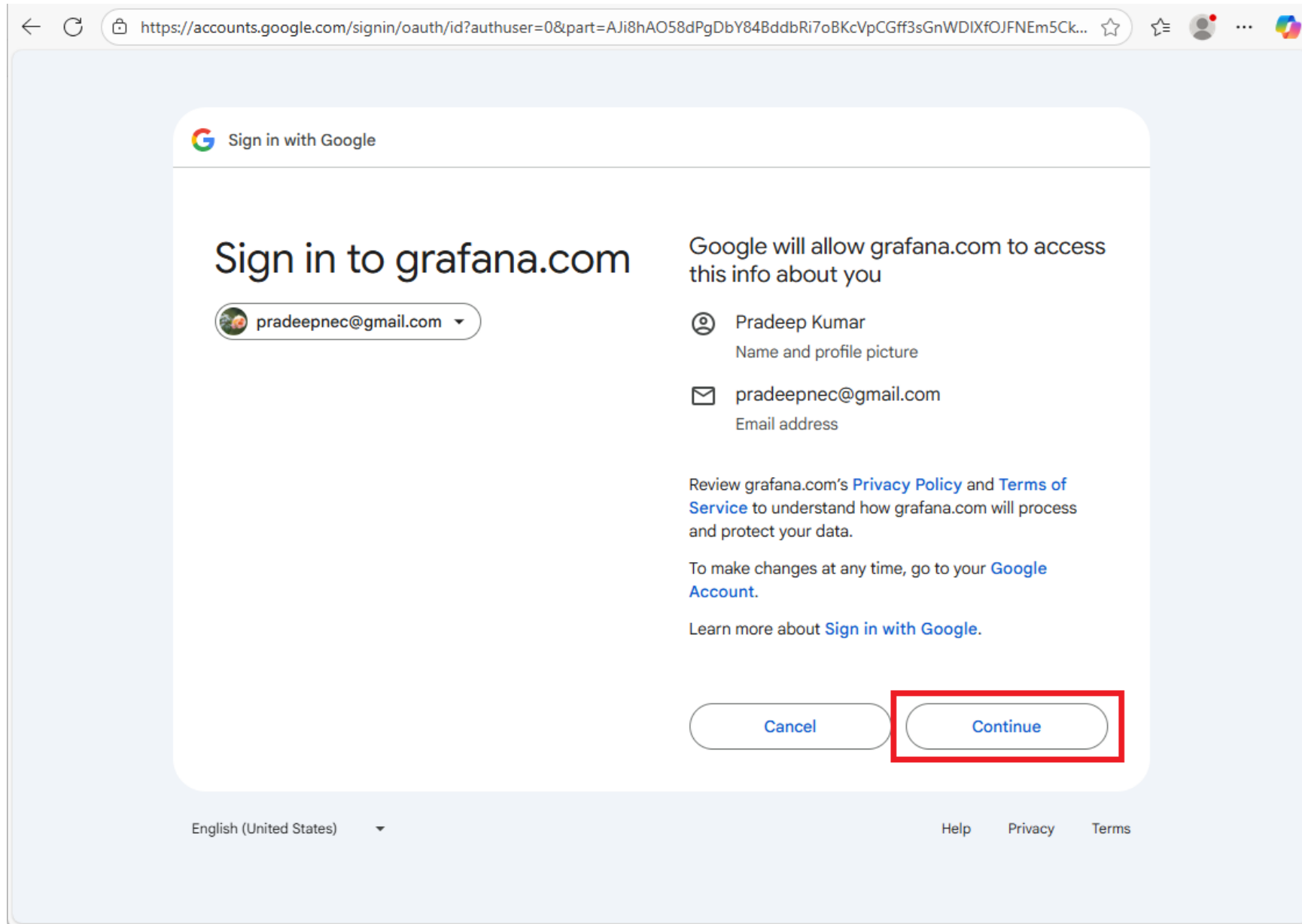
- Here In the Sign In window, select the *Google* option to continue with your Google account.



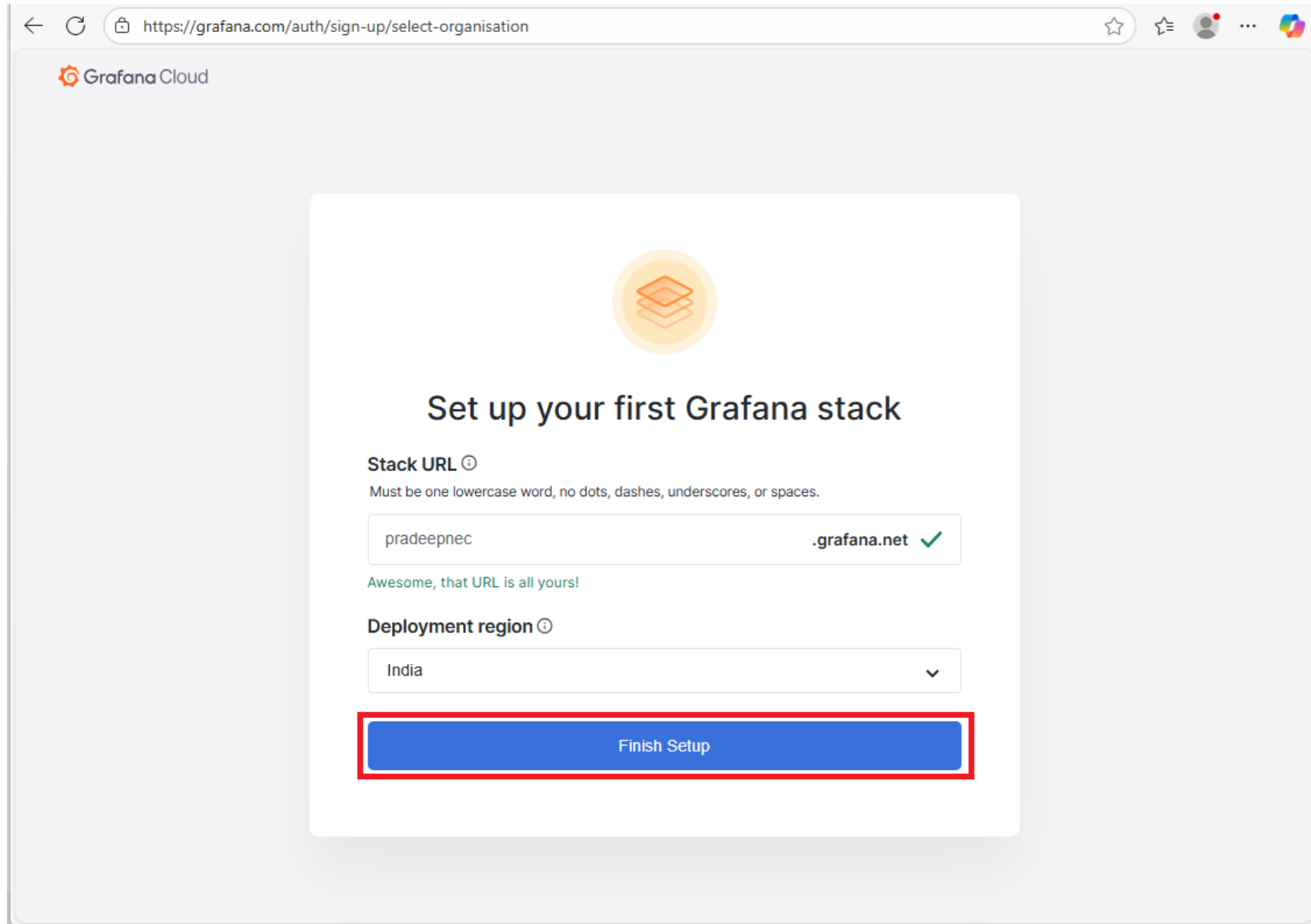
- Then choose your Google account if you are already logged in to Google.



- After selecting your Google account, click *Continue* to proceed.



- Click on the *Finish setup* button to complete the configuration and create your first Grafana stack.



← ↻ https://grafana.com/auth/sign-up/select-organisation

Grafana Cloud

Set up your first Grafana stack

Stack URL ⓘ
Must be one lowercase word, no dots, dashes, underscores, or spaces.

pradeepnec .grafana.net ✓

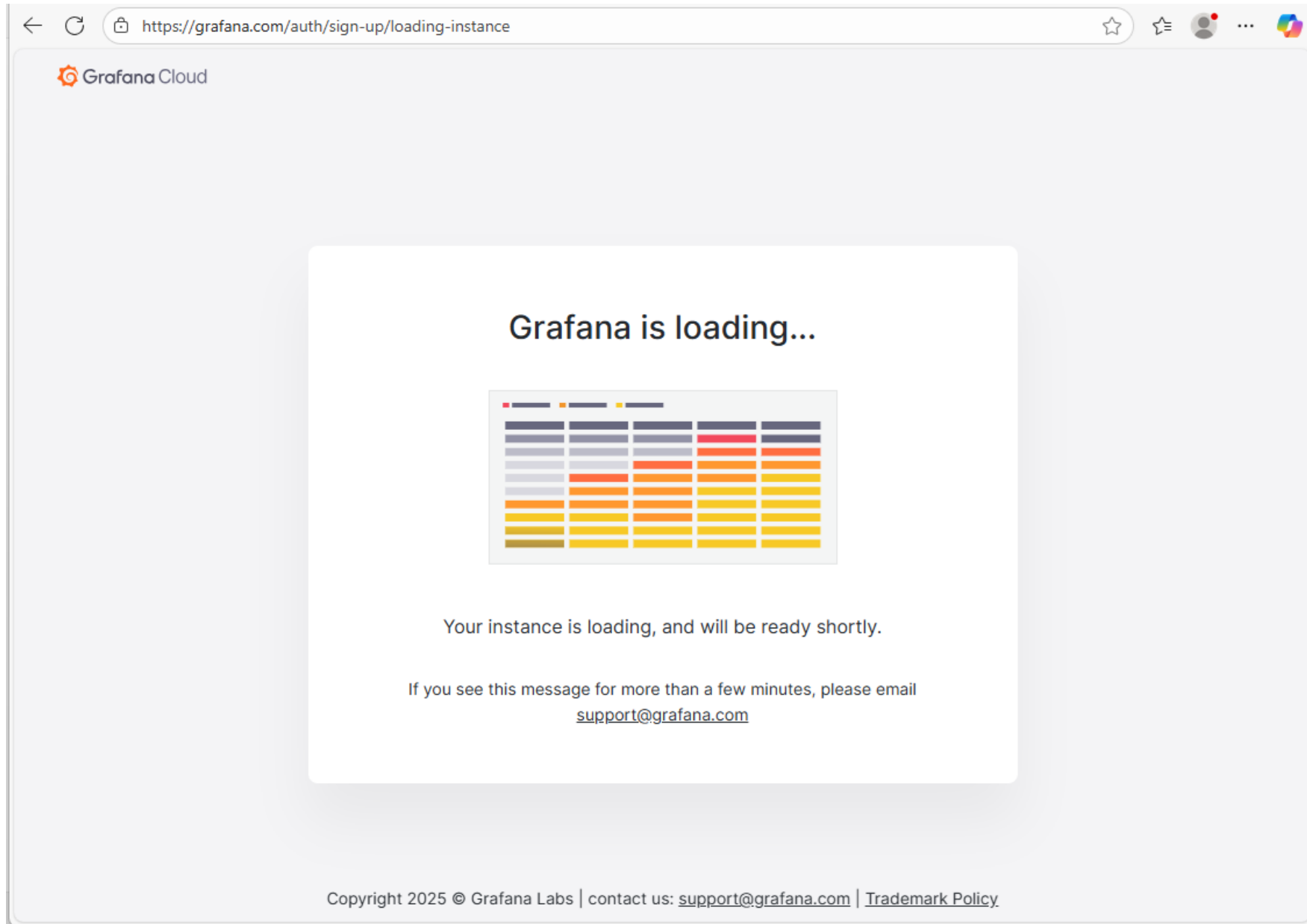
Awesome, that URL is all yours!

Deployment region ⓘ

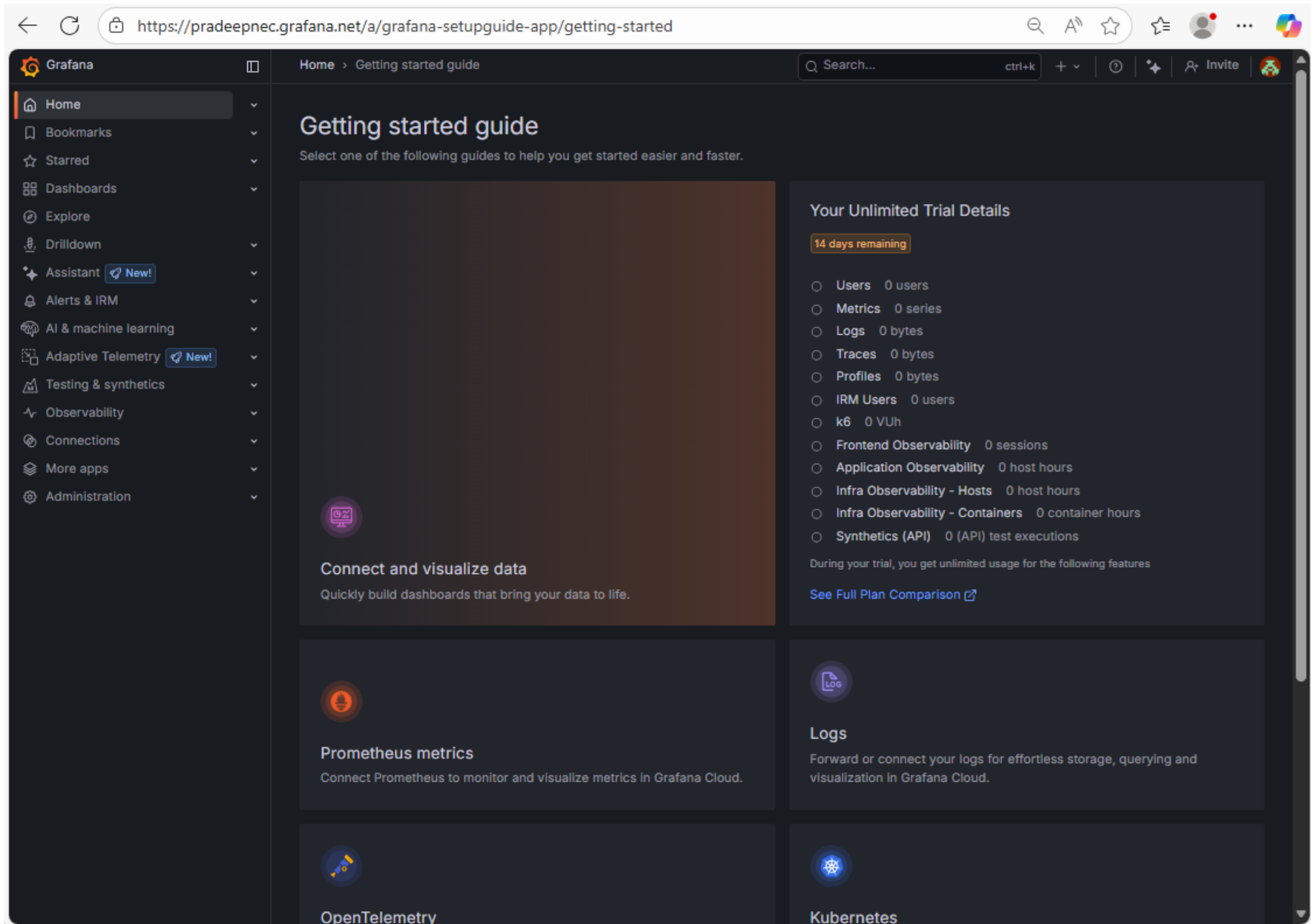
India ▼

Finish Setup

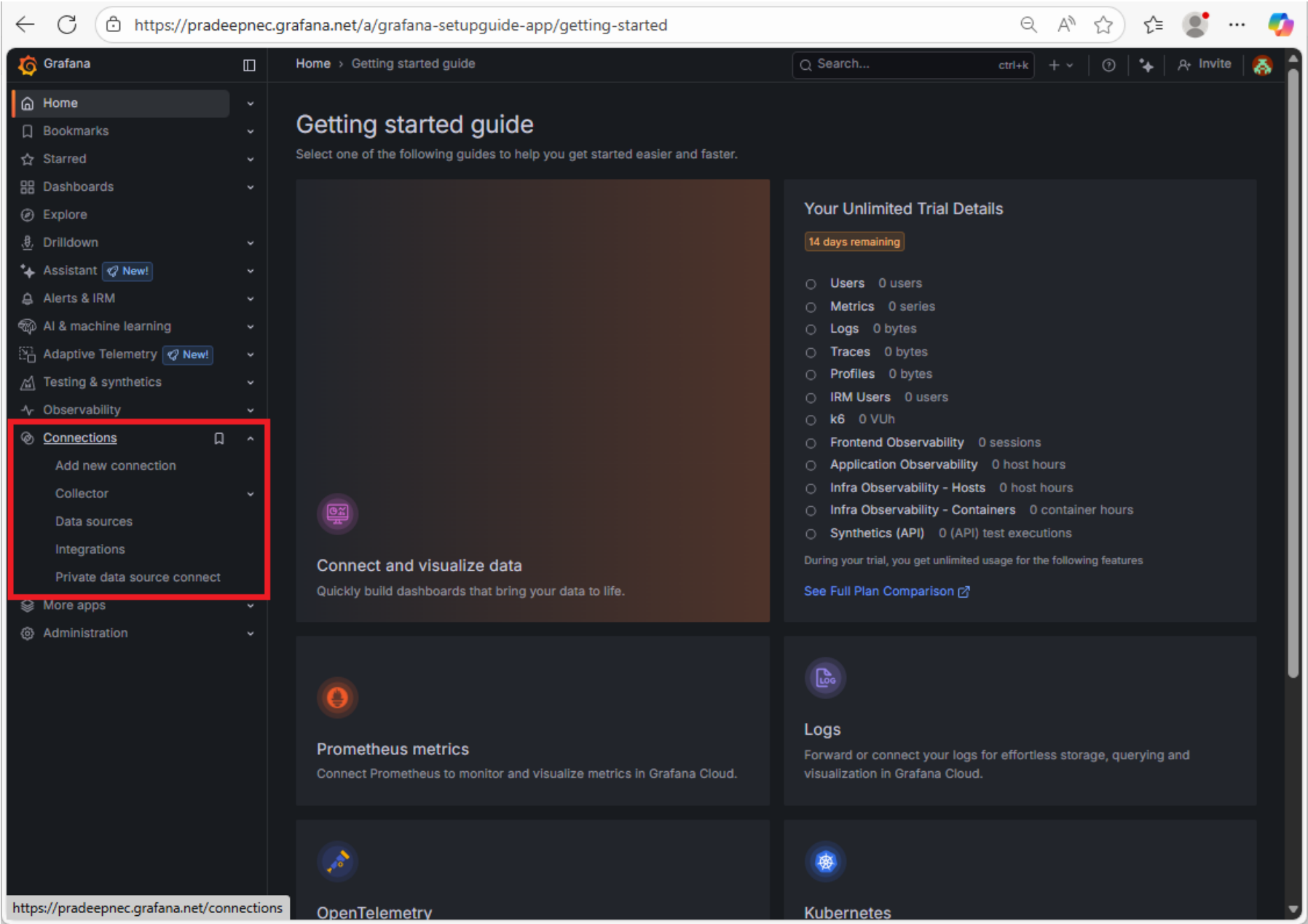
- Once you click the *Finish setup* button, Grafana begins to load your instance.



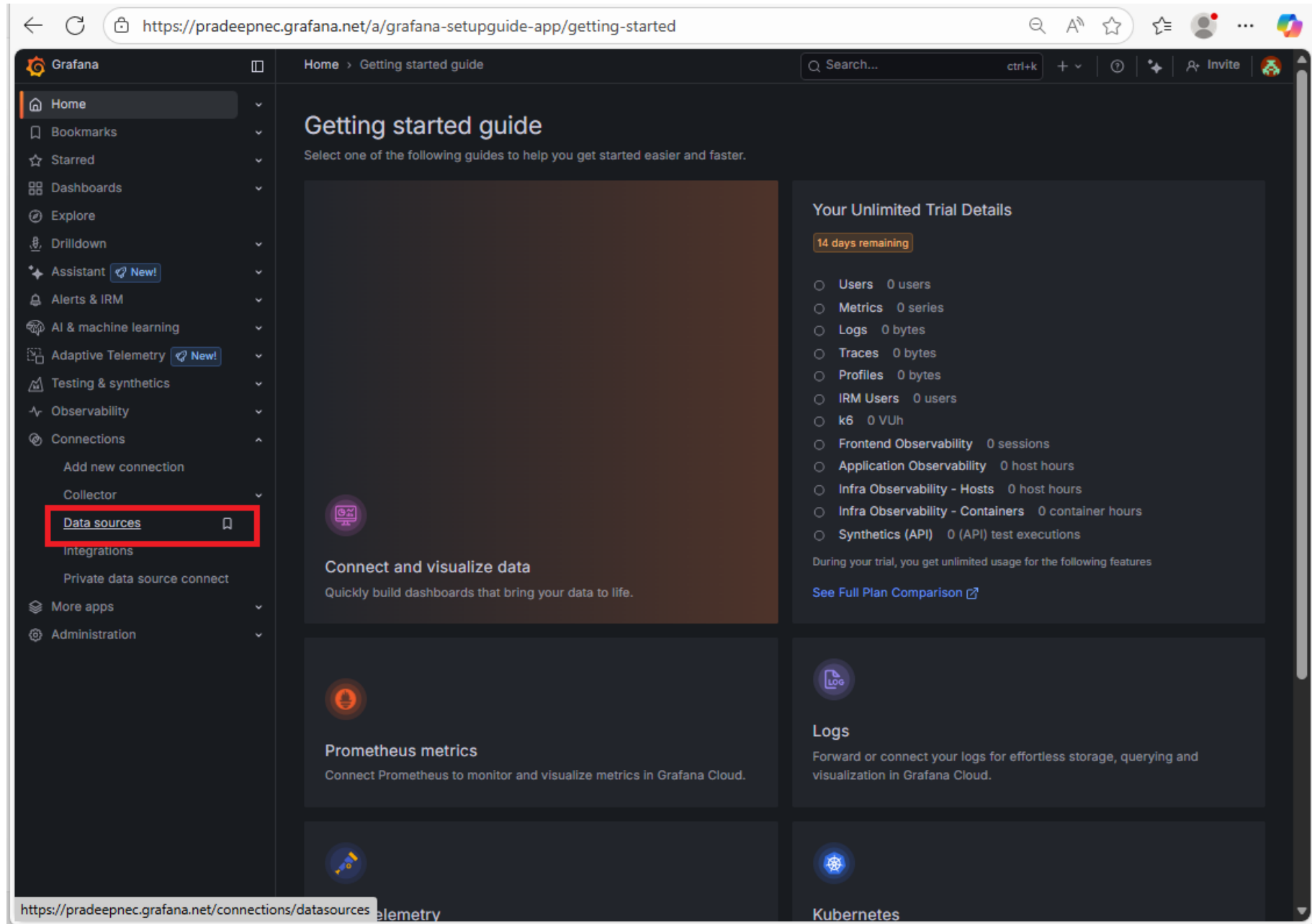
- After the loading process is complete, the Grafana dashboard window will open.



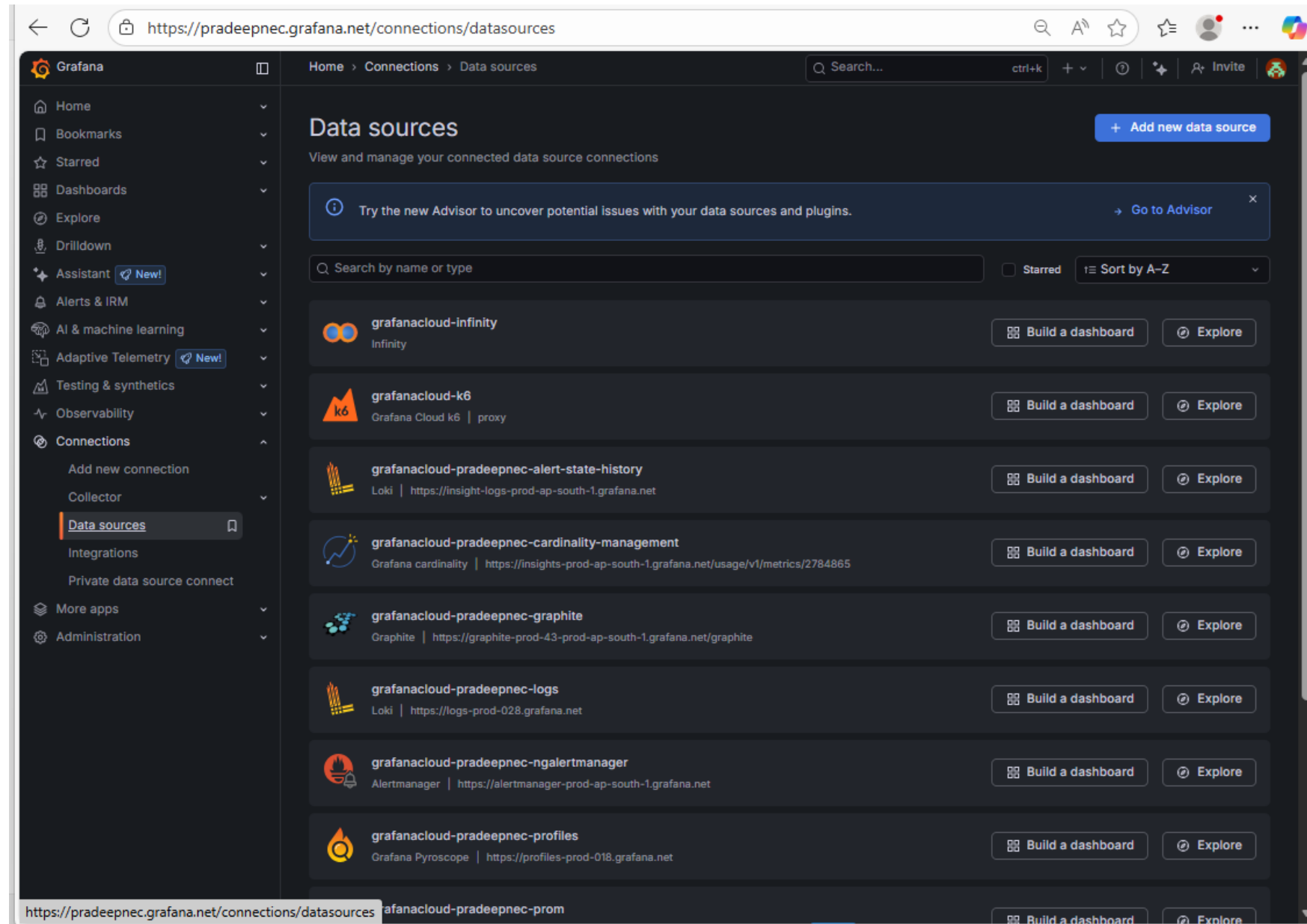
- From the left-side navigation panel, click on *Connections*.



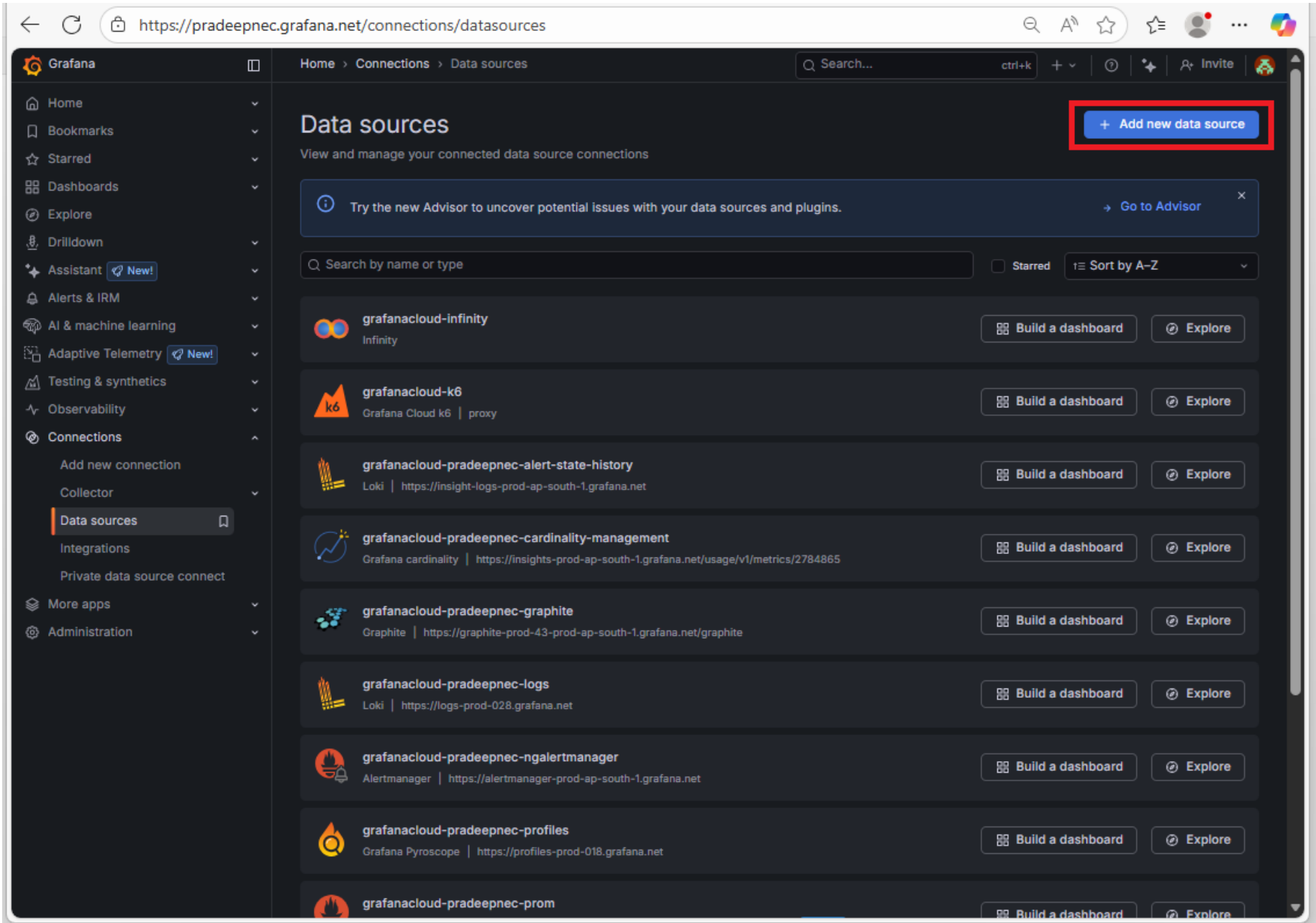
- From the Connections section, click on *Data Sources* to configure a new data source.



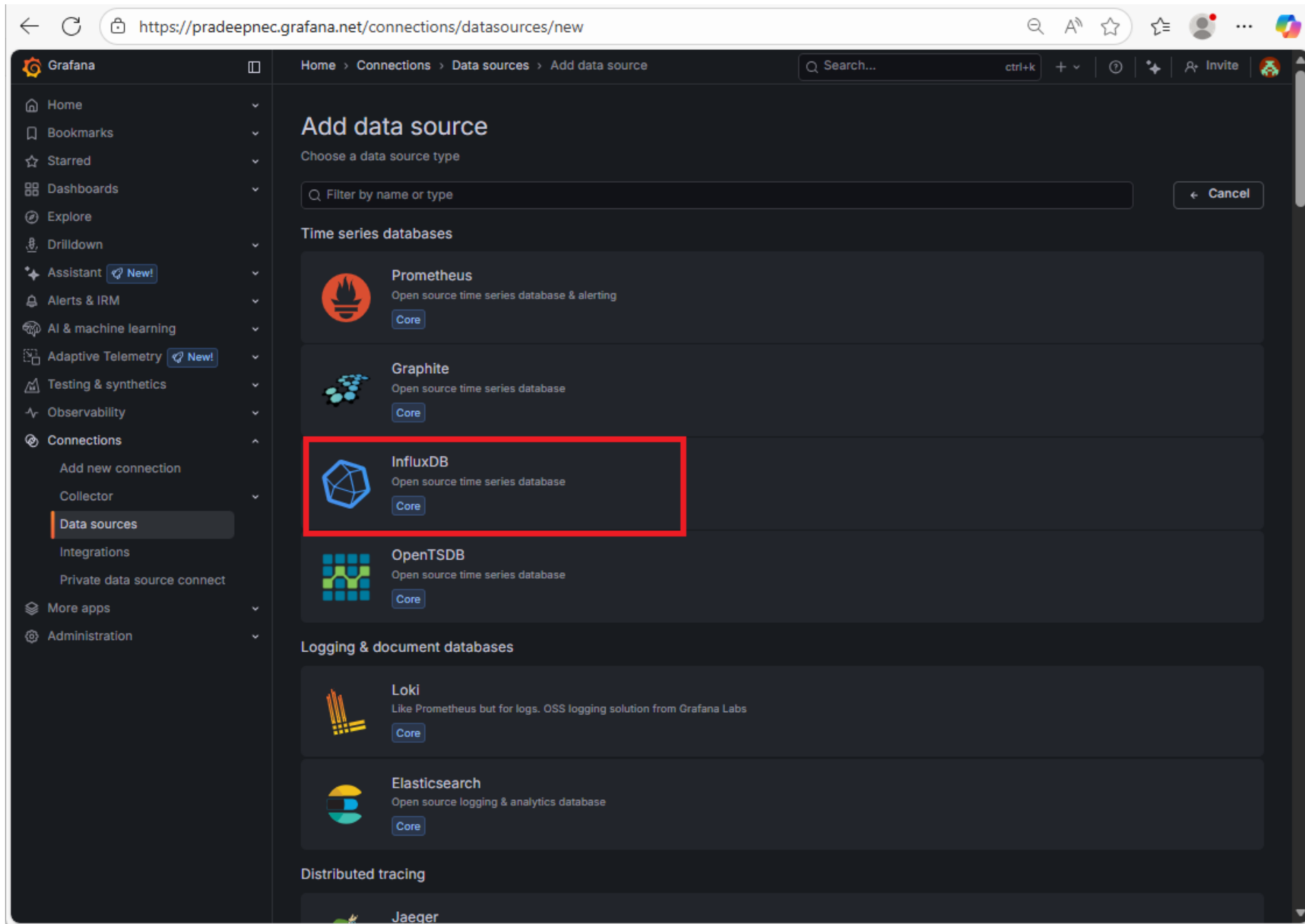
- Once you click on *Data Sources*, the window for adding and configuring a new data source will open.



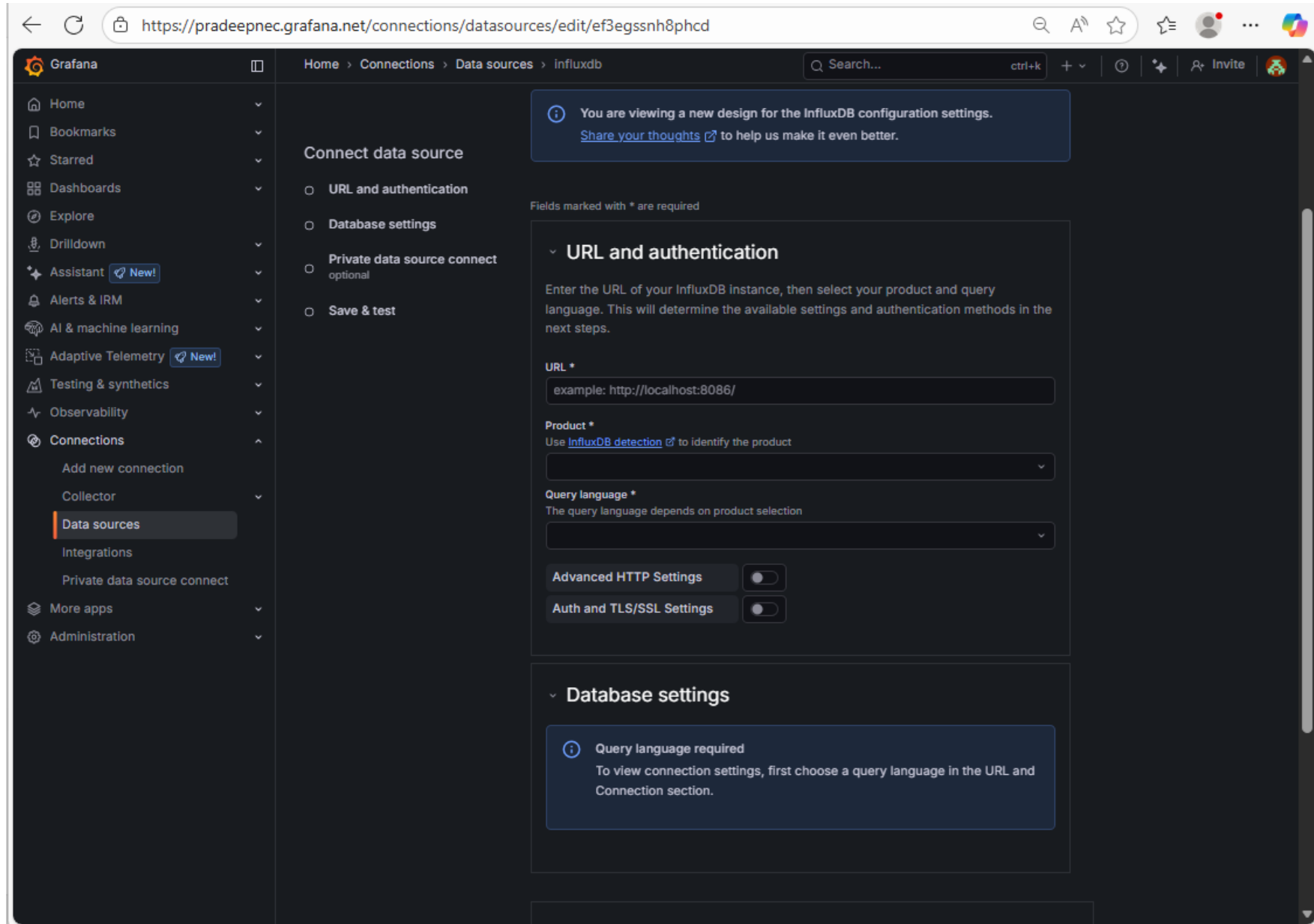
- Click the *Add data source* button on the Data Sources window.



- From the *Add data source* window, select *InfluxDB*.



- After selecting the InfluxDB data source, the data source configuration window will open.



Data Source Configurattion

Url and Authendication

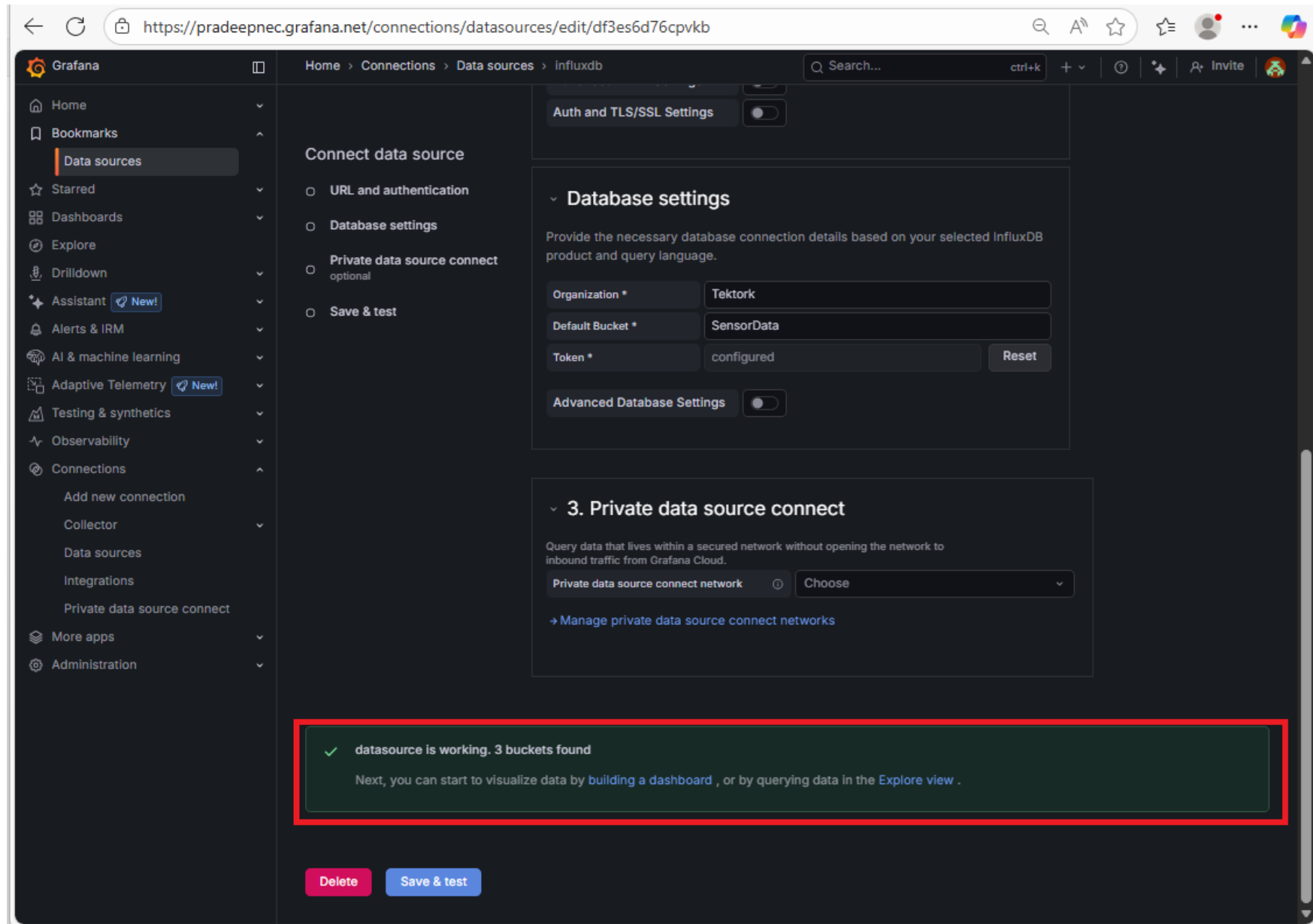
- Paste the InfluxDB Cloud URL into the *URL* field.
- In the *Product* field, select *InfluxDB Cloud Serverless* from the drop-down list.
- In the 'Query Language' field, select 'Flux' from the drop-down list.

Database Settings

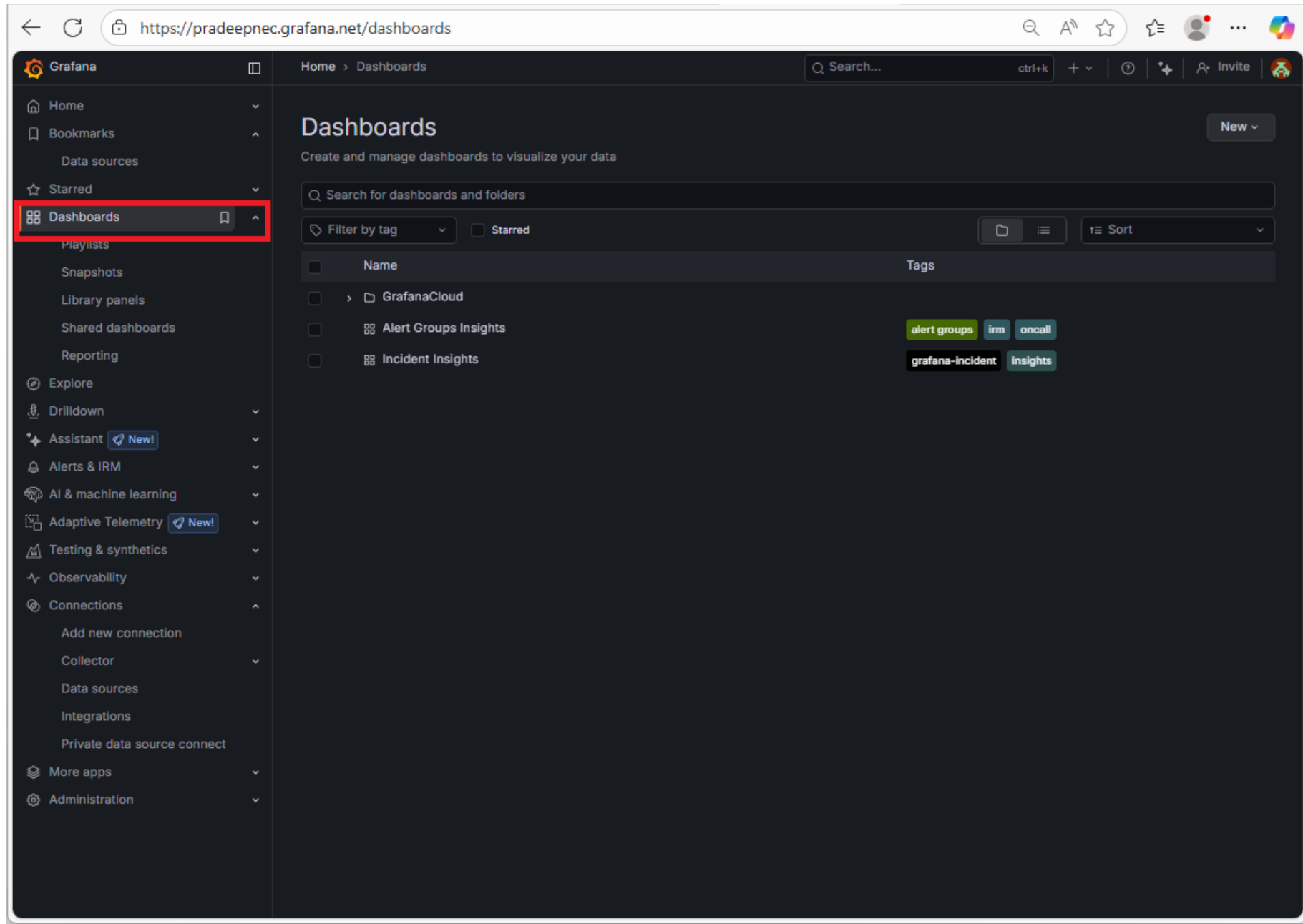
- In the 'Organization' field, enter the organization name of your InfluxDB account.
- In the 'Bucket Name' field, enter the bucket name of your InfluxDB account
- In the 'Token' field, enter the API token from your InfluxDB account.

*Leave the 'Private Data Source Connect' option unchanged.

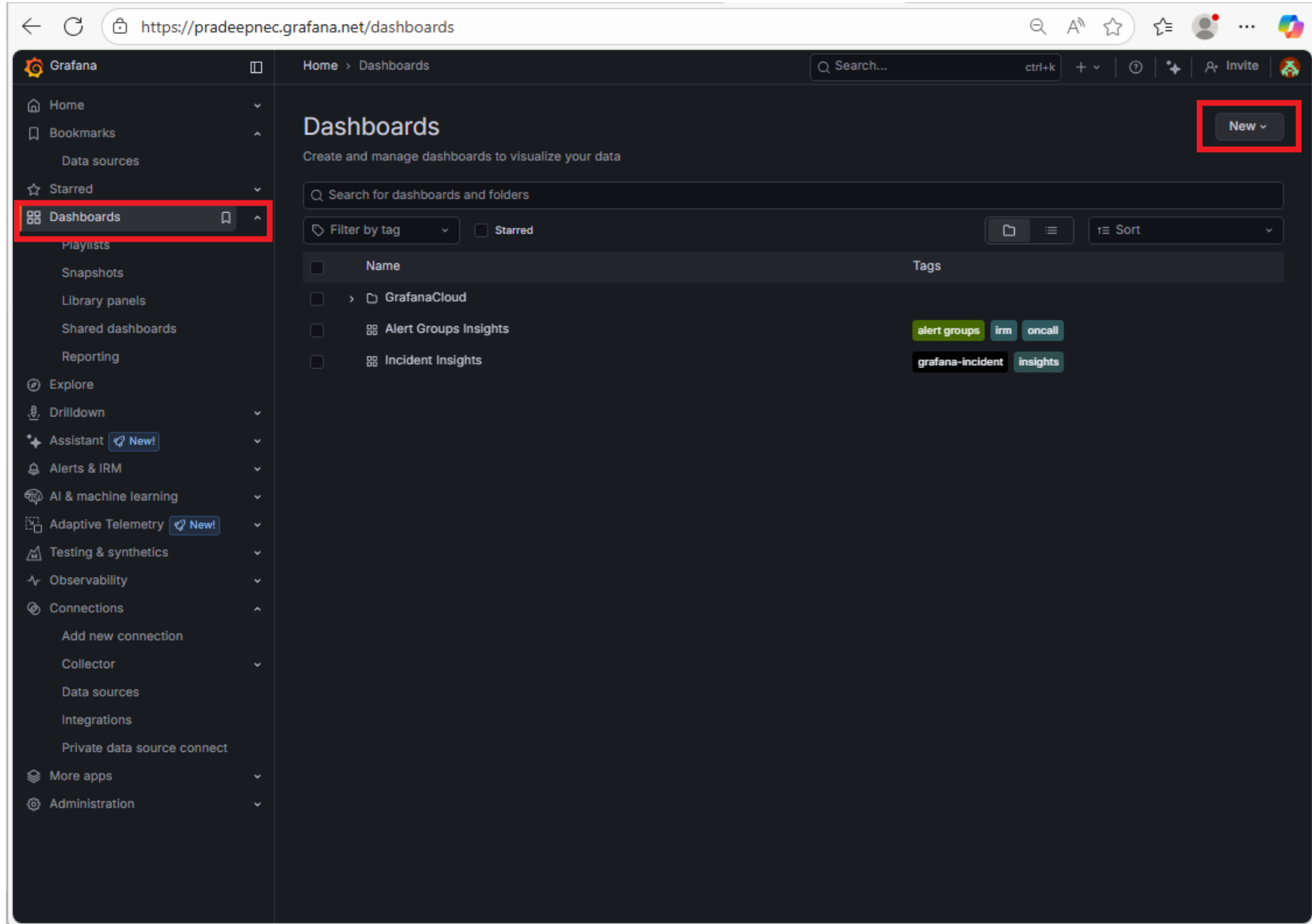
- Once the data source is configured, press the 'Test & Save' button to verify and save your settings.



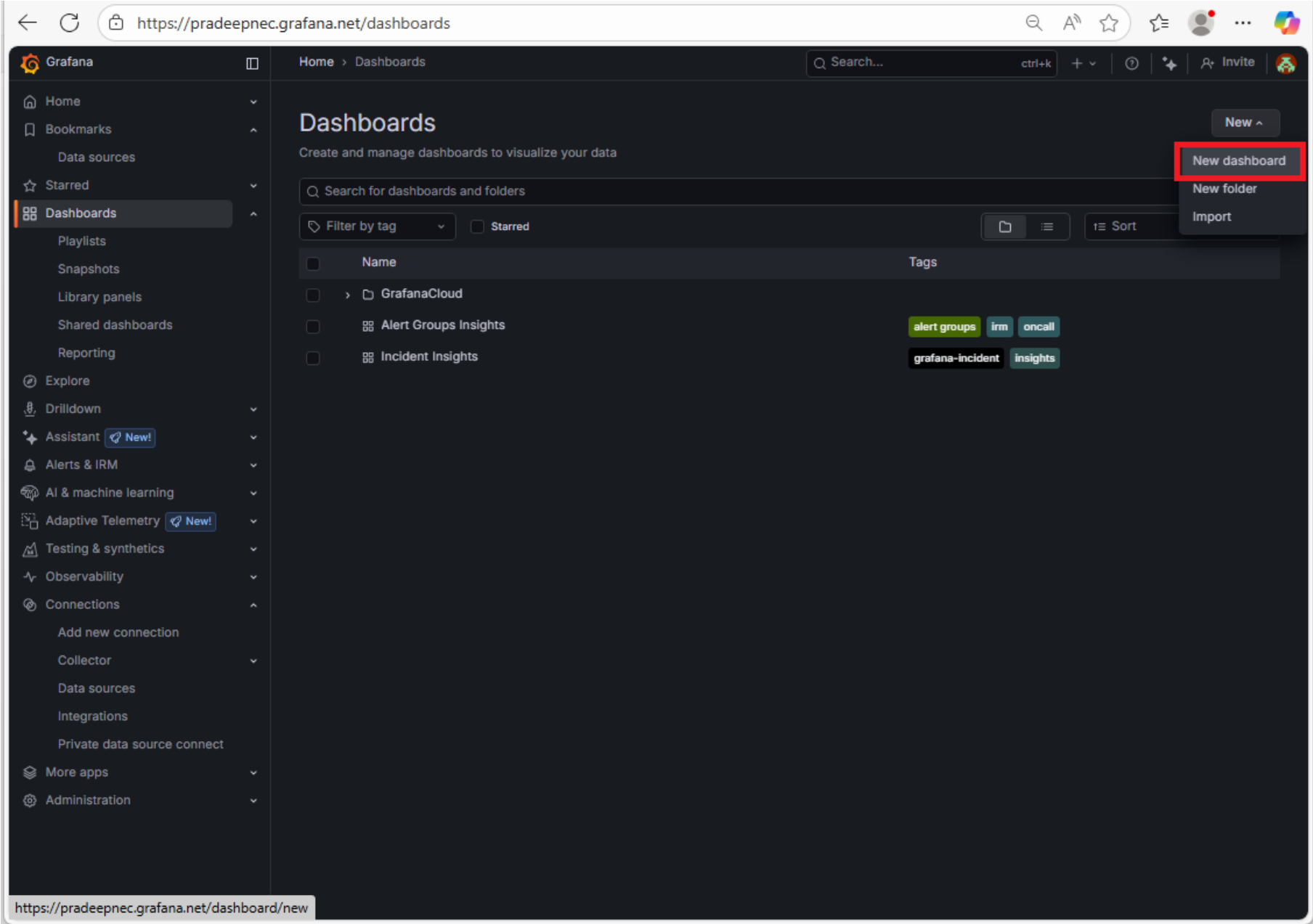
- Go to the left sidebar and click *Dashboards*.



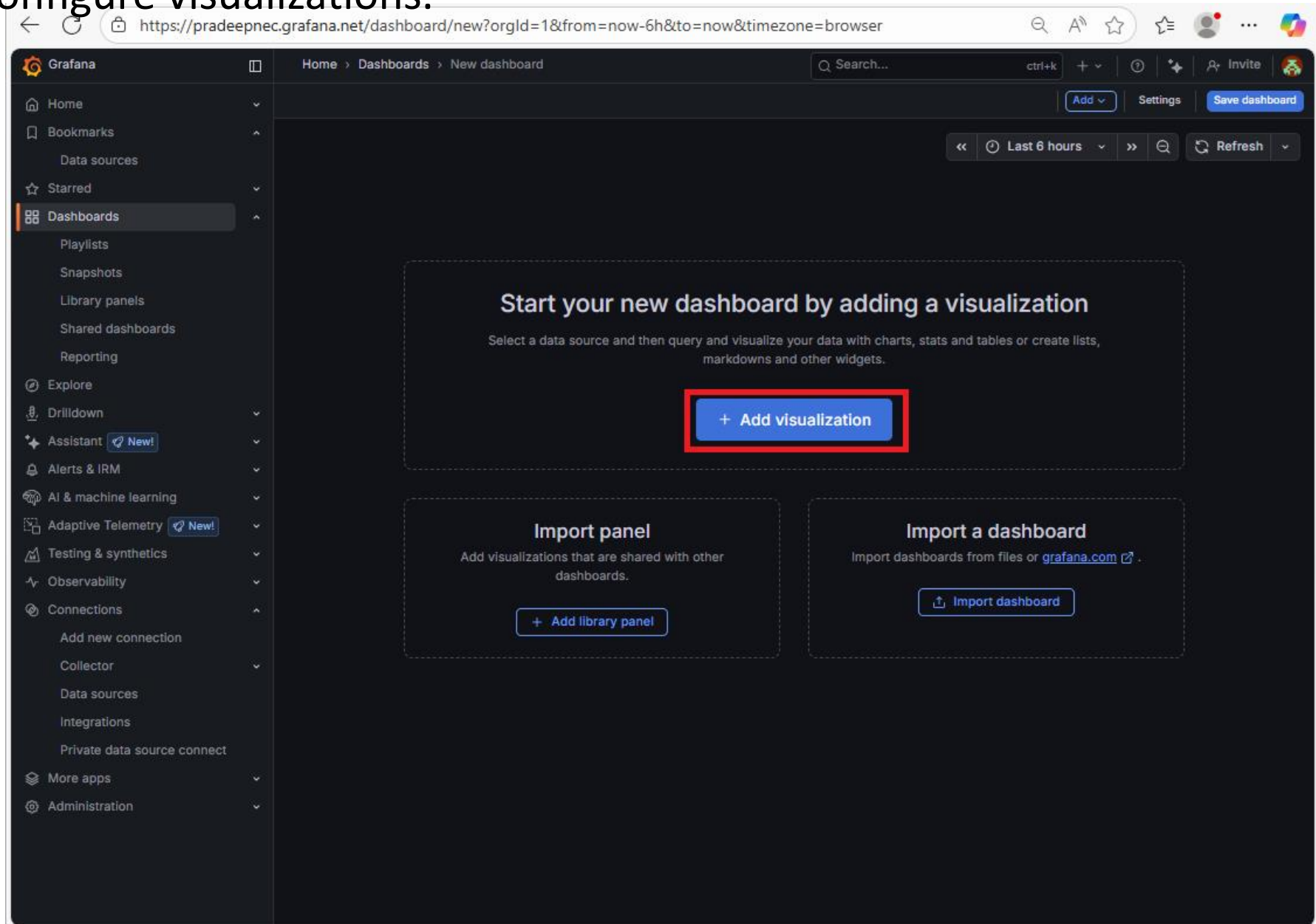
- Select *+ New* to create a new dashboard.



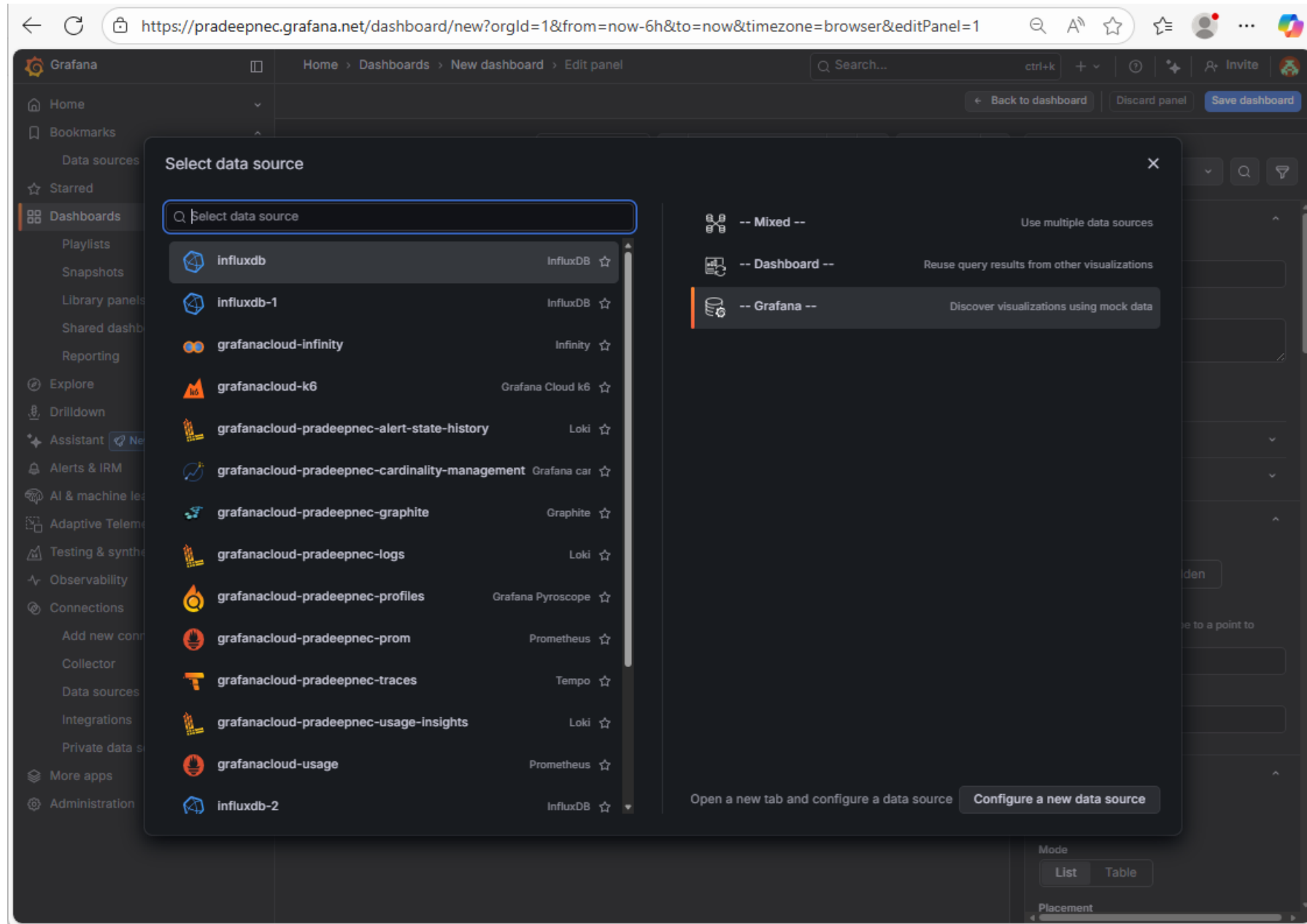
Then click *New dashboard*.



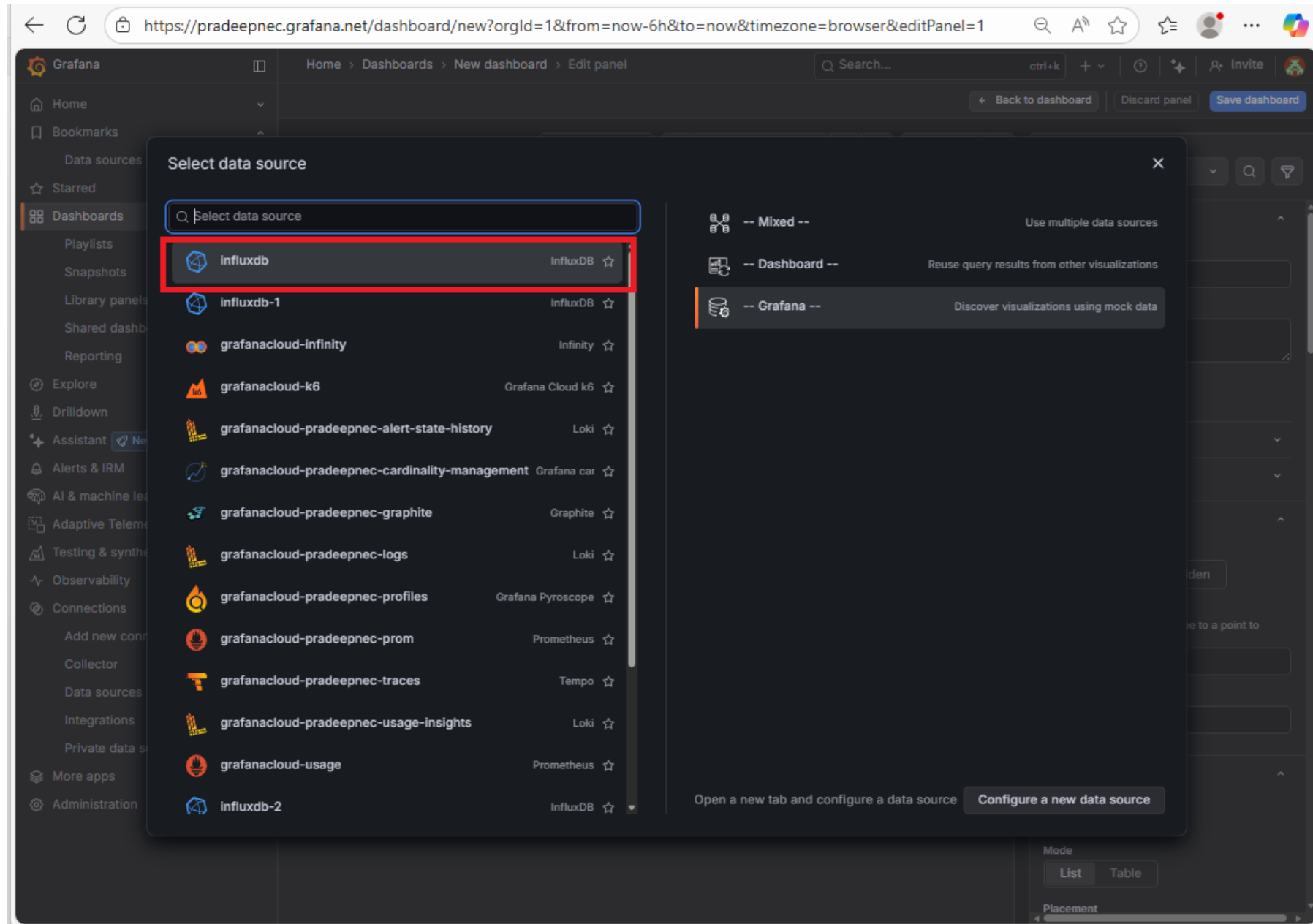
- After clicking *New dashboard*, the panel editor (visualization window) will open, allowing you to add and configure visualizations.



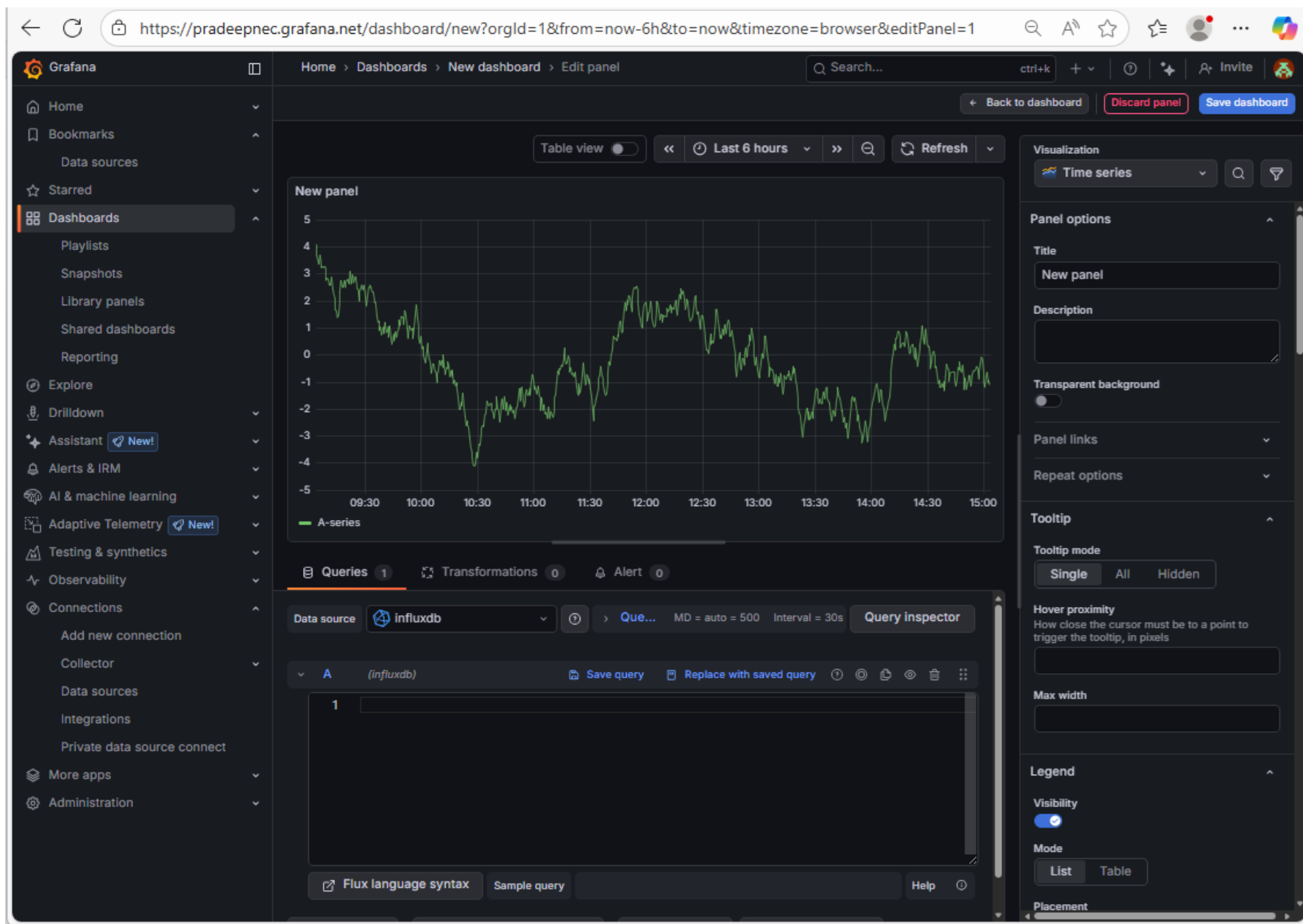
- Click *Add visualization*; the data source selection window will open.



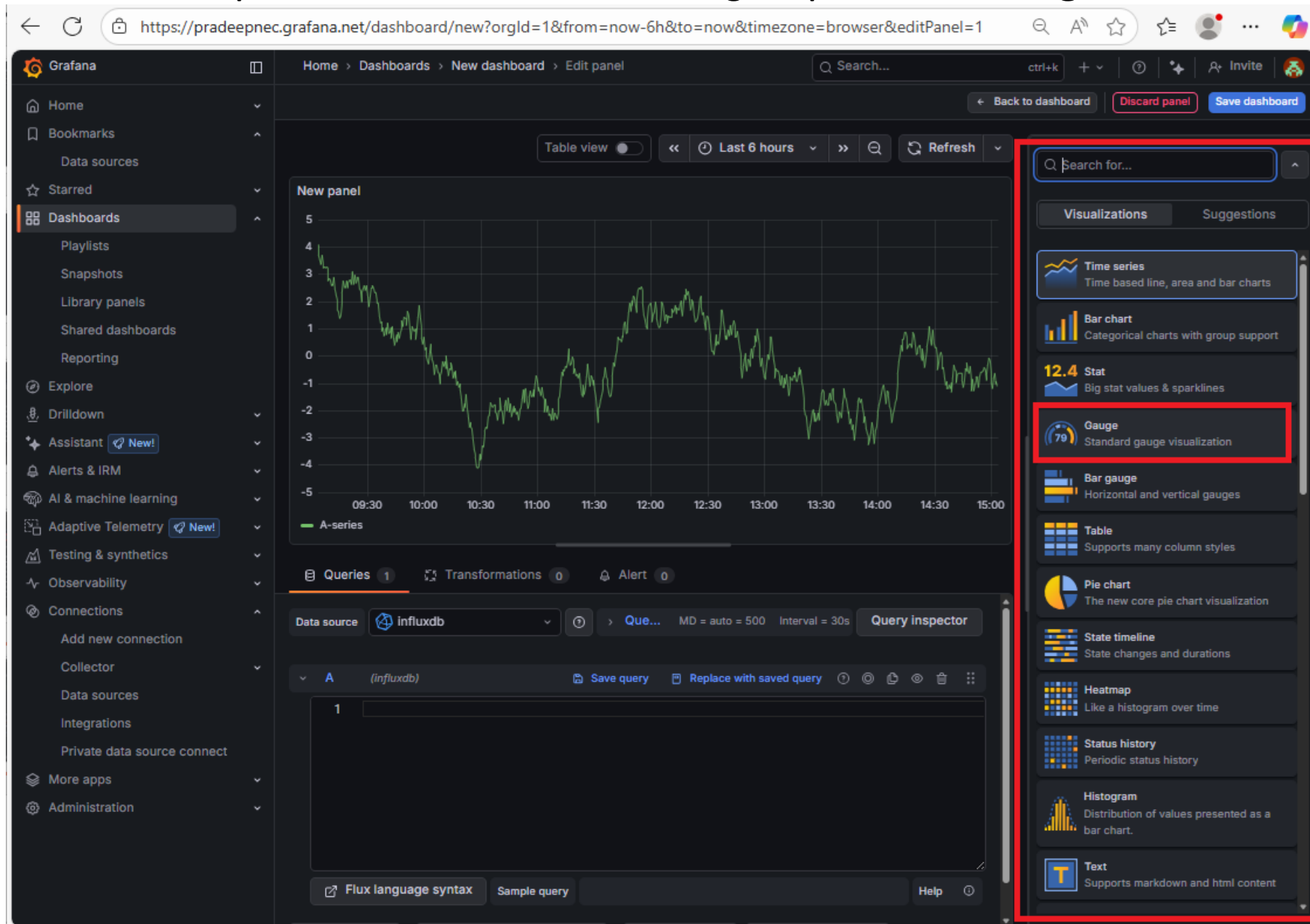
- From the data source window, select *InfluxDB* to proceed.



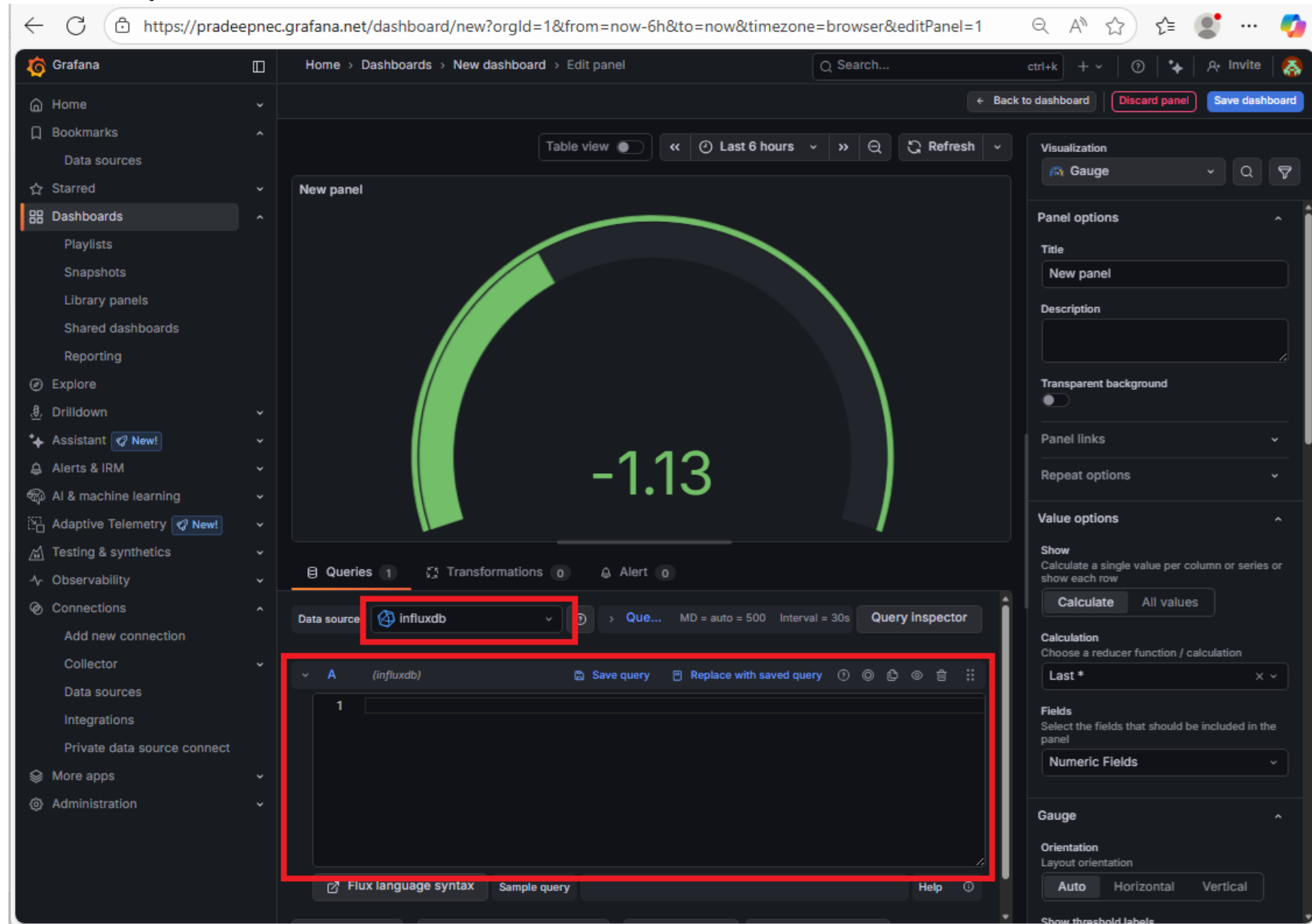
- Once you select the InfluxDB data source, the panel editor window will open automatically.



- In the new panel window, go to the right side and click the drop-down arrow of the *Time series* visualization option, then select the *Gauge* option to change the visualization type



- The Gauge visualization window will open. Verify that the data source is set to InfluxDB in the data source option.



Flux query for LDR

```
from(bucket: "SensorData")  
  |> range(start: -6h)  
  |> filter(fn: (r) => r._measurement == "esp32_sensor_data")  
  |> filter(fn: (r) => r._field == "light_percent")
```

- In the Flux query editor, copy and paste the query into the query input area. Then click *Save query* to save the query.

The screenshot shows the Grafana web interface in 'Edit panel' mode. The main area displays a gauge visualization with a green needle pointing to the value -1.13. The left sidebar contains navigation links for Home, Bookmarks, Data sources, Starred, Dashboards, Playlists, Snapshots, Library panels, Shared dashboards, Reporting, Explore, Drilldown, Assistant, Alerts & IRM, AI & machine learning, Adaptive Telemetry, Testing & synthetics, Observability, Connections, Add new connection, Collector, Data sources, Integrations, Private data source connect, More apps, and Administration.

The right sidebar contains configuration options for the visualization, including Title, Description, Transparent background, Panel links, Repeat options, Value options, Calculation, Fields, Gauge, Orientation, and Show threshold labels.

The bottom section is the Flux query editor, which includes a data source dropdown set to 'influxdb', a query input area, and a 'Save query' button. The query input area contains the following Flux query:

```
1 from(bucket: "SensorData")
2   |> range(start: -6h)
3   |> filter(fn: (r) => r._measurement == "esp32_sensor_data")
4   |> filter(fn: (r) => r._field == "light_percent")
5
```

The 'Save query' button is highlighted with a red box. Below the query input area, there are links for 'Flux language syntax', 'Sample query', and 'Help'.

- After clicking the *Save query* option, a window will open where you can enter the query name. Then click *Save* to confirm.

The screenshot shows the Grafana web interface with the 'Saved queries' dialog box open. The dialog box is titled 'Saved queries' and has tabs for 'All', 'Favorites', and 'Recent'. It contains a search bar and filters for data source, user name, and tag. A list of saved queries is shown, with the first one titled 'LDR query' highlighted. The 'LDR query' entry has a 'New' button next to it. The dialog box also has a 'New query' button. The 'LDR query' entry is expanded, showing the query text:

```
from(bucket: "SensorData")
  |> range(start: -6h)
  |> filter(fn: (r) => r._measurement == "esp32_sensor_data")
  |> filter(fn: (r) => r._field == "light_percent")
```

 The 'Data source' is set to 'influxdb'. The 'Description' field is empty. The 'Tags' field has a 'New tag (enter key to add)' button and an 'Add' button. The 'Author' field is set to 'Pradeep Kumar'. The 'Date added' field is set to 'Sat Nov 08 2025 15:14 GMT+0530'. The 'Share query with all users' checkbox is checked. At the bottom of the dialog box, there are 'Cancel' and 'Save' buttons. The 'Save' button is highlighted with a red box.

https://pradeepnec.grafana.net/dashboard/new?orgId=1&from=now-6h&to=now&timezone=browser&editPanel=1

Grafana

Home > Dashboards > New dashboard > Edit panel

Table view

New panel

Queries 1 Transformations 0 Alert 0

Data source Influxdb

Save query

1 from(bucket: "SensorData")
2 |> range(start: -6h)
3 |> filter(fn: (r) => r._measurement ==
4 |> filter(fn: (r) => r._field == "light
5

Flux language syntax Sample query

Saved queries

All Favorites Recent

Give feedback

Search by data source, query content, title, or description Newest first

Filter by data source Filter by user name Filter by tag

LDR query New

New query

Title LDR query

from(bucket: "SensorData")
|> range(start: -6h)
|> filter(fn: (r) => r._measurement ==
"esp32_sensor_data")
|> filter(fn: (r) => r._field == "light_percent")

Data source influxdb

Description

Tags New tag (enter key to add) Add

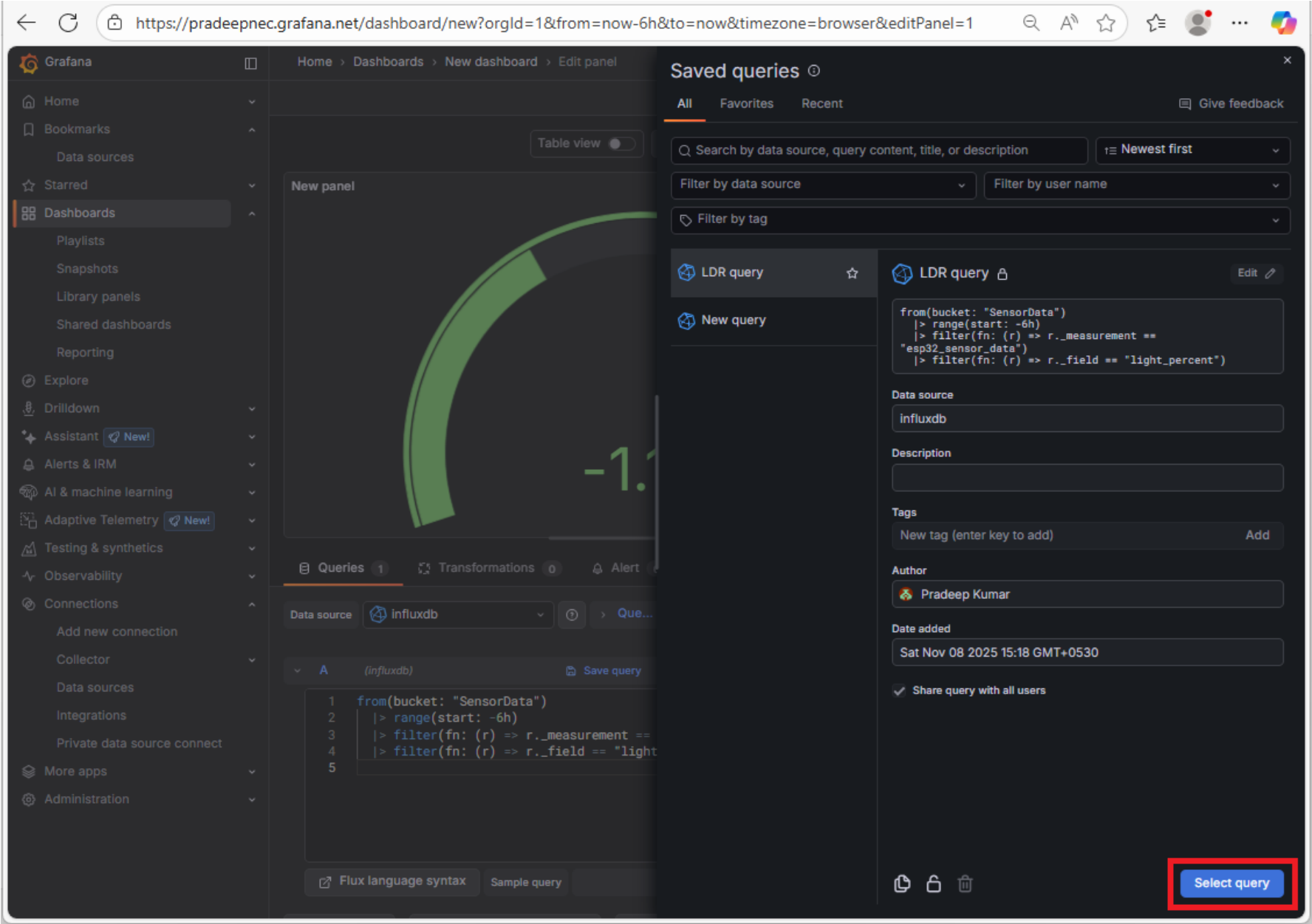
Author Pradeep Kumar

Date added Sat Nov 08 2025 15:14 GMT+0530

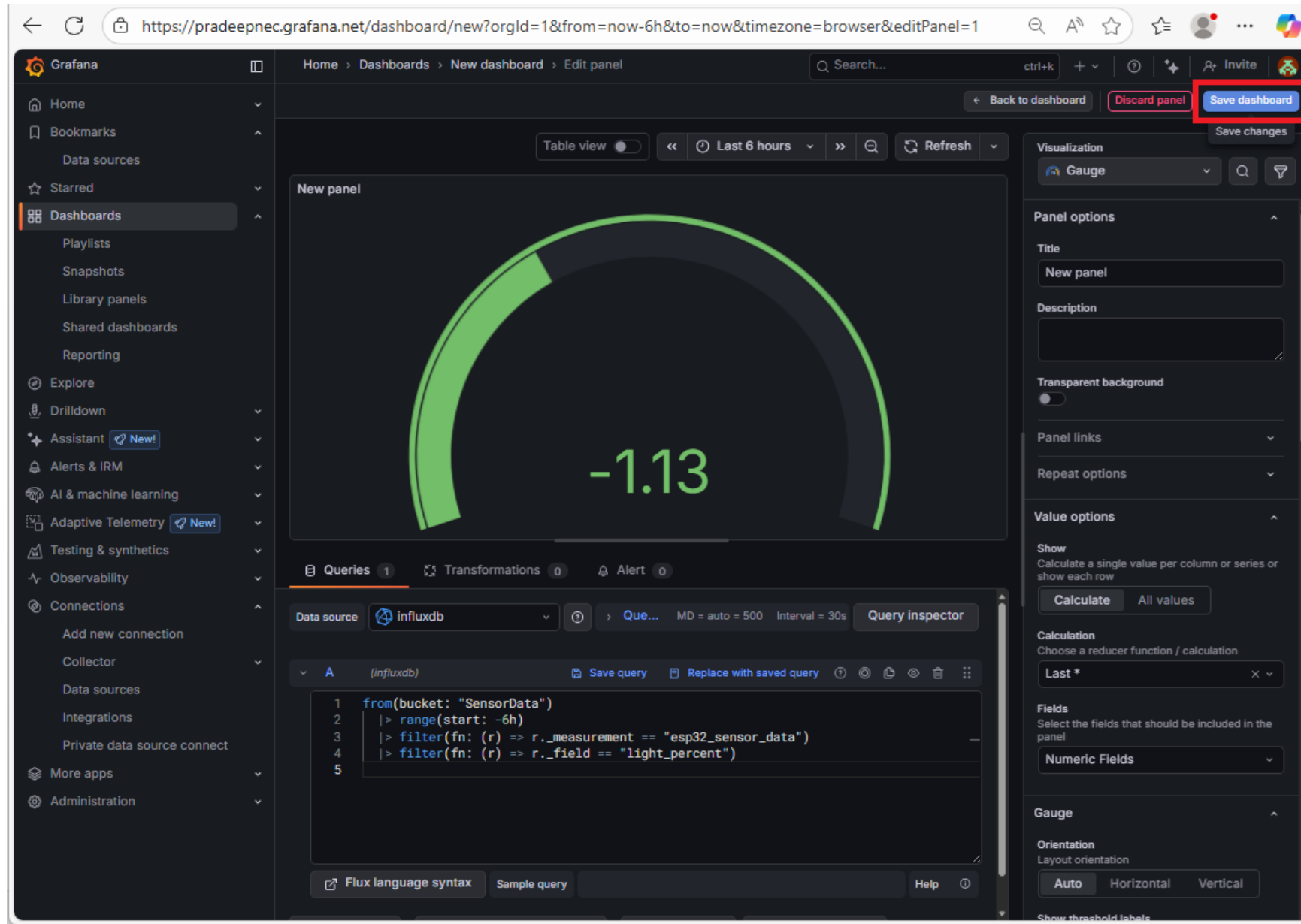
☒ Share query with all users

Cancel Save

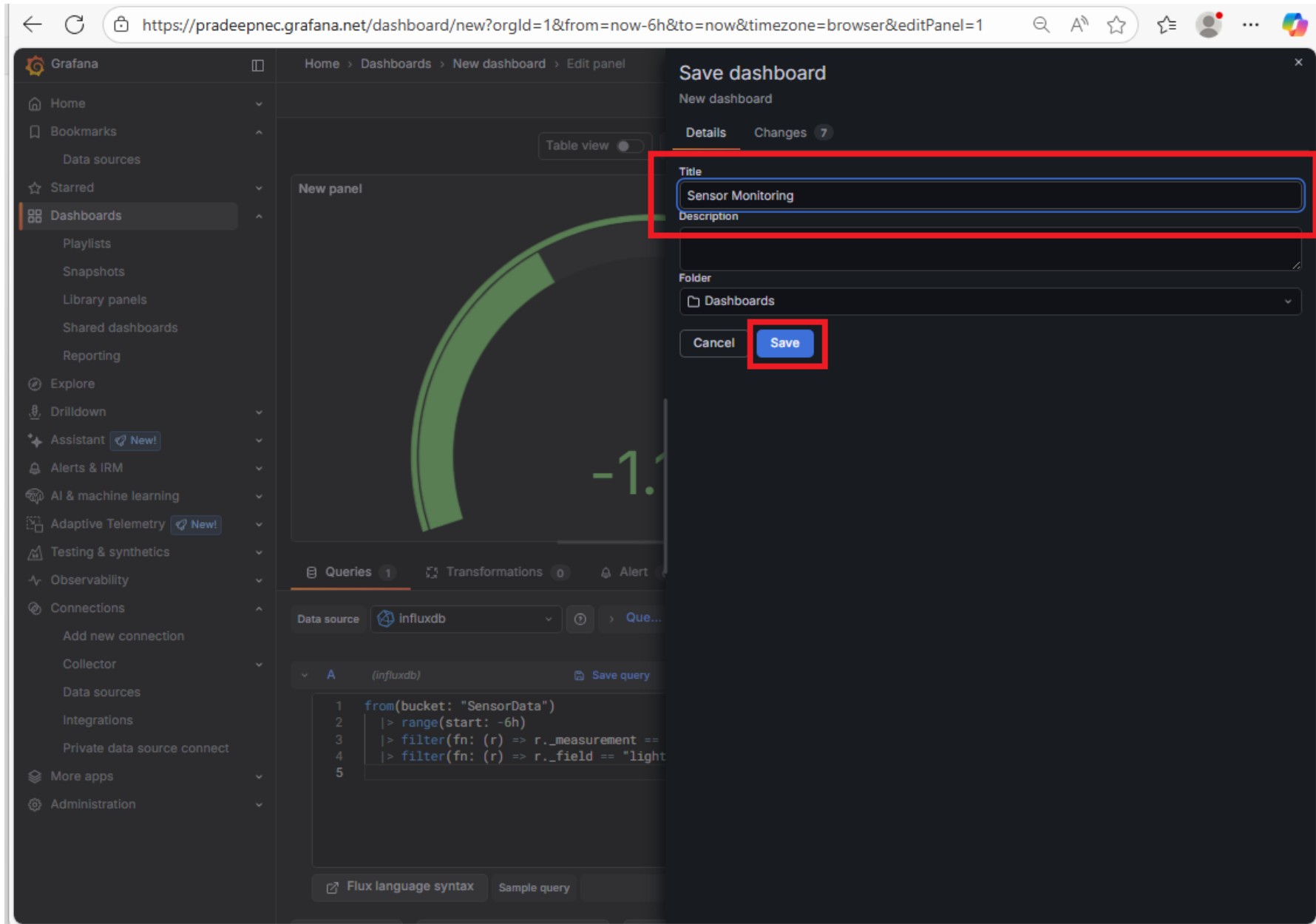
- After saving the query, the query selection window will open. Click the *Select query* option.



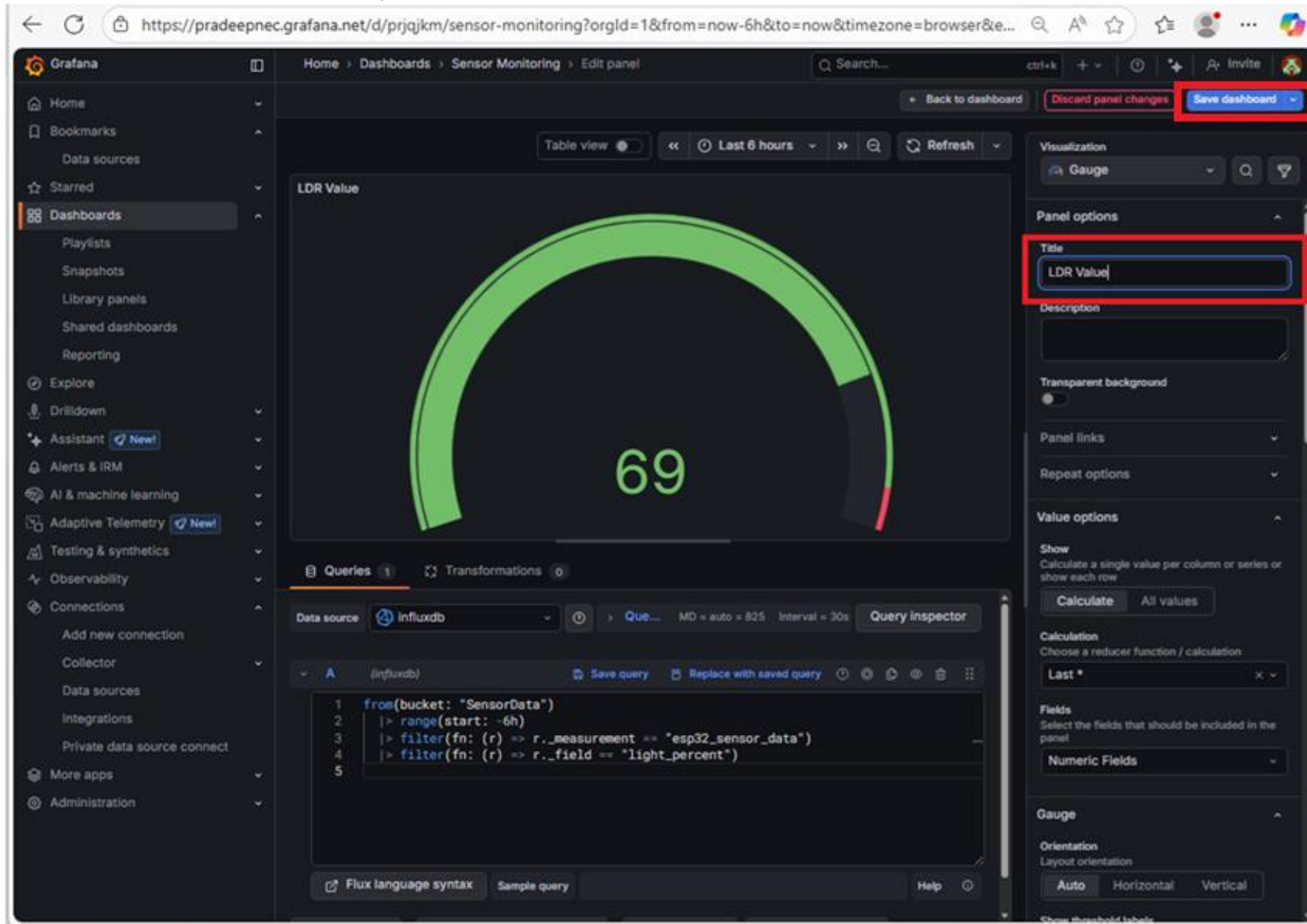
- Then click *Save dashboard* in the upper-right corner of the Grafana window.



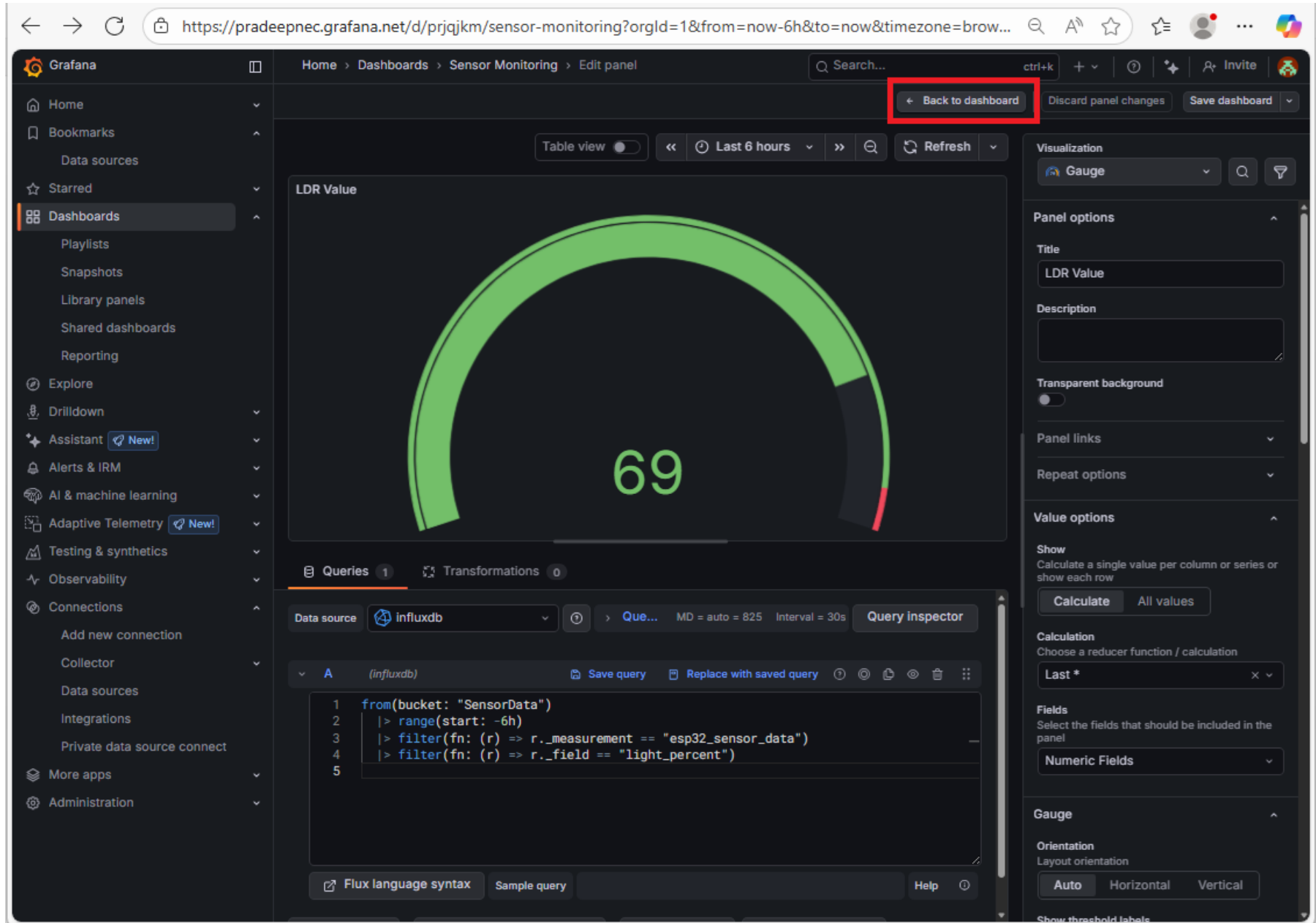
- After clicking the *Save dashboard* option, the dashboard saving window will open. Enter a dashboard name and click *Save*.



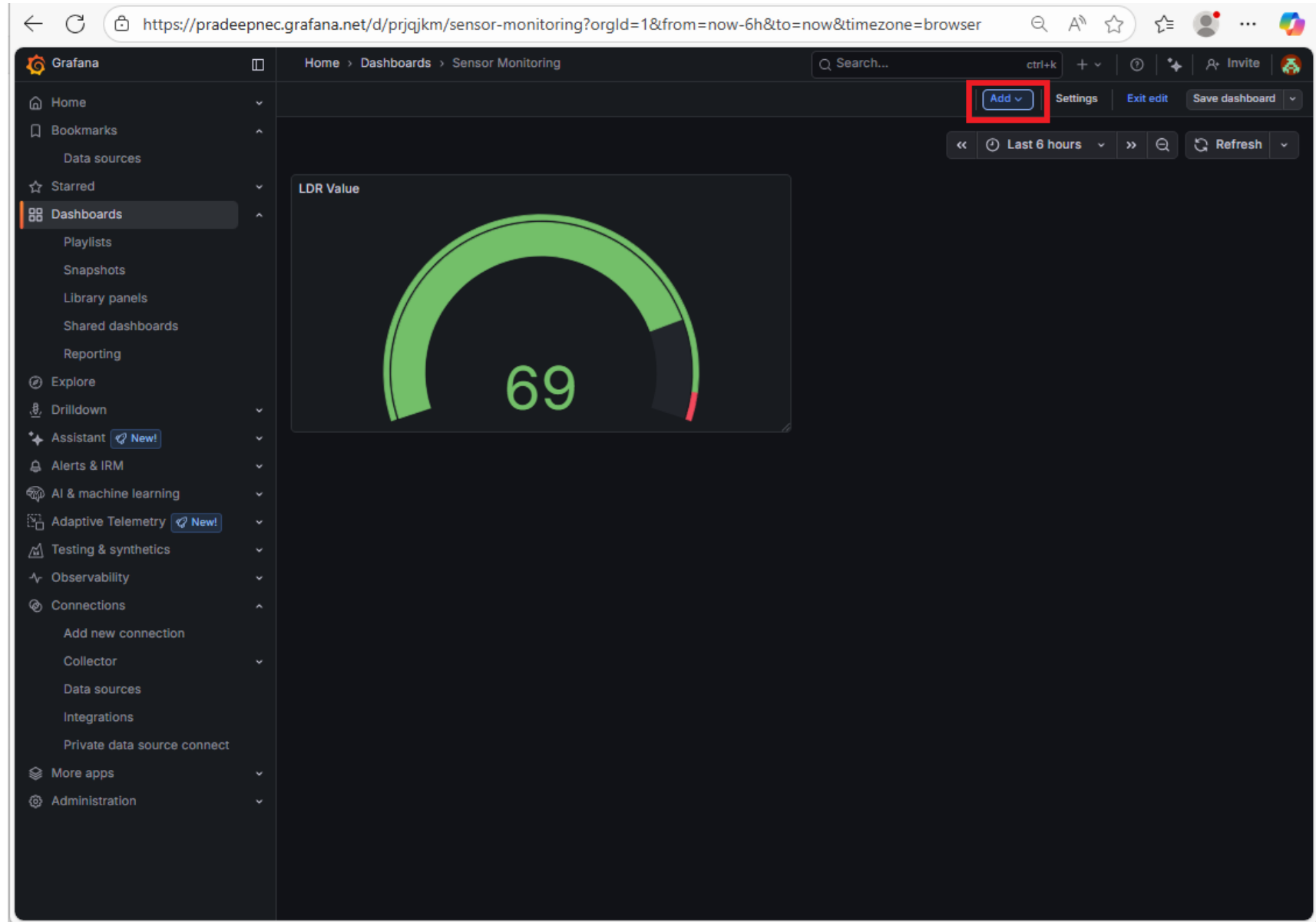
- To rename the visualization panel, go to the Panel options on the right-side panel, enter the new name under the *Title* field, and then click *Save dashboard*.



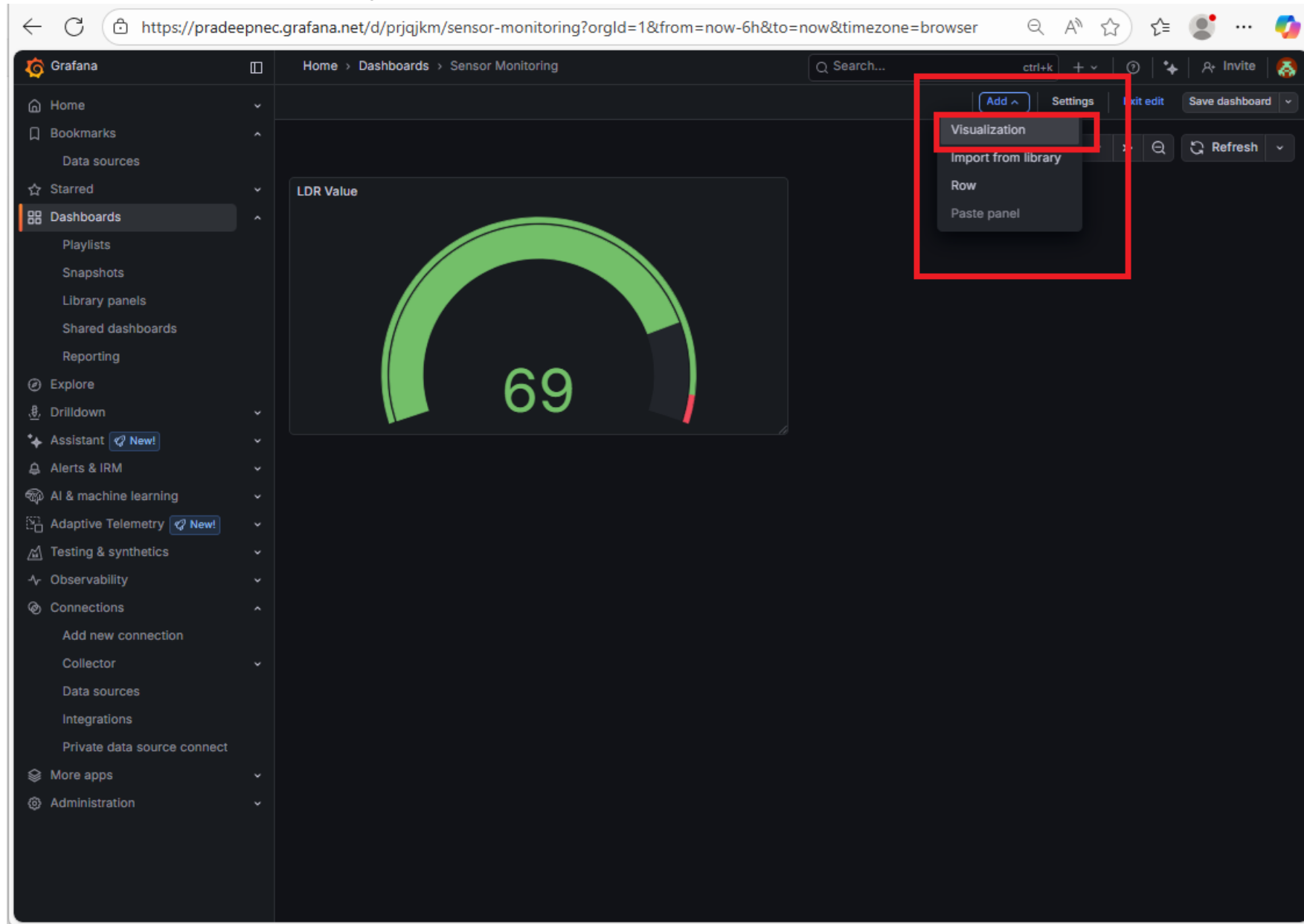
- Then click *Back to dashboard* to return to the dashboard and add a new panel.



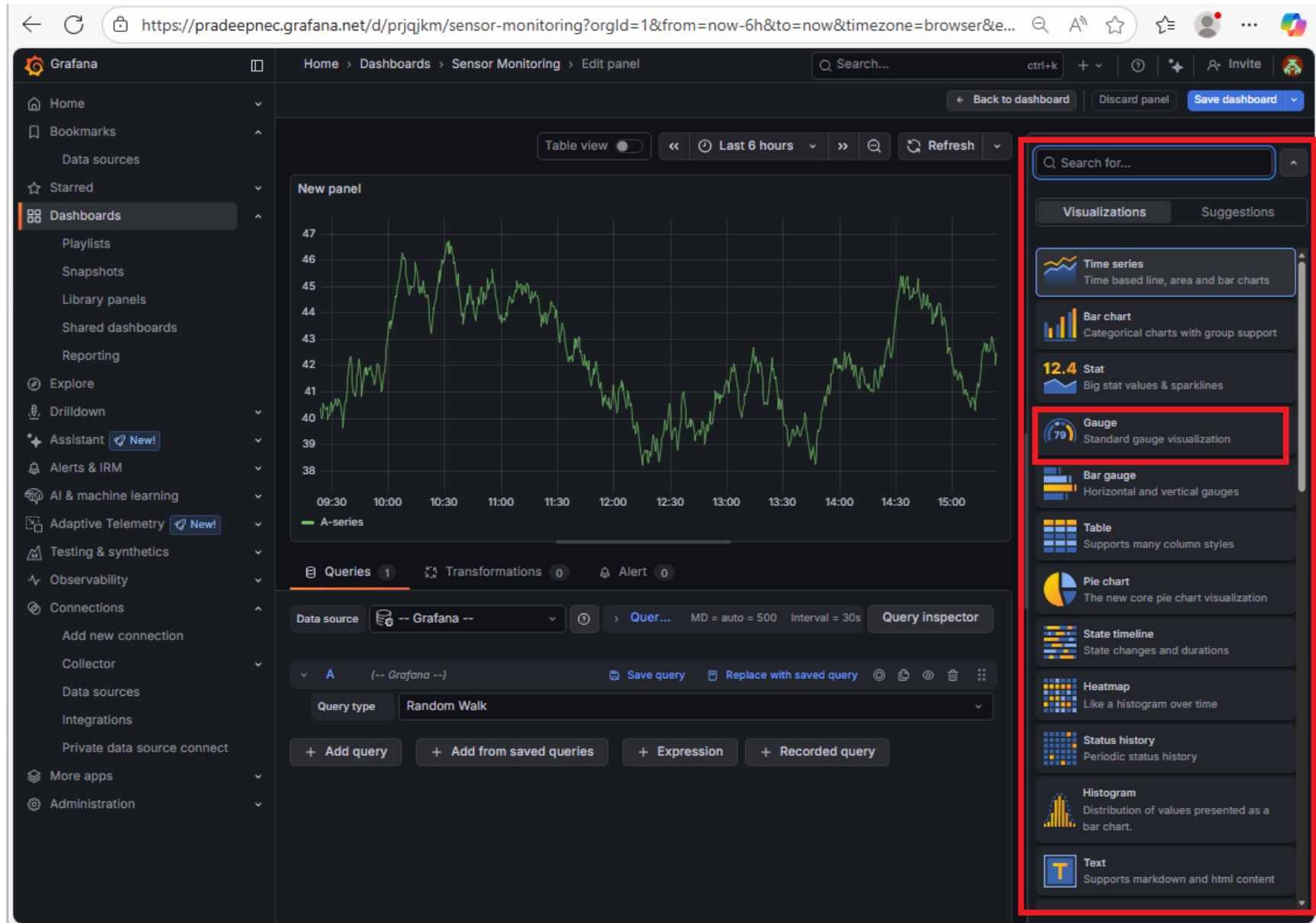
- In the Sensor Monitoring dashboard, click the *Add* option to create a new panel for visualization.



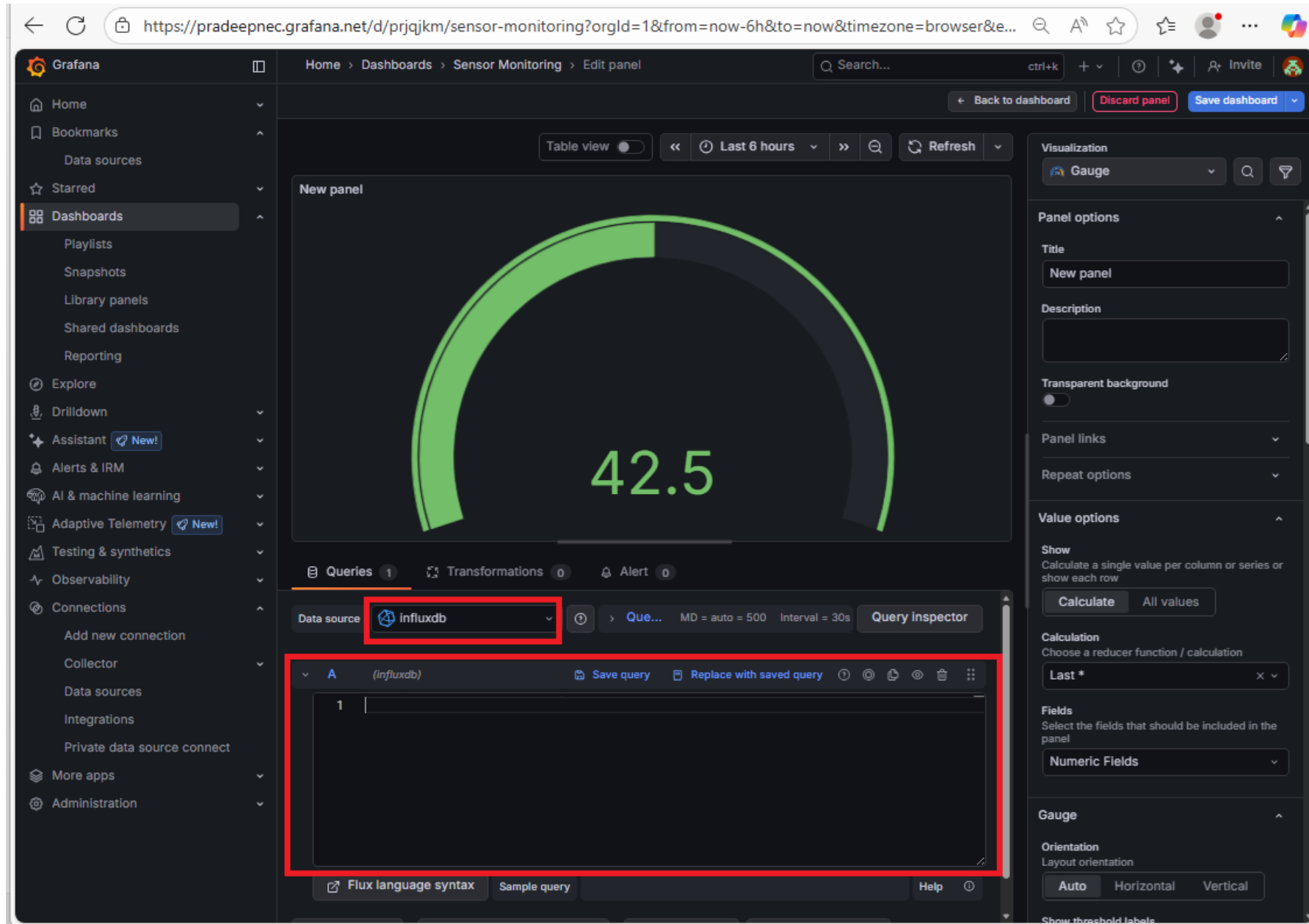
- In the Sensor Monitoring dashboard, go to the top-right corner and click *Add*, then select *Visualization* to create a new panel for data visualization.



- On the right-side panel of the New Panel window, scroll down under the *Visualization* section. Locate the *Time series* option, click the drop-down arrow, and select *Gauge*.



- The Gauge visualization window will open. Verify that the data source is set to InfluxDB in the data source option.



Flux query for Potentiometer

```
from(bucket: "SensorData")  
  |> range(start: -6h)  
  |> filter(fn: (r) => r._measurement == "esp32_sensor_data")  
  |> filter(fn: (r) => r._field == "pot_percent")
```


- In the Flux query editor, copy and paste the query into the query input area. Then click *Save query* to save the query.

The screenshot shows the Grafana web interface. The main panel displays a gauge visualization with a green needle pointing to the value 42.5. The gauge is titled 'New panel'. Below the gauge, the 'Queries' tab is active, showing a single query for the 'Influxdb' data source. The query is highlighted with a red box and is as follows:

```
1 from(bucket: "SensorData")
2   |> range(start: -6h)
3   |> filter(fn: (r) => r._measurement == "esp32_sensor_data")
4   |> filter(fn: (r) => r._field == "pot_percent")
5
```

Below the query editor, there are buttons for 'Flux language syntax', 'Sample query', and 'Help'. On the right side of the interface, the 'Visualization' panel is open, showing options for the gauge, including 'Panel options', 'Value options', and 'Gauge' settings.

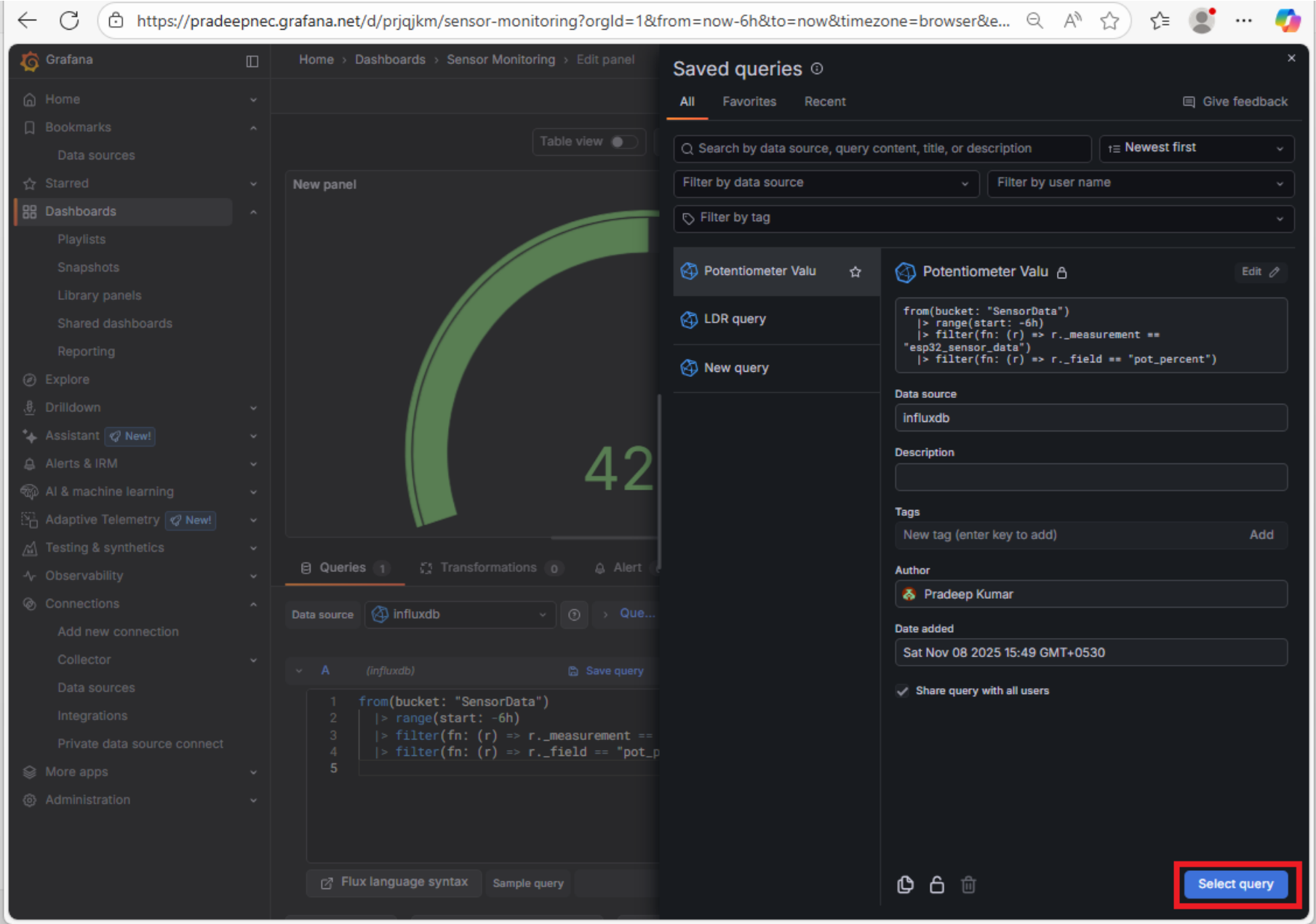
- After clicking the *Save query* option, a window will open where you can enter the query name. Then click *Save* to confirm.

The screenshot shows the Grafana web interface with a 'Saved queries' modal window open. The modal has a dark theme and contains the following elements:

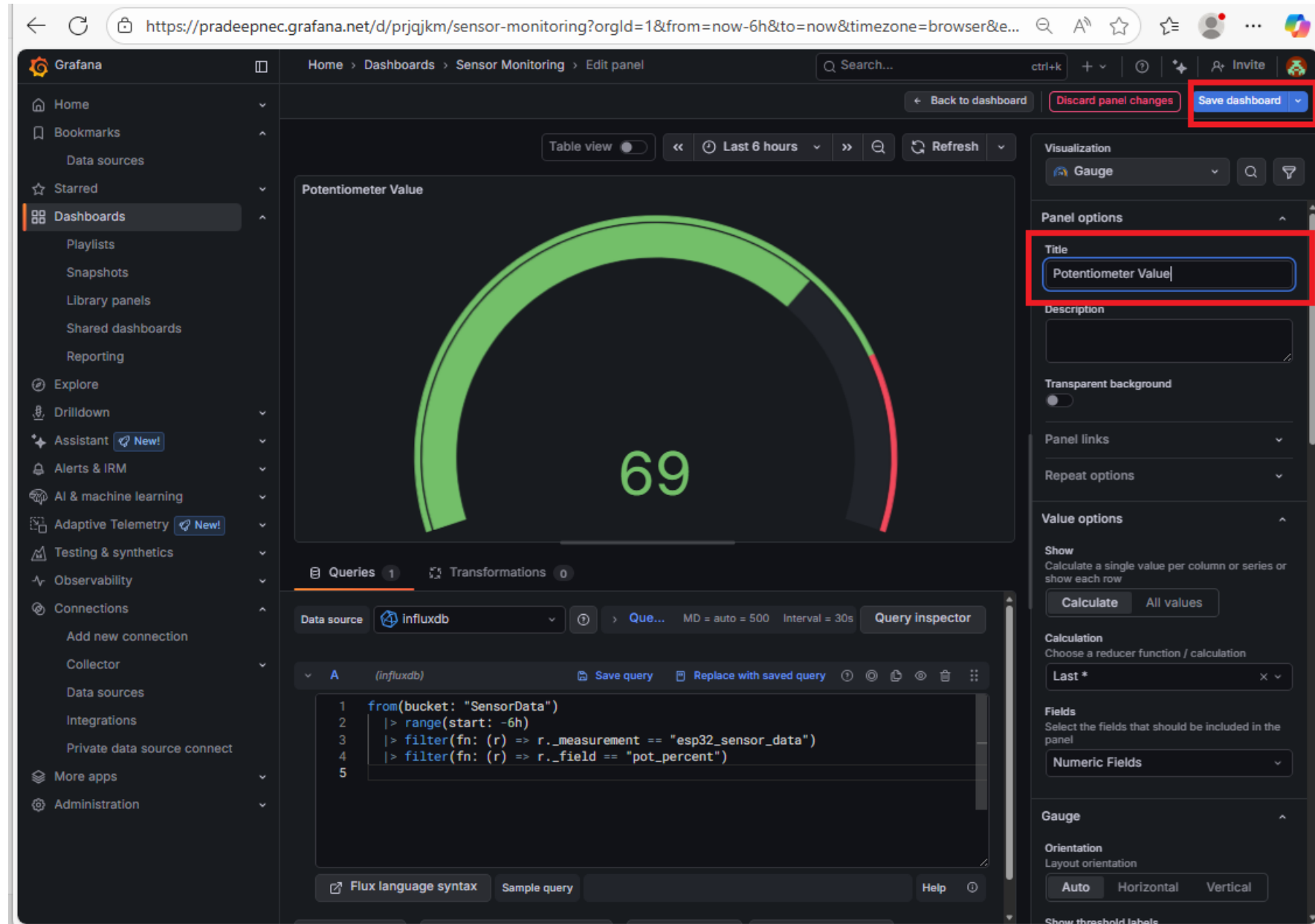
- Title:** A text input field containing 'Potentiometer Valu', which is highlighted with a red rectangle.
- Data source:** A dropdown menu set to 'influxdb'.
- Description:** An empty text input field.
- Tags:** A section with a 'New tag (enter key to add)' input and an 'Add' button.
- Author:** A text input field containing 'Pradeep Kumar'.
- Date added:** A text input field containing 'Sat Nov 08 2025 15:47 GMT+0530'.
- Share query with all users:** A checked checkbox.
- Buttons:** 'Cancel' and 'Save' buttons at the bottom right, with the 'Save' button highlighted by a red rectangle.

The background shows a Grafana dashboard with a 'Sensor Monitoring' panel. The panel displays a green arc gauge and the number '42'. The left sidebar contains navigation links like 'Home', 'Dashboards', 'Explore', and 'Administration'. The top of the dashboard shows the URL: `https://pradeepnec.grafana.net/d/prjqkm/sensor-monitoring?orgId=1&from=now-6h&to=now&timezone=browser&e...`

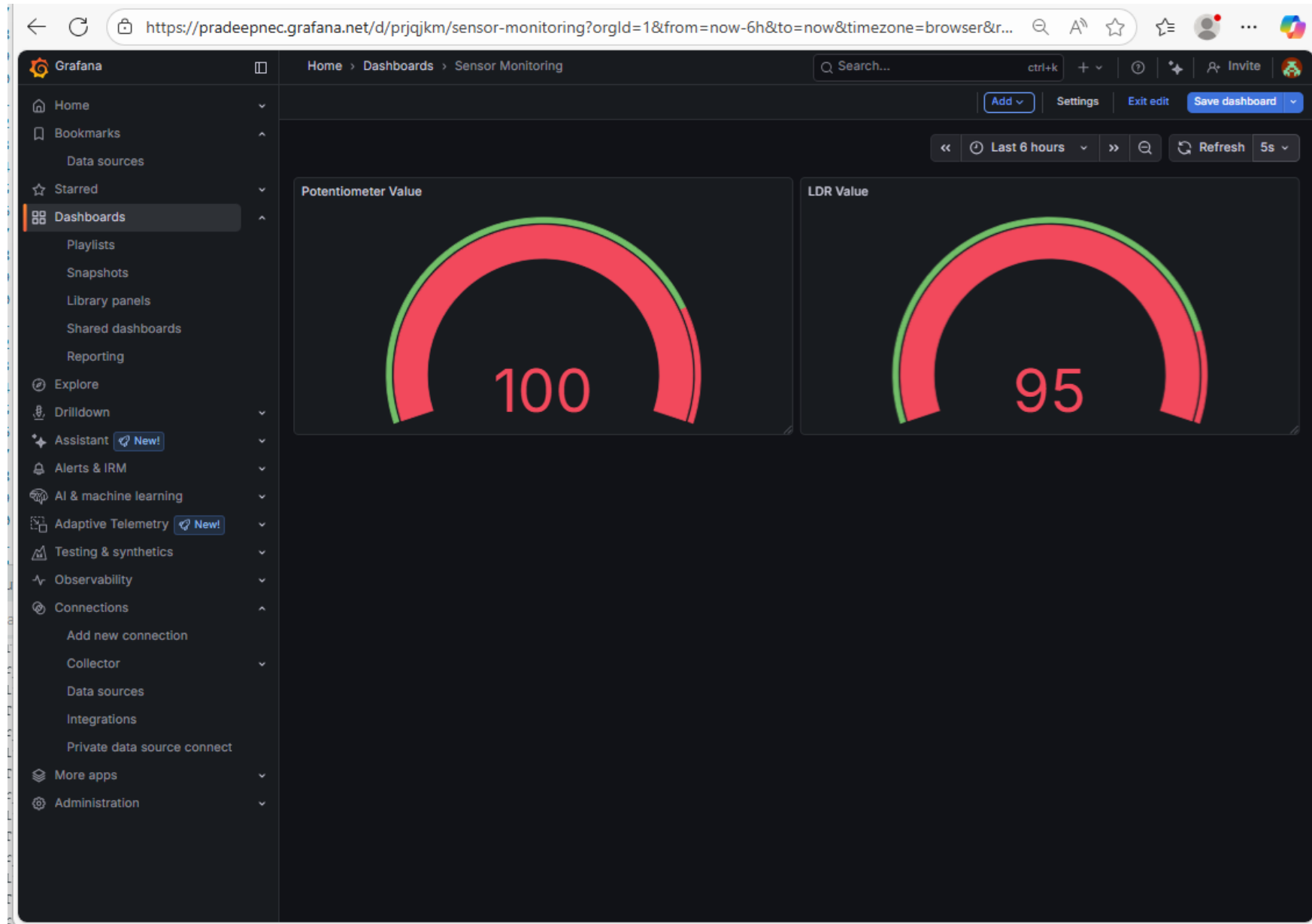
- After saving the query, the query selection window will open. Click the *Select query* option.



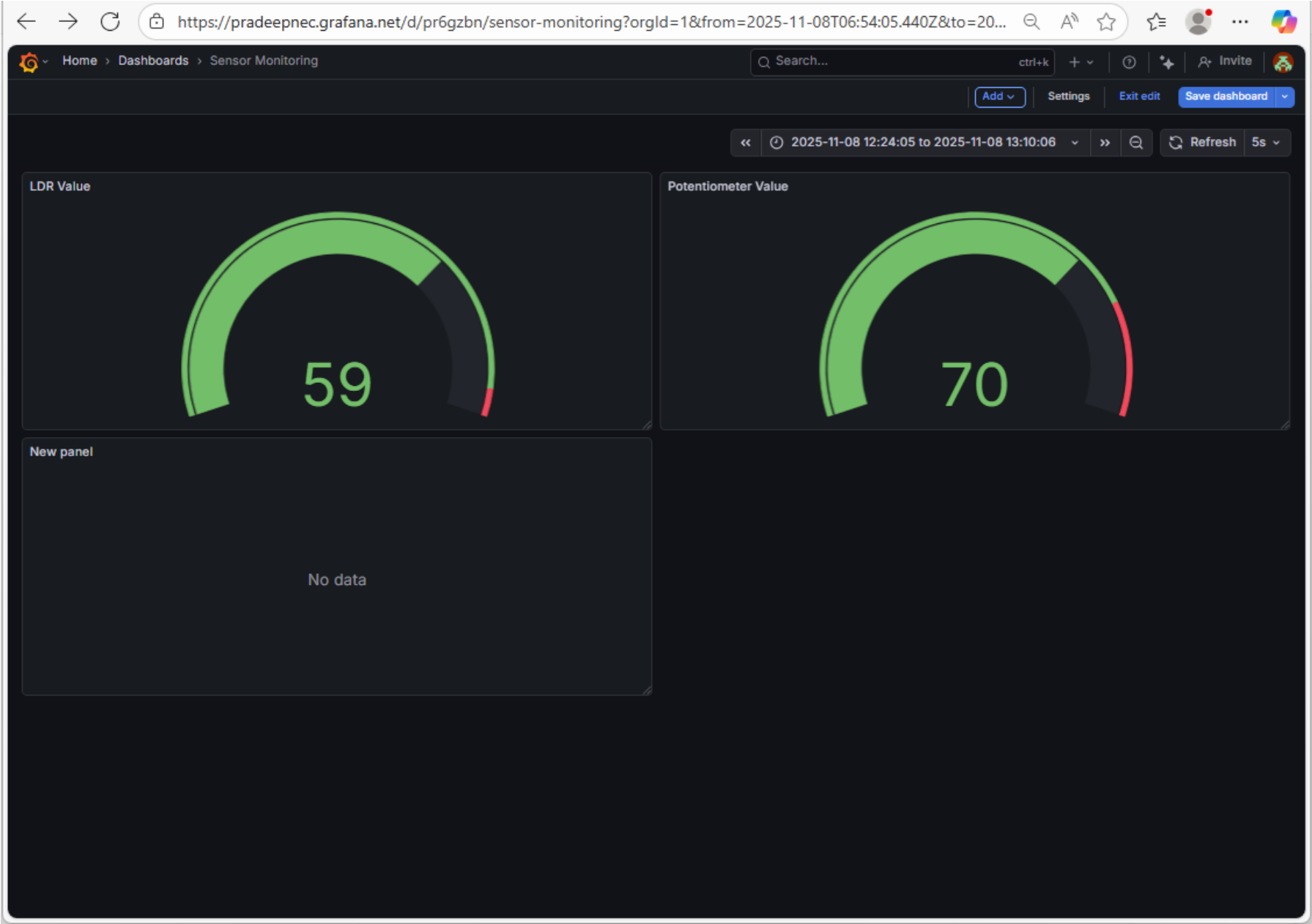
- To rename the visualization panel, go to the Panel options on the right-side panel, enter the new name under the *Title* field, and then click *Save dashboard*.



- After selecting the Gauge visualization, Grafana will fetch the sensor values from the InfluxDB data source and display them on the panel.



Full Screen Dashboard



Thank You